

Ordinary Differential Equations

Unit II — Lecture Notes

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1 Differential Equations: Basic Concepts

- An equation which contains differential coefficients (derivatives) is known as a *differential equation*.
- If the differential coefficients involve only ordinary derivatives, then it is called an *ordinary differential equation* (ODE).
- This chapter deals with the solution of different types of ODEs.

Definition 1.1 (Order). The order of the highest-order derivative in the given differential equation is known as the *order* of the differential equation.

Definition 1.2 (Degree). The power of the highest-order derivative (after the equation is made free of radicals and fractions in the derivatives) is known as the *degree* of the differential equation.

Based on order and degree, let us see some equations:

	Equation	Order	Degree
(1)	$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 4y = 0$	2	1
(2)	$\left(\frac{dy}{dx}\right)^3 + 3\left(\frac{dy}{dx}\right) + 4y = \sin x$	1	3
(3)	$\rho = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{d^2y/dx^2}$	2	2

For (3), squaring and clearing the fraction gives $\left(\rho \frac{d^2y}{dx^2}\right)^2 = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^3$, so the highest derivative $\frac{d^2y}{dx^2}$ appears with power 2.

2 Linear Differential Equations of the First Order

A differential equation of order 1 and degree 1 which is linear in y takes the general form

$$\frac{dy}{dx} + Py = Q, \quad P, Q \text{ functions of } x \text{ only.}$$

Solution:

$$y e^{\int P dx} = \int e^{\int P dx} Q dx + c,$$

where $e^{\int P dx}$ is called the *integrating factor* (I.F.) of the given differential equation.

3 Bernoulli's Differential Equation

- A differential equation of the form $\frac{dy}{dx} + Py = y^n Q$ is known as *Bernoulli's differential equation*.
- To get the solution, divide the above equation by y^n :

$$y^{-n} \frac{dy}{dx} + Py^{1-n} = Q.$$

- Then put $z = y^{1-n}$; find $\frac{dz}{dx} = (1-n)y^{-n} \frac{dy}{dx}$. The above equation is converted into a linear differential equation in z , which is solved as in the previous section.

3.1 Problems

Problem 3.1. Solve $x \frac{dy}{dx} + y = y^2 \log x$.

Solution. Dividing by x :

$$\frac{dy}{dx} + \frac{y}{x} = y^2 \frac{\log x}{x} \quad (\text{Bernoulli's equation}).$$

Dividing by y^2 :

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = \frac{\log x}{x}. \quad (1)$$

Put $z = y^{-1}$. Then

$$\frac{dz}{dx} = -1 \cdot y^{-2} \frac{dy}{dx} = -\frac{1}{y^2} \frac{dy}{dx}.$$

So (1) becomes

$$-\frac{dz}{dx} + \frac{z}{x} = \frac{\log x}{x} \Rightarrow \frac{dz}{dx} + \left(-\frac{1}{x}\right)z = -\frac{\log x}{x},$$

which is linear in z with $P = -\frac{1}{x}$, $Q = -\frac{\log x}{x}$.

Integrating factor: $e^{\int P dx} = e^{-\int \frac{1}{x} dx} = e^{-\log x} = e^{\log x^{-1}} = x^{-1} = \frac{1}{x}$.

The solution is $z e^{\int P dx} = \int e^{\int P dx} Q dx + c$:

$$z x^{-1} = \int x^{-1} \left(-\frac{\log x}{x}\right) dx + c \Rightarrow \frac{z}{x} = -\int \frac{\log x}{x^2} dx + c = -\int \underbrace{\log x}_u \underbrace{x^{-2}}_v dx + c.$$

Using integration by parts, $\int uv dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx\right) dx$:

$$\begin{aligned} \int \log x \cdot x^{-2} dx &= \log x \int x^{-2} dx - \int \left(\frac{d}{dx}(\log x) \int x^{-2} dx\right) dx = \log x \left(-\frac{1}{x}\right) - \int \frac{1}{x} \left(-\frac{1}{x}\right) dx \\ &= -\frac{\log x}{x} + \int x^{-2} dx = -\frac{\log x}{x} - \frac{1}{x} = -\left(\frac{\log x + 1}{x}\right). \end{aligned}$$

Therefore

$$\frac{z}{x} = -\left(-\frac{\log x + 1}{x}\right) + c = \frac{\log x}{x} + \frac{1}{x} + c \Rightarrow z = \log x + 1 + cx.$$

Since $z = y^{-1}$:

$$\frac{1}{y} = \log x + 1 + cx \Rightarrow \boxed{y = \frac{1}{\log x + 1 + cx}}.$$

Problem 3.2. Solve $(1 - x^2) \frac{dy}{dx} + xy = y^3 \sin x$.

Solution. Dividing by $(1 - x^2)$:

$$\frac{dy}{dx} + \frac{xy}{1 - x^2} = y^3 \frac{\sin x}{1 - x^2}.$$

Dividing by y^3 :

$$y^{-3} \frac{dy}{dx} + \frac{xy}{(1 - x^2)y^3} = \frac{\sin x}{1 - x^2} \Rightarrow y^{-3} \frac{dy}{dx} + \frac{xy^{-2}}{1 - x^2} = \frac{\sin x}{1 - x^2}. \quad (1)$$

Put $z = y^{-2}$; then $\frac{dz}{dx} = -2y^{-3} \frac{dy}{dx}$, i.e. $y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dz}{dx}$. Equation (1) becomes

$$-\frac{1}{2} \frac{dz}{dx} + \frac{x}{1 - x^2} z = \frac{\sin x}{1 - x^2} \Rightarrow \frac{dz}{dx} - \frac{2x}{1 - x^2} z = -\frac{2 \sin x}{1 - x^2},$$

which is linear in z with $P = -\frac{2x}{1 - x^2}$, $Q = -\frac{2 \sin x}{1 - x^2}$.

Using $\int \frac{f'(x)}{f(x)} dx = \log |f(x)|$:

$$\int P dx = \int \frac{-2x}{1 - x^2} dx = \log(1 - x^2), \quad e^{\int P dx} = e^{\log(1 - x^2)} = 1 - x^2.$$

The solution is

$$\begin{aligned} z e^{\int P dx} &= \int e^{\int P dx} Q dx + c \Rightarrow z(1 - x^2) = \int (1 - x^2) \left(\frac{-2 \sin x}{1 - x^2} \right) dx + c \\ &\Rightarrow z(1 - x^2) = -2 \int \sin x dx + c = -2(-\cos x) + c = 2 \cos x + c \\ &\Rightarrow z = \frac{2 \cos x}{1 - x^2} + \frac{c}{1 - x^2} = \frac{2 \cos x + c}{1 - x^2}. \end{aligned}$$

Since $z = y^{-2}$:

$$\frac{1}{y^2} = \frac{2 \cos x + c}{1 - x^2} \Rightarrow \boxed{y^2 = \frac{1 - x^2}{2 \cos x + c}}.$$

4 Exact Differential Equations

The differential equation of the form

$$M dx + N dy = 0, \quad M, N \text{ functions of } x \text{ and } y,$$

is said to be *exact* if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

We can solve an exact differential equation by the following steps:

- (i) Integrate M with respect to x (keeping y constant).
- (ii) In N , choose the terms which are free from x , and integrate them with respect to y .
- (iii) Then the solution is (i) + (ii) = constant.

4.1 Problems

Problem 4.1. Check whether the differential equation $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$ is exact. Hence solve it.

Solution.

$$\begin{aligned} \frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} &\Rightarrow (\sin x + x \cos y + x) dy = -(y \cos x + \sin y + y) dx \\ &\Rightarrow (y + y \cos x + \sin y) dx + (x + x \cos y + \sin x) dy = 0, \end{aligned}$$

which is of the form $M dx + N dy = 0$ with

$$\begin{aligned} M &= y + y \cos x + \sin y, & N &= x + x \cos y + \sin x. \\ \frac{\partial M}{\partial y} &= 1 + \cos x + \cos y, & \frac{\partial N}{\partial x} &= 1 + \cos y + \cos x. \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the given differential equation is exact.

Integrate M with respect to x , keeping y constant:

$$\int M dx = \int (y + y \cos x + \sin y) dx = xy + y \sin x + x \sin y. \quad (1)$$

In N , choose the terms which are free from x and integrate with respect to y : there are no terms free from x , so

$$\int N dy = 0. \quad (2)$$

(1) + (2) = c :

$$\boxed{y \sin x + x \sin y + xy = c}.$$

Problem 4.2. Solve $(y^2 e^{xy^2} + 4x^3) dx + (2xy e^{xy^2} - 3y^2) dy = 0$.

Solution. Checking exactness, with $M dx + N dy = 0$:

$$\begin{aligned} M &= y^2 e^{xy^2} + 4x^3, & N &= 2xy e^{xy^2} - 3y^2. \\ \frac{\partial M}{\partial y} &= 2y e^{xy^2} + y^2 e^{xy^2} \cdot 2xy = 2y e^{xy^2} + 2xy^3 e^{xy^2}, \\ \frac{\partial N}{\partial x} &= 2y e^{xy^2} \cdot 1 + 2xy e^{xy^2} \cdot y^2 = 2y e^{xy^2} + 2xy^3 e^{xy^2}. \end{aligned}$$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so the equation is exact.

$$\int M dx = \int (y^2 e^{xy^2} + 4x^3) dx = y^2 \frac{e^{xy^2}}{y^2} + 4 \cdot \frac{x^4}{4} = e^{xy^2} + x^4. \quad (1)$$

Terms of N free from x : $-3y^2$.

$$\int (-3y^2) dy = -3 \frac{y^3}{3} = -y^3. \quad (2)$$

(1) + (2) = c :

$$\boxed{e^{xy^2} + x^4 - y^3 = c}.$$

5 First-Order Differential Equations of Higher Degree

The general form is

$$a_0 \left(\frac{dy}{dx} \right)^n + a_1 \left(\frac{dy}{dx} \right)^{n-1} + \cdots + a_{n-1} \left(\frac{dy}{dx} \right) + a_n = f(x),$$

where $a_0, a_1, \dots, a_n$ may be constants or functions of x . If $p = \frac{dy}{dx}$, then

$$a_0 p^n + a_1 p^{n-1} + \cdots + a_{n-1} p + a_n = f(x).$$

The above equation can be solved by any one of the following methods:

1. Equations solvable for p ;
2. Equations solvable for y ;
3. Equations solvable for x ;
4. Clairaut's equation.

5.1 Equations solvable for p

- Suppose the given differential equation can be considered as a polynomial equation of degree 2 or 3 (etc.) in p ; then we can use this procedure.
- In this method, let $\alpha_1, \alpha_2, \dots, \alpha_n$ be the roots of the polynomial equation in p . Then

$$(p - \alpha_1)(p - \alpha_2)(p - \alpha_3) \cdots (p - \alpha_n) = 0,$$

so $\frac{dy}{dx} = \alpha_1$ or $\frac{dy}{dx} = \alpha_2$ or $\dots$ or $\frac{dy}{dx} = \alpha_n$. On integrating,

$$y = \alpha_1 x + c \quad \text{or} \quad y = \alpha_2 x + c \quad \text{or} \quad \dots \quad \text{or} \quad y = \alpha_n x + c.$$

General solution: $(y - \alpha_1 x - c)(y - \alpha_2 x - c) \cdots (y - \alpha_n x - c) = 0$.

Since the equation is of order 1, only one arbitrary constant c is used.

Problem 5.1. Solve $p^2 - 9p + 18 = 0$, where $p = \frac{dy}{dx}$.

Solution. This is a quadratic equation in p , so it is solvable for p .

$$p^2 - 9p + 18 = 0 \Rightarrow (p - 3)(p - 6) = 0 \quad (\text{roots } 3, 6).$$

$$p = \frac{dy}{dx} = 3 \Rightarrow y = 3x + c; \quad p = \frac{dy}{dx} = 6 \Rightarrow y = 6x + c.$$

General solution:

$$(y - 3x - c)(y - 6x - c) = 0.$$

Problem 5.2. Solve $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$, where $p = \frac{dy}{dx}$.

Solution. This is a quadratic equation in p ; using $p = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$\begin{aligned} p &= \frac{-(3x^2 - 2y^2) \pm \sqrt{(3x^2 - 2y^2)^2 - 4(xy)(-6xy)}}{2xy} \\ &= \frac{-(3x^2 - 2y^2) \pm \sqrt{9x^4 + 4y^4 - 12x^2y^2 + 24x^2y^2}}{2xy} \\ &= \frac{-(3x^2 - 2y^2) \pm \sqrt{9x^4 + 4y^4 + 12x^2y^2}}{2xy} \\ &= \frac{-(3x^2 - 2y^2) \pm \sqrt{(3x^2 + 2y^2)^2}}{2xy} \\ &= \frac{-3x^2 + 2y^2 \pm (3x^2 + 2y^2)}{2xy}. \end{aligned}$$

(1) Taking +:

$$p = \frac{-3x^2 + 2y^2 + 3x^2 + 2y^2}{2xy} = \frac{4y^2}{2xy} = \frac{2y}{x} \Rightarrow \frac{dy}{dx} = \frac{2y}{x} \Rightarrow \frac{dy}{y} = 2 \frac{dx}{x}.$$

Integrating: $\log y = 2 \log x + \log c = \log x^2 + \log c \Rightarrow y = cx^2$.

(2) Taking -:

$$p = \frac{-3x^2 + 2y^2 - 3x^2 - 2y^2}{2xy} = \frac{-6x^2}{2xy} = -\frac{3x}{y} \Rightarrow \frac{dy}{dx} = -\frac{3x}{y} \Rightarrow y dy = -3x dx.$$

Integrating: $\frac{y^2}{2} = -\frac{3x^2}{2} + c \Rightarrow y^2 = -3x^2 + 2c$.

General solution:

$$(y - cx^2)(y^2 + 3x^2 - 2c) = 0.$$

Problem 5.3. Solve $4p^2 - 8p + 3 = 0$, where $p = \frac{dy}{dx}$.

Solution. Splitting the middle term $(-6p - 2p)$:

$$4p^2 - 6p - 2p + 3 = 0 \Rightarrow 2p(2p - 3) - 1(2p - 3) = 0 \Rightarrow (2p - 3)(2p - 1) = 0,$$

so $p = \frac{3}{2}$ or $p = \frac{1}{2}$.

$$\frac{dy}{dx} = \frac{3}{2} \Rightarrow y = \frac{3}{2}x + c; \quad \frac{dy}{dx} = \frac{1}{2} \Rightarrow y = \frac{1}{2}x + c.$$

General solution:

$$\left(y - \frac{3}{2}x - c\right) \left(y - \frac{1}{2}x - c\right) = 0.$$

Problem 5.4. Solve $p(p + 2y) = x(x + 2y)$, where $p = \frac{dy}{dx}$.^a

^aThe notes write the right-hand side as $x(x + y)$, but the working (a perfect square under the root, giving $p = x$ and $p = -x - 2y$) corresponds to $x(x + 2y)$, which is used here.

Solution.

$$p^2 + 2yp = x^2 + 2xy \Rightarrow p^2 + 2yp - (x^2 + 2xy) = 0.$$

$$\begin{aligned} p &= \frac{-2y \pm \sqrt{4y^2 - 4(1)(-(x^2 + 2xy))}}{2(1)} = \frac{-2y \pm \sqrt{4y^2 + 4x^2 + 8xy}}{2} \\ &= \frac{-2y \pm 2\sqrt{x^2 + y^2 + 2xy}}{2} = -y \pm \sqrt{(x + y)^2} = -y \pm (x + y). \end{aligned}$$

(1) $p = -y + (x + y) = x$:

$$\frac{dy}{dx} = x \Rightarrow \int dy = \int x dx \Rightarrow y = \frac{x^2}{2} + c \Rightarrow 2y = x^2 + 2c.$$

(2) $p = -y - (x + y) = -2y - x$:

$$\frac{dy}{dx} = -2y - x \Rightarrow \frac{dy}{dx} + 2y = -x,$$

which is linear with $P = 2$, $Q = -x$. I.F. = $e^{\int 2 dx} = e^{2x}$.

$$y e^{2x} = \int e^{2x}(-x) dx + c = -\int x e^{2x} dx + c.$$

By parts (ILATE, $u = x$, $v = e^{2x}$):

$$\int x e^{2x} dx = x \int e^{2x} dx - \int \left(\frac{d}{dx}(x) \int e^{2x} dx \right) dx = \frac{x e^{2x}}{2} - \int \frac{e^{2x}}{2} dx = \frac{x e^{2x}}{2} - \frac{e^{2x}}{4}.$$

$$y e^{2x} = -\frac{e^{2x}}{2} \left(x - \frac{1}{2} \right) + c \Rightarrow y = -\frac{x}{2} + \frac{1}{4} + c e^{-2x} \Rightarrow 4y + 2x - 1 = 4c e^{-2x}.$$

General solution:¹

$$\boxed{(2y - x^2 - 2c)(4y + 2x - 1 - 4c e^{-2x}) = 0}.$$

5.2 Equations solvable for y

- Suppose the given differential equation can be written as

$$y = f(x, p), \quad p = \frac{dy}{dx}; \quad (1)$$

then we can consider it as *solvable for y* (only one “ y ” term appears in the equation).

- To get the solution, differentiate (1) with respect to x :

$$\frac{dy}{dx} = g\left(x, p, \frac{dp}{dx}\right) \Rightarrow p = g\left(x, p, \frac{dp}{dx}\right) \Rightarrow \phi\left(x, p, \frac{dp}{dx}\right) = 0.$$

¹The notes lose a sign in the last step of case (2); the result here follows from $y e^{2x} = -\frac{e^{2x}}{2} \left(x - \frac{1}{2} \right) + c$.

- By solving the above equation we get the values of p , and hence correspondingly the solution of the given equation.

Note. • Obtaining p values from the above equation, we may get p with an arbitrary constant c , and p without an arbitrary constant c .

- Solving with the p that contains the arbitrary constant c produces the *general solution*; the solution obtained using the p without an arbitrary constant is known as the *singular solution*.
- The singular solution is the *envelope* of the family of curves given by the general solution.

5.3 Equations solvable for x

- Suppose the given differential equation can be written in the form

$$x = f(y, p), \quad p = \frac{dy}{dx}; \quad (1)$$

then we can adopt the solvable-for- x procedure (only one “ x ” term appears).

- Differentiating (1) with respect to y , we get

$$\frac{dx}{dy} = g\left(y, p, \frac{dp}{dy}\right) \Rightarrow \frac{1}{p} = g\left(y, p, \frac{dp}{dy}\right) \Rightarrow \phi\left(y, p, \frac{dp}{dy}\right) = 0.$$

- Solving the above expression we get p values; substituting the values of p in (1), we get the general solution as well as the singular solution.

Problems

Problem 5.5. Solve $y = 2px + p^4x^2$, where $p = \frac{dy}{dx}$.

Solution. This is of the form $y = f(x, p)$, so we use the solvable-for- y method.

$$y = 2px + p^4x^2. \quad (1)$$

Differentiating with respect to x (product rule):

$$\frac{dy}{dx} = 2p(1) + 2x \frac{dp}{dx} + p^4(2x) + x^2 4p^3 \frac{dp}{dx} \Rightarrow p = 2p + 2x \frac{dp}{dx} + 2xp^4 + 4p^3x^2 \frac{dp}{dx}.$$

$$0 = p + 2x \frac{dp}{dx} + 2xp^4 + 4p^3x^2 \frac{dp}{dx} \Rightarrow 0 = \frac{dp}{dx} (2x + 4p^3x^2) + (p + 2xp^4)$$

$$\Rightarrow 0 = \frac{dp}{dx} (2x) (1 + 2p^3x) + p (1 + 2xp^3) \Rightarrow 0 = \left(2x \frac{dp}{dx} + p\right) (1 + 2xp^3) \quad (\text{common factor}).$$

Hence $1 + 2xp^3 = 0$ or $2x \frac{dp}{dx} + p = 0$.

General solution. From $2x \frac{dp}{dx} + p = 0$:

$$p = -2x \frac{dp}{dx} \Rightarrow -\frac{dx}{x} = 2 \frac{dp}{p}.$$

Integrating: $-\log x + \log c = 2 \log p \Rightarrow \log \frac{c}{x} = \log p^2 \Rightarrow p^2 = \frac{c}{x} \Rightarrow p = \sqrt{\frac{c}{x}}$ (with an arbitrary constant).

Substituting in (1):

$$y = 2\frac{\sqrt{c}}{\sqrt{x}}x + \frac{c^2}{x^2}x^2 \Rightarrow \boxed{y = 2\sqrt{cx} + c^2} \quad (\text{general solution}).$$

Singular solution. From $1 + 2xp^3 = 0$: $p^3 = -\frac{1}{2x}$ (no arbitrary constant). Writing (1) as $y = p(2x + p^3x^2)$ and cubing:

$$\begin{aligned} y^3 &= p^3 (2x + p^3x^2)^3 \\ &= -\frac{1}{2x} \left(2x - \frac{1}{2x}x^2\right)^3 \\ &= -\frac{1}{2x} \left(2x - \frac{x}{2}\right)^3 \\ &= -\frac{1}{2x} \left(\frac{3x}{2}\right)^3 \\ &= -\frac{1}{2x} \cdot \frac{27x^3}{8} = -\frac{27x^2}{16}. \end{aligned}$$

$$\boxed{16y^3 + 27x^2 = 0} \quad (\text{singular solution}).$$

Note. The above singular solution is the envelope of the family of curves given by the general solution of (1).

Problem 5.6. Solve $p^2x + py - y^4 = 0$, where $p = \frac{dy}{dx}$.

Solution.

$$p^2x + py - y^4 = 0 \quad (1)$$

$$\Rightarrow x = \frac{y^4}{p^2} - \frac{py}{p^2} = \frac{y^4}{p^2} - \frac{y}{p} \quad (x = f(y, p) : \text{solvable for } x). \quad (2)$$

Differentiating with respect to y (quotient rule $(\frac{u}{v})' = \frac{vu' - uv'}{v^2}$):

$$\begin{aligned} \frac{dx}{dy} &= \frac{p^2(4y^3) - y^4 \left(2p \frac{dp}{dy}\right)}{p^4} - \frac{p(1) - y \frac{dp}{dy}}{p^2} \\ &\Rightarrow \frac{1}{p} = \frac{4y^3}{p^2} - \frac{2y^4}{p^3} \frac{dp}{dy} - \frac{1}{p} + \frac{y}{p^2} \frac{dp}{dy} \\ &\Rightarrow \frac{1}{p} + \frac{1}{p} - \frac{4y^3}{p^2} + \frac{2y^4}{p^3} \frac{dp}{dy} - \frac{y}{p^2} \frac{dp}{dy} = 0 \Rightarrow \frac{2}{p} \left(1 - \frac{2y^3}{p}\right) + \frac{dp}{dy} \frac{y}{p^2} \left(\frac{2y^3}{p} - 1\right) = 0 \\ &\Rightarrow \frac{2}{p} \left(1 - \frac{2y^3}{p}\right) - \frac{y}{p^2} \frac{dp}{dy} \left(1 - \frac{2y^3}{p}\right) = 0 \Rightarrow \left(1 - \frac{2y^3}{p}\right) \left(\frac{2}{p} - \frac{y}{p^2} \frac{dp}{dy}\right) = 0. \end{aligned}$$

General solution. From $\frac{2}{p} - \frac{y}{p^2} \frac{dp}{dy} = 0$:

$$\frac{2}{p} = \frac{y}{p^2} \frac{dp}{dy} \Rightarrow 2 = \frac{y}{p} \frac{dp}{dy} \Rightarrow 2 \frac{dy}{y} = \frac{dp}{p}.$$

Integrating: $2 \log y + \log c = \log p \Rightarrow p = y^2c$. Substituting in (1):

$$y^4c^2x + y^2cy - y^4 = 0 \Rightarrow y^4(c^2x - 1) + y^3c = 0 \Rightarrow y(c^2x - 1) + c = 0$$

$$\Rightarrow yc^2x - y + c = 0 \Rightarrow yc^2x = y - c \Rightarrow x = \frac{y}{yc^2} - \frac{c}{yc^2} \Rightarrow \boxed{x = \frac{1}{c^2} - \frac{1}{cy}} \quad (\text{general solution}).$$

Singular solution. From $1 - \frac{2y^3}{p} = 0$: $p = 2y^3$. Substituting in (1):

$$4y^6x + 2y^4 - y^4 = 0 \Rightarrow 4xy^6 + y^4 = 0 \Rightarrow y^4(4xy^2 + 1) = 0,$$

$$\boxed{4xy^2 + 1 = 0}, \quad \text{i.e. } x = -\frac{1}{4y^2} \quad (\text{singular solution}).$$

Problem 5.7. Solve the differential equations (i) $xp^2 + x = 2yp$; (ii) $p^3 - 2xyp + 4y^2 = 0$, where $p = \frac{dy}{dx}$.

Solution. (i) $xp^2 + x = 2yp$. Dividing by $2p$:

$$y = \frac{xp}{2} + \frac{x}{2p} \quad (\text{solvable for } y). \quad (1)$$

Differentiating with respect to x :

$$\frac{dy}{dx} = \frac{1}{2} \left((x) \frac{dp}{dx} + (p)(1) \right) + \frac{1}{2} \left(\frac{p(1) - x \frac{dp}{dx}}{p^2} \right) \Rightarrow p = \frac{1}{2} x \frac{dp}{dx} + \frac{p}{2} + \frac{1}{2p} - \frac{x}{2p^2} \frac{dp}{dx}.$$

$$p - \frac{p}{2} - \frac{1}{2p} = \frac{x}{2} \frac{dp}{dx} \left(1 - \frac{1}{p^2} \right) \Rightarrow \frac{p}{2} \left(1 - \frac{1}{p^2} \right) = \frac{x}{2} \frac{dp}{dx} \left(1 - \frac{1}{p^2} \right) \\ \Rightarrow \left(\frac{p}{2} - \frac{x}{2} \frac{dp}{dx} \right) \left(1 - \frac{1}{p^2} \right) = 0.$$

General solution. $\frac{p}{2} - \frac{x}{2} \frac{dp}{dx} = 0 \Rightarrow p = x \frac{dp}{dx} \Rightarrow \frac{dx}{x} = \frac{dp}{p}$. Integrating: $\log x + \log c = \log p \Rightarrow p = cx$. Substituting in $xp^2 + x = 2yp$:

$$x(cx)^2 + x = 2y(cx) \Rightarrow c^2x^3 + x = 2cxy \Rightarrow c^2x^2 + 1 = 2cy \Rightarrow \boxed{y = \frac{cx^2}{2} + \frac{1}{2c}}.^2$$

Singular solution. $1 - \frac{1}{p^2} = 0 \Rightarrow p^2 = 1 \Rightarrow p = \pm 1$.

- $p = 1$: $x(1) + x = 2y(1) \Rightarrow 2x = 2y \Rightarrow x = y$.
- $p = -1$: $x(1) + x = 2y(-1) \Rightarrow 2x = -2y \Rightarrow x = -y$.

Singular solution: $x = \pm y$, i.e. $y^2 = x^2$.

$$(ii) \quad p^3 - 2xyp + 4y^2 = 0 \Rightarrow 2xyp = p^3 + 4y^2 \Rightarrow x = \frac{p^3}{2yp} + \frac{4y^2}{2yp}:$$

$$x = \frac{p^2}{2y} + \frac{2y}{p} \quad (\text{solvable for } x). \quad (2)$$

Differentiating with respect to y :

$$\frac{dx}{dy} = \frac{1}{2} \left(\frac{y \cdot 2p \frac{dp}{dy} - p^2(1)}{y^2} \right) + 2 \left(\frac{p(1) - y \frac{dp}{dy}}{p^2} \right) \Rightarrow \frac{1}{p} = \frac{p}{y} \frac{dp}{dy} - \frac{p^2}{2y^2} + \frac{2}{p} - \frac{2y}{p^2} \frac{dp}{dy}.$$

²The notes substitute $p = xc$ as $x^2c + x = 2yxc$ (dropping the square on p), giving $y = \frac{x}{2} + \frac{1}{2c}$; the correct substitution gives $c^2x^3 + x = 2cxy$.

$$\begin{aligned} \frac{1}{p} - \frac{2}{p} + \frac{p^2}{2y^2} &= \frac{p}{y} \frac{dp}{dy} - \frac{2y}{p^2} \frac{dp}{dy} \Rightarrow -\frac{1}{p} + \frac{p^2}{2y^2} = \frac{dp}{dy} \left(\frac{p}{y} - \frac{2y}{p^2} \right) \\ \Rightarrow \frac{p}{2y} \left(\frac{p}{y} - \frac{2y}{p^2} \right) &= \frac{dp}{dy} \left(\frac{p}{y} - \frac{2y}{p^2} \right) \Rightarrow \left(\frac{p}{y} - \frac{2y}{p^2} \right) \left(\frac{p}{2y} - \frac{dp}{dy} \right) = 0. \end{aligned}$$

General solution. $\frac{p}{2y} = \frac{dp}{dy} \Rightarrow \frac{1}{2} \frac{dy}{y} = \frac{dp}{p}$. Integrating: $\log y^{1/2} + \log c^{1/2} = \log p \Rightarrow p^2 = cy$,
i.e. $p = \sqrt{cy}$. Substituting $p = c^{1/2}y^{1/2}$ in $p^3 - 2xyp + 4y^2 = 0$:

$$c^{3/2}y^{3/2} - 2xy c^{1/2}y^{1/2} + 4y^2 = 0.$$

Dividing by $c^{1/2}y^{3/2}$:

$$c - 2x + \frac{4y^{1/2}}{c^{1/2}} = 0 \Rightarrow 4\sqrt{y} = \sqrt{c}(2x - c) \Rightarrow \boxed{16y = c(2x - c)^2} \quad (\text{general solution}).$$

Singular solution. $\frac{p}{y} = \frac{2y}{p^2} \Rightarrow p^3 = 2y^2$. Substituting in $p^3 - 2xyp + 4y^2 = 0$ with $p = 2^{1/3}y^{2/3}$:

$$2y^2 - 2xy \cdot 2^{1/3}y^{2/3} + 4y^2 = 0 \Rightarrow 6y^2 = 2^{4/3}xy^{5/3} \Rightarrow 6y^{1/3} = 2^{4/3}x.$$

Cubing: $216y = 16x^3$, i.e.

$$\boxed{27y = 2x^3} \quad (\text{singular solution}).^3$$

6 Clairaut's Equation

The differential equation of the form

$$y = px + f(p) \quad (1)$$

is known as *Clairaut's form*.

To get the solution, differentiate (1) with respect to x :

$$\frac{dy}{dx} = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx} \Rightarrow p = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx} \Rightarrow x \frac{dp}{dx} + f'(p) \frac{dp}{dx} = 0 \Rightarrow \frac{dp}{dx} (x + f'(p)) = 0.$$

- $\frac{dp}{dx} = 0 \Rightarrow p = c$ (constant) $\Rightarrow$ **general solution** $y = cx + f(c)$.
- $x + f'(p) = 0 \Rightarrow$ **singular solution** (eliminate p using (1)).

Substitute the p values in (1) to get the solutions. The general solution of Clairaut's equation is also known as the *primitive* of the given differential equation.

6.1 Problems

³The notes stop this step at a decimal approximation ($3y^2 - 1.25xy^{5/3} = 0$); the exact elimination is shown here.

Problem 6.1. Solve $y = xy' - (y')^2$, where $y' = \frac{dy}{dx} = p$.

Solution. $y = xp - p^2$, which is of Clairaut's form with $f(p) = -p^2$, $f'(p) = -2p$.

General solution (primitive). $p = c$: $y = cx - c^2$.

Singular solution. $x + f'(p) = 0 \Rightarrow x - 2p = 0 \Rightarrow p = \frac{x}{2}$. Substituting in $y = xp - p^2$:

$$y = x \cdot \frac{x}{2} - \frac{x^2}{4} = \frac{x^2}{2} - \frac{x^2}{4} = \frac{x^2}{4} \Rightarrow x^2 = 4y.$$

Problem 6.2. Solve $p = \log(px - y)$.

Solution. $e^p = px - y \Rightarrow y = px - e^p$, which is of Clairaut's form with $f(p) = -e^p$, $f'(p) = -e^p$.

General solution. $p = c$: $y = cx - e^c$.

Singular solution. $x + f'(p) = 0 \Rightarrow x - e^p = 0 \Rightarrow e^p = x \Rightarrow \log(e^p) = \log x \Rightarrow p = \log x$. Substituting in $y = px - e^p$:

$$y = x \log x - e^{\log x} = x \log x - x \Rightarrow y = x(\log x - 1).$$

Problem 6.3. Solve $(px - y)(py + x) = 2p$ by reducing it to Clairaut's form through the substitution $x^2 = X$, $y^2 = Y$.

Solution. Let $X = x^2$, $Y = y^2$ and $P = \frac{dY}{dX}$. Then $\frac{dX}{dx} = 2x$, $\frac{dY}{dy} = 2y$ and

$$P = \frac{dY}{dX} = \frac{dY}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{dX} = 2y \cdot p \cdot \frac{1}{2x} = \frac{py}{x} \Rightarrow p = \frac{xP}{y}.$$

Substituting in $(px - y)(py + x) = 2p$:

$$\left(\frac{xP}{y}x - y\right)\left(\frac{xP}{y}y + x\right) = 2\frac{xP}{y} \Rightarrow \left(\frac{x^2P - y^2}{y}\right)(xP + x) = 2\frac{xP}{y}$$

$$\Rightarrow (x^2P - y^2)(xP + x) = 2xP \Rightarrow (x^2P - y^2)x(P + 1) = 2xP \Rightarrow (x^2P - y^2)(P + 1) = 2P.$$

With $x^2 = X$, $y^2 = Y$:

$$(XP - Y)(P + 1) = 2P \Rightarrow XP - Y = \frac{2P}{P + 1} \Rightarrow Y = PX - \frac{2P}{P + 1} \quad (\text{Clairaut's equation}).$$

General solution ($P = c$): $Y = cX - \frac{2c}{c + 1}$. Since $Y = y^2$, $X = x^2$:

$$y^2 = cx^2 - \frac{2c}{c + 1}.$$

(Conversion problems ask only for the general solution.)

Problem 6.4. Solve $(p-1)e^{3x} + p^3e^{2y} = 0$ by converting it into Clairaut's equation through the substitution $X = e^x$, $Y = e^y$.

Solution. $X = e^x \Rightarrow \frac{dX}{dx} = e^x$; $Y = e^y \Rightarrow \frac{dY}{dy} = e^y$. With $P = \frac{dY}{dX}$:

$$P = \frac{dY}{dX} = \frac{dY}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{dX} = e^y \cdot p \cdot \frac{1}{e^x} \Rightarrow P = p e^{(y-x)} \Rightarrow p = P \frac{e^x}{e^y}.$$

Substituting in $(p-1)e^{3x} + p^3e^{2y} = 0$:

$$\begin{aligned} \left(P \frac{e^x}{e^y} - 1\right) e^{3x} + P^3 \frac{e^{3x}}{e^{3y}} e^{2y} = 0 &\Rightarrow \left(\frac{Pe^x - e^y}{e^y}\right) e^{3x} + P^3 \frac{e^{3x}}{e^y} = 0 \Rightarrow e^{3x} \left(\frac{Pe^x - e^y}{e^y} + \frac{P^3}{e^y}\right) = 0. \\ &\Rightarrow \frac{Pe^x - e^y}{e^y} + \frac{P^3}{e^y} = 0. \end{aligned}$$

With $e^x = X$, $e^y = Y$:

$$\begin{aligned} \frac{PX - Y}{Y} + \frac{P^3}{Y} &= 0 \\ \Rightarrow \frac{PX - Y + P^3}{Y} &= 0 \Rightarrow PX - Y + P^3 = 0 \\ \Rightarrow Y &= PX + P^3 \quad (\text{Clairaut's form}). \end{aligned}$$

General solution ($P = c$): $Y = cX + c^3$, i.e.

$$\boxed{e^y = ce^x + c^3}.$$

Problem 6.5. Solve $e^{4x}(p-1) + e^{2y}p^2 = 0$ using the substitution $X = e^{2x}$, $Y = e^{2y}$.

Solution. $\frac{dX}{dx} = 2e^{2x}$, $\frac{dY}{dy} = 2e^{2y}$.

$$P = \frac{dY}{dX} = \frac{dY}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{dX} = 2e^{2y} \cdot p \cdot \frac{1}{2e^{2x}} = p \frac{e^{2y}}{e^{2x}} \Rightarrow \boxed{p = P \frac{e^{2x}}{e^{2y}}}.$$

Substituting:

$$\begin{aligned} e^{4x} \left(P \frac{e^{2x}}{e^{2y}} - 1\right) + e^{2y} P^2 \frac{e^{4x}}{e^{4y}} &= 0 \Rightarrow e^{4x} \left(\frac{Pe^{2x} - e^{2y}}{e^{2y}}\right) + \frac{P^2 e^{4x}}{e^{2y}} = 0 \\ \Rightarrow \frac{e^{4x}}{e^{2y}} (Pe^{2x} - e^{2y} + P^2) &= 0 \Rightarrow Pe^{2x} - e^{2y} + P^2 = 0. \end{aligned}$$

With $e^{2x} = X$, $e^{2y} = Y$: $PX - Y + P^2 = 0 \Rightarrow Y = PX + P^2$ (Clairaut's equation).

General solution: $Y = cX + c^2$, i.e.

$$\boxed{e^{2y} = ce^{2x} + c^2}.$$

Problem 6.6. Solve $a^2p = (py + x)(px - y)$ using the substitution $X = x^2$, $Y = y^2$.

Solution. $\frac{dX}{dx} = 2x$, $\frac{dY}{dy} = 2y$.

$$P = \frac{dY}{dX} = \frac{dY}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{dX} = 2y \cdot p \cdot \frac{1}{2x} = \frac{y}{x} p \Rightarrow \boxed{p = \frac{x}{y} P}.$$

Substituting in $a^2p = (py + x)(px - y)$:

$$a^2 \frac{x}{y} P = \left(\frac{x}{y} P y + x \right) \left(\frac{x}{y} P x - y \right) = x(P + 1) \frac{Px^2 - y^2}{y} \Rightarrow a^2 P = (P + 1)(Px^2 - y^2).$$

With $x^2 = X$, $y^2 = Y$:

$$a^2 P = (P + 1)(PX - Y) \Rightarrow \frac{a^2 P}{P + 1} = PX - Y \Rightarrow Y = PX - \frac{a^2 P}{P + 1}.$$

General solution:

$$Y = cX - \frac{a^2 c}{c + 1} \Rightarrow \boxed{y^2 = cx^2 - \frac{a^2 c}{c + 1}}.$$

Homework

1. Solve $y = xp + \frac{a}{2p}$. Find the general and singular solutions.
2. Solve $y = px + 2p^2$.

Problem 6.7. Solve $y = px + \frac{a}{2p}$; find the general and singular solutions.

Solution. Clairaut's form with $f(p) = \frac{a}{2p}$, $f'(p) = -\frac{a}{2p^2}$.

General solution. $p = c$: $\boxed{y = cx + \frac{a}{2c}}$.

Singular solution. $x + f'(p) = 0 \Rightarrow x - \frac{a}{2} \cdot \frac{1}{p^2} = 0 \Rightarrow x = \frac{a}{2p^2} \Rightarrow p^2 = \frac{a}{2x} \Rightarrow p = \sqrt{\frac{a}{2x}}$.

$$y = \frac{\sqrt{a}}{\sqrt{2}\sqrt{x}} x + \frac{a}{2} \cdot \frac{\sqrt{2}\sqrt{x}}{\sqrt{a}} = \frac{\sqrt{a}\sqrt{x}}{\sqrt{2}} + \frac{\sqrt{a}\sqrt{x}}{\sqrt{2}} = \frac{2\sqrt{ax}}{\sqrt{2}} = \sqrt{2ax},$$

$$\boxed{y = \sqrt{2ax}}, \quad \text{i.e. } y^2 = 2ax.$$

Problem 6.8. Solve $y = px + 2p^2$.

Solution. Clairaut's form with $f(p) = 2p^2$, $f'(p) = 4p$.

General solution. $p = c$: $y = cx + 2c^2$.

Singular solution. $x + f'(p) = 0 \Rightarrow x + 4p = 0 \Rightarrow p = -\frac{x}{4}$.

$$y = -\frac{x}{4} \cdot x + 2 \cdot \frac{x^2}{16} = -\frac{x^2}{4} + \frac{x^2}{8} = -\frac{x^2}{8} \Rightarrow x^2 = -8y.$$

7 Higher-Order Linear Differential Equations with Constant Coefficients

The general form of a higher-order, first-degree differential equation with constant coefficients is

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1} \frac{dy}{dx} + a_n y = f(x). \quad (1)$$

It can also be written as

$$(a_0 D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n) y = f(x), \quad \text{i.e.} \quad \phi(D) y = f(x), \quad (2)$$

where $D = \frac{d}{dx}$ and $D^n = \frac{d^n}{dx^n}$.

General solution of the above differential equation:

$$y = \text{C.F.} + \text{P.I.},$$

where C.F. is the *complementary function* and P.I. is the *particular integral*.

7.1 Finding the complementary function

- The C.F. is the solution of the differential equation obtained by treating the R.H.S. $f(x) = 0$.
- To get the C.F., construct the *auxiliary equation* (A.E.) by replacing D by m . The auxiliary equation of (2) is

$$a_0 m^n + a_1 m^{n-1} + \cdots + a_{n-1} m + a_n = 0.$$

Let $m_1, m_2, m_3, \dots, m_n$ be the n roots of the above equation.

Case 1. If the roots are real and distinct:

$$\text{C.F.} = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots$$

Case 2. If some roots are real and equal:

(i) $m_1 = m_2$: $\text{C.F.} = (c_1 + c_2 x) e^{m_1 x} + c_3 e^{m_3 x} + \cdots$

(ii) $m_1 = m_2 = m_3$: $\text{C.F.} = (c_1 + c_2 x + c_3 x^2) e^{m_1 x} + c_4 e^{m_4 x} + \cdots$

Case 3. If some roots are imaginary, $m_1 = \alpha + i\beta$, $m_2 = \alpha - i\beta$:

$$\text{C.F.} = e^{\alpha x} (k_1 \cos \beta x + k_2 \sin \beta x) + c_3 e^{m_3 x} + \cdots$$

Derivation of Case 3. Using $e^{i\theta} = \cos \theta + i \sin \theta$,

$$\begin{aligned} c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x} &= e^{\alpha x} (c_1 (\cos \beta x + i \sin \beta x) + c_2 (\cos \beta x - i \sin \beta x)) \\ &= e^{\alpha x} (c_1 \cos \beta x + i c_1 \sin \beta x + c_2 \cos \beta x - i c_2 \sin \beta x) \\ &= e^{\alpha x} (\cos \beta x (c_1 + c_2) + \sin \beta x (i c_1 - i c_2)) = e^{\alpha x} (k_1 \cos \beta x + k_2 \sin \beta x), \end{aligned}$$

where $k_1 = c_1 + c_2$ and $k_2 = i(c_1 - c_2)$.

Note. In the given differential equation, if the R.H.S. is 0, the C.F. itself is the general solution.

Problems

Problem 7.1. Solve $\frac{d^4 y}{dx^4} - y = 0$, if it represents the deflection $y(x)$ of a beam fixed at the two ends.

Solution. R.H.S. = 0, so the C.F. is the general solution. $(D^4 - 1)y = 0$.

A.E.: $m^4 - 1 = 0 \Rightarrow (m^2 - 1)(m^2 + 1) = 0$:

$$m^2 = 1 \Rightarrow m = \pm 1; \quad m^2 = -1 \Rightarrow m = \pm i = \alpha \pm i\beta \quad (\alpha = 0, \beta = 1).$$

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) + c_3 e^{m_3 x} + c_4 e^{m_4 x} = e^0 (c_1 \cos x + c_2 \sin x) + c_3 e^x + c_4 e^{-x},$$

$$y = c_1 \cos x + c_2 \sin x + c_3 e^x + c_4 e^{-x}.$$

Problem 7.2. Solve $\frac{d^2 y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.

Solution. $(D^2 - 5D + 6)y = 0$. A.E.: $m^2 - 5m + 6 = 0 \Rightarrow (m - 2)(m - 3) = 0 \Rightarrow m = 2, 3$.

$$y = c_1 e^{2x} + c_2 e^{3x}.$$

Problem 7.3. The equation for the bending of a strut is $EI \frac{d^2 y}{dx^2} + Py = 0$, where EI and P are constants. Find the displacement of the strut.

Solution.

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = 0 \Rightarrow (D^2 + k^2)y = 0, \quad k^2 = \frac{P}{EI}.$$

A.E.: $m^2 + k^2 = 0 \Rightarrow m^2 = -k^2 \Rightarrow m = \pm \sqrt{-k^2} = \pm ik$ ($\alpha \pm i\beta$ form with $\alpha = 0, \beta = k = \sqrt{P/EI}$).

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) = e^0 \left(c_1 \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x \right),$$

$$y = c_1 \cos \sqrt{\frac{P}{EI}} x + c_2 \sin \sqrt{\frac{P}{EI}} x.$$

7.2 Finding the particular integral

If the R.H.S. contains some $f(x)$, we have to find the P.I. For standard functions such as exponential, sine, cosine or algebraic polynomials, we can follow the short-cut procedures below.

From (2), $\phi(D)y = f(x) \Rightarrow y = \frac{1}{\phi(D)}f(x)$.

Type 1: $f(x) = e^{ax}$.

$$\text{P.I.} = \frac{1}{\phi(D)}e^{ax}.$$

Replace D by a : $\text{P.I.} = \frac{1}{\phi(a)}e^{ax}$, provided $\phi(a) \neq 0$.

- If $\phi(a) = 0$: put one x in the numerator and differentiate the denominator $\phi(D)$ with respect to D; then replace D by a :

$$\text{P.I.} = \frac{x}{\phi'(a)}e^{ax}, \quad \text{provided } \phi'(a) \neq 0.$$

- If $\phi'(a) = 0$: put one more x in the numerator, differentiate the denominator again and replace D by a :

$$\text{P.I.} = \frac{x^2}{\phi''(a)}e^{ax}, \quad \text{provided } \phi''(a) \neq 0.$$

- Proceed with this process until the denominator $\neq 0$.

Problem 7.4. Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = \sinh x$.

Solution. $(D^2 - 2D + 2)y = \sinh x$.

C.F. A.E.: $m^2 - 2m + 2 = 0$,

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{2 \pm \sqrt{4 - 8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i \quad (\alpha = 1, \beta = 1).$$

C.F. $= e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x) = e^x(c_1 \cos x + c_2 \sin x)$.

P.I. Using $\sinh x = \frac{e^x - e^{-x}}{2}$:

$$\text{P.I.} = \frac{1}{D^2 - 2D + 2} \sinh x = \frac{1}{2} \left(\frac{e^x}{D^2 - 2D + 2} - \frac{e^{-x}}{D^2 - 2D + 2} \right).$$

For e^x replace D by 1; for e^{-x} replace D by -1 :

$$\text{P.I.} = \frac{1}{2} \left(\frac{e^x}{1^2 - 2(1) + 2} - \frac{e^{-x}}{(-1)^2 - 2(-1) + 2} \right) = \frac{1}{2} \left(\frac{e^x}{1} - \frac{e^{-x}}{5} \right) = \frac{5e^x - e^{-x}}{10}.$$

General solution:

$$y = e^x(c_1 \cos x + c_2 \sin x) + \frac{1}{10}(5e^x - e^{-x}).$$

Problem 7.5. Solve $(D^3 + 1)y = e^{-x}$.

Solution. **C.F.** A.E.: $m^3 + 1 = 0$. $m = -1$ is a root. By synthetic division,

$$\begin{array}{r|rrrr} -1 & 1 & 0 & 0 & 1 \\ & & -1 & 1 & -1 \\ \hline & 1 & -1 & 1 & 0 \end{array} \Rightarrow m^2 - m + 1 = 0,$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{1 \pm \sqrt{1 - 4}}{2} = \frac{1 \pm \sqrt{3}i}{2} \quad \left(\alpha = \frac{1}{2}, \beta = \frac{\sqrt{3}}{2} \right).$$

$$\text{C.F.} = c_1 e^{-x} + e^{x/2} \left(c_2 \cos \frac{\sqrt{3}}{2} x + c_3 \sin \frac{\sqrt{3}}{2} x \right).$$

P.I. P.I. = $\frac{e^{-x}}{D^3 + 1}$. Replacing D by -1 : $\frac{e^{-x}}{-1 + 1}$; the denominator is 0. So put x in the numerator and differentiate the denominator with respect to D:

$$\text{P.I.} = \frac{x e^{-x}}{3D^2} \xrightarrow{D \rightarrow -1} \frac{x e^{-x}}{3}.$$

General solution:

$$y = c_1 e^{-x} + e^{x/2} \left(c_2 \cos \frac{\sqrt{3}}{2} x + c_3 \sin \frac{\sqrt{3}}{2} x \right) + \frac{1}{3} x e^{-x}.$$

Problem 7.6. Solve $(D - 2)^3 y = e^{2x}$.

Solution. **C.F.** A.E.: $(m - 2)^3 = 0 \Rightarrow m = 2, 2, 2$. **C.F.** = $(c_1 + c_2 x + c_3 x^2) e^{2x}$.

P.I. P.I. = $\frac{e^{2x}}{(D - 2)^3}$. Replace D by 2: denominator = 0. Put x in the numerator and differentiate the denominator with respect to D:

$$\begin{aligned} &= \frac{x e^{2x}}{3(D - 2)^2} \quad (D \rightarrow 2 : \text{denominator} = 0) \\ \Rightarrow &= \frac{x^2 e^{2x}}{6(D - 2)} \quad (D \rightarrow 2 : \text{denominator} = 0) \\ \Rightarrow &= \frac{x^3 e^{2x}}{6(1 - 0)}. \\ \text{P.I.} &= \frac{x^3 e^{2x}}{6}. \end{aligned}$$

General solution:

$$y = (c_1 + c_2 x + c_3 x^2) e^{2x} + \frac{x^3 e^{2x}}{6}.$$

Type 2: $f(x) = \cos ax$ or $\sin ax$.

$$\text{P.I.} = \frac{1}{\phi(D)} \cos ax \quad (\text{or } \sin ax).$$

Replace D^2 by $-a^2$.

- If the denominator is of the form $lD + m$, multiply both numerator and denominator by $lD - m$ and again replace D^2 by $-a^2$ in the denominator. In this case the D in the numerator stands for $\frac{d}{dx}$.
- We apply the same procedure as in Type 1 (put x in the numerator and differentiate the denominator) if the denominator becomes 0.

Problem 7.7. Solve $(D^2 - 4D + 1)y = \cos x \cos 2x$.

Solution. **C.F.** A.E.: $m^2 - 4m + 1 = 0$,

$$m = \frac{4 \pm \sqrt{16 - 4}}{2} = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}.$$

$$\text{C.F.} = c_1 e^{(2+\sqrt{3})x} + c_2 e^{(2-\sqrt{3})x}.$$

$$\text{P.I. Using } \cos a \cos b = \frac{\cos(a+b) + \cos(a-b)}{2}:$$

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 - 4D + 1} \cos x \cos 2x \\ &= \frac{1}{D^2 - 4D + 1} \cdot \frac{1}{2} (\cos 3x + \cos x) \\ &= \frac{1}{2} \left(\frac{\cos 3x}{D^2 - 4D + 1} + \frac{\cos x}{D^2 - 4D + 1} \right). \end{aligned}$$

Replace D^2 by -9 in the first term and by -1 in the second:

$$= \frac{1}{2} \left(\frac{\cos 3x}{-9 - 4D + 1} + \frac{\cos x}{-1 - 4D + 1} \right) = \frac{1}{2} \left(\frac{\cos 3x}{-4D - 8} + \frac{\cos x}{-4D} \right) = -\frac{1}{8} \left(\underbrace{\frac{\cos 3x}{D+2}}_{(1)} + \underbrace{\frac{\cos x}{D}}_{(2)} \right).$$

$$(2): \text{ Since } \frac{1}{D} = \int dx, \quad \frac{\cos x}{D} = \int \cos x dx = \sin x.$$

(1): The denominator is of the form $lD + m$; multiply and divide by its conjugate $D - 2$:

$$\begin{aligned} \frac{\cos 3x}{D+2} \cdot \frac{D-2}{D-2} &= \frac{(D-2) \cos 3x}{D^2 - 4} \xrightarrow{D^2 \rightarrow -3^2} \frac{(D-2) \cos 3x}{-9 - 4} = -\frac{1}{13} (D \cos 3x - 2 \cos 3x) \\ &= -\frac{1}{13} (-3 \sin 3x - 2 \cos 3x) = \frac{1}{13} (3 \sin 3x + 2 \cos 3x). \end{aligned}$$

Hence

$$\text{P.I.} = -\frac{1}{8} \left(\frac{1}{13} (3 \sin 3x + 2 \cos 3x) + \sin x \right) = -\frac{1}{8} \left(\frac{3}{13} \sin 3x + \frac{2}{13} \cos 3x + \sin x \right).$$

General solution:

$$y = c_1 e^{(2+\sqrt{3})x} + c_2 e^{(2-\sqrt{3})x} - \frac{1}{8} \left(\frac{3}{13} \sin 3x + \frac{2}{13} \cos 3x + \sin x \right).$$

Problem 7.8. Solve $(D^2 + 4)y = \cos^2 x$.

Solution. **C.F.** A.E.: $m^2 + 4 = 0 \Rightarrow m^2 = -4 \Rightarrow m = \pm 2i$ ($\alpha = 0, \beta = 2$).

$$\text{C.F.} = e^0(c_1 \cos 2x + c_2 \sin 2x) = c_1 \cos 2x + c_2 \sin 2x.$$

P.I. Using $\cos^2 x = \frac{1 + \cos 2x}{2}$:

$$\text{P.I.} = \frac{\cos^2 x}{D^2 + 4} = \frac{1}{2} \left(\frac{1 + \cos 2x}{D^2 + 4} \right) = \frac{1}{2} \left(\frac{1}{D^2 + 4} + \frac{\cos 2x}{D^2 + 4} \right).$$

In the first term $1 = e^{0x}$, so replace D^2 by $-0^2 = 0$: $\frac{1}{0 + 4} = \frac{1}{4}$.

In the second term, replacing D^2 by -2^2 gives $\frac{\cos 2x}{-4 + 4}$; the denominator is 0. So put x in the numerator and differentiate the denominator with respect to D :

$$\frac{\cos 2x}{D^2 + 4} = \frac{x \cos 2x}{2D} = \frac{x}{2} \int \cos 2x \, dx = \frac{x}{2} \cdot \frac{1}{2} \sin 2x = \frac{x}{4} \sin 2x.$$

(The x brought into the numerator is not involved in the integration.)

$$\text{P.I.} = \frac{1}{2} \left(\frac{1}{4} + \frac{1}{4} x \sin 2x \right) = \frac{1}{2} \cdot \frac{1}{4} (1 + x \sin 2x) = \frac{1}{8} (1 + x \sin 2x).$$

General solution:

$$y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{8} + \frac{x \sin 2x}{8}.$$

Type 3: $f(x)$ an algebraic polynomial x^n .

$$\text{P.I.} = \frac{1}{\phi(D)} x^n.$$

In $\phi(D)$ take the least power of D (mostly the constant) as a common factor, so that

$$\text{P.I.} = \frac{x^n}{g(D)(1 + \theta(D))} = \frac{(1 + \theta(D))^{-1}}{g(D)} x^n.$$

Further, apply binomial series such as $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots$, $(1 - x)^{-1} = 1 + x + x^2 + \dots$, $(1 + x)^{-2}, \dots$; stop the series when the power of D equals the degree of the polynomial, and find the particular integral.

Problem 7.9. Solve $(D^2 + 5D + 6)y = x^2 + 3x + 1$.

Solution. **C.F.** A.E.: $m^2 + 5m + 6 = 0 \Rightarrow (m + 2)(m + 3) = 0 \Rightarrow m = -2, -3$. **C.F.** = $c_1 e^{-2x} + c_2 e^{-3x}$.

P.I.

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 5D + 6}(x^2 + 3x + 1) \\ &= \frac{1}{6\left(\frac{D^2}{6} + \frac{5D}{6} + 1\right)}(x^2 + 3x + 1) \\ &= \frac{1}{6}\left(1 + \frac{D^2 + 5D}{6}\right)^{-1}(x^2 + 3x + 1). \end{aligned}$$

Using $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots$ (maximum power of D needed = 2 = degree of the polynomial):

$$\begin{aligned} \text{P.I.} &= \frac{1}{6}\left(1 - \left(\frac{D^2 + 5D}{6}\right) + \left(\frac{D^2 + 5D}{6}\right)^2 - \dots\right)(x^2 + 3x + 1) \\ &= \frac{1}{6}\left(1 - \frac{D^2}{6} - \frac{5D}{6} + \frac{25D^2}{36} + \dots\right)(x^2 + 3x + 1) \quad (\text{neglecting higher powers of D}) \\ &= \frac{1}{6}\left(1 - \frac{5}{6}D + \frac{-6 + 25}{36}D^2\right)(x^2 + 3x + 1) = \frac{1}{6}\left(1 - \frac{5}{6}D + \frac{19}{36}D^2\right)(x^2 + 3x + 1). \end{aligned}$$

Using $D(x^2 + 3x + 1) = 2x + 3$ and $D^2(x^2 + 3x + 1) = 2$:

$$\begin{aligned} \text{P.I.} &= \frac{1}{6}\left[(x^2 + 3x + 1) - \frac{5}{6}(2x + 3) + \frac{19}{36}(2)\right] = \frac{1}{6}\left[x^2 + 3x + 1 - \frac{5}{3}x - \frac{5}{2} + \frac{19}{18}\right] \\ &= \frac{1}{6}\left[x^2 + \frac{4}{3}x + \frac{18 - 45 + 19}{18}\right] = \frac{1}{6}\left(x^2 + \frac{4}{3}x - \frac{4}{9}\right). \end{aligned}$$

General solution:

$$y = c_1 e^{-2x} + c_2 e^{-3x} + \frac{1}{6}\left(x^2 + \frac{4}{3}x - \frac{4}{9}\right).$$

Problem 7.10. Solve $(D^2 - 2D + 2)y = x^2$.

Solution. **C.F.** A.E.: $m^2 - 2m + 2 = 0 \Rightarrow m = \frac{2 \pm \sqrt{4 - 8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$ ($\alpha = 1, \beta = 1$).

$$\text{C.F.} = e^x(c_1 \cos x + c_2 \sin x).$$

P.I.

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 - 2D + 2}x^2 = \frac{1}{2\left(\frac{D^2 - 2D}{2} + 1\right)}x^2 = \frac{1}{2}\left(1 + \frac{D^2 - 2D}{2}\right)^{-1}x^2 \\ &= \frac{1}{2}\left(1 - \left(\frac{D^2 - 2D}{2}\right) + \left(\frac{D^2 - 2D}{2}\right)^2 - \dots\right)x^2 = \frac{1}{2}\left(1 - \frac{D^2}{2} + D + \frac{4D^2}{4} + \dots\right)x^2 \\ &= \frac{1}{2}\left(1 + D + \frac{D^2}{2}\right)x^2 \quad (\text{neglecting higher powers}) \end{aligned}$$

$$= \frac{1}{2} \left(x^2 + 2x + \frac{1}{2} \cdot 2 \right) = \frac{1}{2} (x^2 + 2x + 1) = \frac{1}{2} (x+1)^2.$$

General solution:

$$y = e^x (c_1 \cos x + c_2 \sin x) + \frac{1}{2} (x^2 + 2x + 1).$$

Type 4: $f(x) = e^{ax}V$, where $V = \cos bx$, $\sin bx$ or x^n .

$$\text{P.I.} = \frac{1}{\phi(D)} e^{ax} V.$$

Take the exponential term out and replace D by $D + a$:

$$\text{P.I.} = e^{ax} \frac{1}{\phi(D+a)} V,$$

then proceed with the Type 2 or Type 3 procedure.

Problem 7.11. Solve $(D^2 - 4D + 3)y = 3e^x \cos 2x$.

Solution. **C.F.** A.E.: $m^2 - 4m + 3 = 0 \Rightarrow (m-3)(m-1) = 0 \Rightarrow m = 3, 1$. **C.F.** $= c_1 e^x + c_2 e^{3x}$.

P.I. The exponential has coefficient $a = 1$: replace D by $D + 1$ and take the exponential term out.

$$\text{P.I.} = \frac{1}{D^2 - 4D + 3} 3e^x \cos 2x = 3e^x \frac{1}{(D+1)^2 - 4(D+1) + 3} \cos 2x.$$

The denominator is $D^2 + 2D + 1 - 4D - 4 + 3 = D^2 - 2D$, so

$$\text{P.I.} = 3e^x \frac{\cos 2x}{D^2 - 2D} \quad (\text{Type 2; replace } D^2 \text{ by } -2^2) = 3e^x \frac{\cos 2x}{-4 - 2D} = -\frac{3}{2} e^x \frac{\cos 2x}{2 + D}.$$

Multiplying by the conjugate:

$$= -\frac{3}{2} e^x \left(\frac{2 - D}{4 - D^2} \right) \cos 2x \xrightarrow{D^2 \rightarrow -4} -\frac{3}{2} e^x \left(\frac{2 \cos 2x - D \cos 2x}{4 + 4} \right).$$

Since $D \cos 2x = \frac{d}{dx} \cos 2x = -2 \sin 2x$:

$$= -\frac{3}{2} e^x \cdot \frac{1}{8} (2 \cos 2x + 2 \sin 2x) = -\frac{3e^x}{16} (2 \cos 2x + 2 \sin 2x) = -\frac{3e^x}{8} (\cos 2x + \sin 2x).$$

General solution:

$$y = c_1 e^x + c_2 e^{3x} - \frac{3e^x}{8} (\cos 2x + \sin 2x).$$

Problem 7.12. A body executes damped forced vibrations given by the equation

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + b^2x = e^{-kt} \sin \omega t, \quad k^2 < b^2.$$

Solve the above differential equation for the cases (i) $\omega^2 \neq b^2 - k^2$ and (ii) $\omega^2 = b^2 - k^2$.

Solution. $(D^2 + 2kD + b^2)x = e^{-kt} \sin \omega t$, where $D = \frac{d}{dt}$.

C.F. A.E.: $m^2 + 2km + b^2 = 0$,

$$m = \frac{-2k \pm \sqrt{4k^2 - 4b^2}}{2} = \frac{-2k \pm 2\sqrt{k^2 - b^2}}{2} = -k \pm \sqrt{k^2 - b^2}.$$

We know $k^2 < b^2$, so $k^2 - b^2 < 0$ (negative) and $b^2 - k^2 > 0$ (positive):

$$m = -k \pm \sqrt{-(b^2 - k^2)} = -k \pm i\sqrt{b^2 - k^2} \quad (\alpha = -k, \beta = \sqrt{b^2 - k^2}).$$

$$\text{C.F.} = e^{-kt} \left(c_1 \cos \sqrt{b^2 - k^2} t + c_2 \sin \sqrt{b^2 - k^2} t \right).$$

P.I. Replace D by $D + (-k)$:

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 2kD + b^2} e^{-kt} \sin \omega t \\ &= e^{-kt} \frac{1}{(D - k)^2 + 2k(D - k) + b^2} \sin \omega t \\ &= e^{-kt} \frac{\sin \omega t}{D^2 + k^2 - 2Dk + 2kD - 2k^2 + b^2} \\ &= e^{-kt} \frac{\sin \omega t}{D^2 - k^2 + b^2} \\ &= e^{-kt} \frac{\sin \omega t}{D^2 + (b^2 - k^2)} \quad (\text{Type 2}). \end{aligned}$$

Replace D^2 by $-\omega^2$:

$$\text{P.I.} = e^{-kt} \frac{\sin \omega t}{-\omega^2 + (b^2 - k^2)}. \quad (1)$$

Case (i) $\omega^2 \neq b^2 - k^2$:

$$\text{P.I.}_1 = e^{-kt} \frac{\sin \omega t}{-\omega^2 + b^2 - k^2}.$$

Case (ii) $\omega^2 = b^2 - k^2$: the denominator in (1) is 0, so put t in the numerator and differentiate the denominator with respect to D :

$$\begin{aligned} \text{P.I.} &= e^{-kt} \frac{t \sin \omega t}{2D + 0} \\ &= \frac{e^{-kt}}{2} t \frac{\sin \omega t}{D} \quad \left(\frac{1}{D} = \text{integration} \right) \\ &= \frac{e^{-kt}}{2} t \int \sin \omega t dt \\ &= \frac{e^{-kt}}{2} t \left(-\frac{\cos \omega t}{\omega} \right), \\ \text{P.I.}_2 &= -\frac{e^{-kt} t \cos \omega t}{2\omega}. \end{aligned}$$

General solution.Case 1 ($\omega^2 \neq b^2 - k^2$):

$$x = e^{-kt} \left(c_1 \cos \sqrt{b^2 - k^2} t + c_2 \sin \sqrt{b^2 - k^2} t \right) + e^{-kt} \frac{\sin \omega t}{-\omega^2 + b^2 - k^2}.$$

Case 2 ($\omega^2 = b^2 - k^2$):

$$x = e^{-kt} \left(c_1 \cos \sqrt{b^2 - k^2} t + c_2 \sin \sqrt{b^2 - k^2} t \right) - e^{-kt} \frac{t \cos \omega t}{2\omega},$$

and since $b^2 - k^2 = \omega^2$,

$$x = e^{-kt} \left(c_1 \cos \omega t + c_2 \sin \omega t - \frac{t \cos \omega t}{2\omega} \right).$$

Problem 7.13. Solve $(D^2 - 3D + 2)y = xe^{3x}$.*Solution.* **C.F.** A.E.: $m^2 - 3m + 2 = 0 \Rightarrow (m - 1)(m - 2) = 0 \Rightarrow m = 1, 2$. **C.F.** $= c_1 e^x + c_2 e^{2x}$.**P.I.** Replace D by D + 3 and take the exponential term out:

$$\text{P.I.} = \frac{1}{D^2 - 3D + 2} x e^{3x} = e^{3x} \frac{x}{(D + 3)^2 - 3(D + 3) + 2}.$$

Denominator: $D^2 + 9 + 6D - 3D - 9 + 2 = D^2 + 3D + 2$. So (Type 3; take 2 common)

$$\begin{aligned} \text{P.I.} &= e^{3x} \frac{x}{2 \left(\frac{D^2 + 3D}{2} + 1 \right)} \\ &= e^{3x} \frac{1}{2} \left(1 + \frac{D^2 + 3D}{2} \right)^{-1} x \\ &= e^{3x} \frac{1}{2} \left(1 - \left(\frac{D^2 + 3D}{2} \right) + \left(\frac{D^2 + 3D}{2} \right)^2 - \dots \right) x \\ &= e^{3x} \frac{1}{2} \left(1 - \frac{3D}{2} + \left(-\frac{D^2}{2} + \frac{9D^2}{4} \right) + \dots \right) x \\ &= e^{3x} \frac{1}{2} \left(1 - \frac{3}{2}D + \frac{7}{4}D^2 \right) x \\ &= \frac{e^{3x}}{2} \left(x - \frac{3}{2}(1) + \frac{7}{4}(0) \right) \\ &= \frac{e^{3x}}{2} \left(x - \frac{3}{2} \right). \end{aligned}$$

General solution:

$$y = c_1 e^x + c_2 e^{2x} + \frac{e^{3x}}{2} \left(x - \frac{3}{2} \right).$$

Type 5: $f(x) = x^n \cos ax$ or $x^n \sin ax$.

$$\text{P.I.} = \frac{1}{\phi(D)} (\cos ax \text{ or } \sin ax) x^n = (\text{R.P. or I.P.}) \frac{1}{\phi(D)} e^{iax} x^n,$$

since $\cos ax = \text{R.P.} (e^{iax})$ (real part) and $\sin ax = \text{I.P.} (e^{iax})$ (imaginary part). Replace D by $D + ai$ and then proceed with the Type 3 / Type 4 procedure.

Problem 7.14. Solve $(D^2 + 4)y = x^2 \cos 2x$.

Solution. **C.F.** A.E.: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$ ($\alpha = 0, \beta = 2$). **C.F.** $= c_1 \cos 2x + c_2 \sin 2x$.

P.I.

$$\text{P.I.} = \frac{1}{D^2 + 4} x^2 \cos 2x = \text{R.P.} \frac{1}{D^2 + 4} e^{i2x} x^2.$$

Replace D by $D + 2i$ and take e^{i2x} out:

$$= \text{R.P.} e^{i2x} \frac{1}{(D + 2i)^2 + 4} x^2 = \text{R.P.} e^{i2x} \frac{1}{D^2 + 4i^2 + 4Di + 4} x^2 = \text{R.P.} e^{i2x} \frac{x^2}{D^2 + 4Di} \quad (\text{Type 3}).$$

Taking the minimum power of D common:

$$= \text{R.P.} \frac{e^{i2x}}{4Di} \left(\frac{D^2}{4Di} + 1 \right)^{-1} x^2 = \text{R.P.} \frac{e^{i2x}}{4Di} \left(1 + \frac{D}{4i} \right)^{-1} x^2.$$

$$\text{Since } \frac{D}{4i} = \frac{Di}{4i^2} = -\frac{Di}{4}:$$

$$= \text{R.P.} \frac{e^{i2x}}{4Di} \left(1 - \frac{Di}{4} \right)^{-1} x^2.$$

Using $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots$:

$$\begin{aligned} &= \text{R.P.} (e^{i2x}) \frac{1}{4Di} \left(1 + \frac{Di}{4} + \frac{D^2 i^2}{16} + \dots \right) x^2 = \text{R.P.} (e^{i2x}) \frac{1}{4Di} \left(1 + \frac{Di}{4} - \frac{D^2}{16} \right) x^2 \\ &= \text{R.P.} (e^{i2x}) \frac{1}{4Di} \left(x^2 + \frac{2xi}{4} - \frac{2}{16} \right) = \text{R.P.} (e^{i2x}) \left(-\frac{i}{4D} \right) \left(x^2 - \frac{1}{8} + \frac{xi}{2} \right) \quad \left(\frac{1}{i} = -i \right). \end{aligned}$$

With $\frac{1}{D} = \int dx$:

$$\begin{aligned} &= \text{R.P.} (e^{i2x}) \left(-\frac{i}{4} \right) \left[\int x^2 dx - \frac{1}{8} \int dx + \frac{i}{2} \int x dx \right] = \text{R.P.} e^{i2x} \left(-\frac{i}{4} \right) \left(\frac{x^3}{3} - \frac{x}{8} + \frac{ix^2}{4} \right) \\ &= \text{R.P.} (e^{i2x}) \frac{1}{4} \left(\left(\frac{x}{8} - \frac{x^3}{3} \right) i + \frac{x^2}{4} \right) \quad (\text{multiplying } -i \text{ inside}). \end{aligned}$$

Using $e^{i\theta} = \cos \theta + i \sin \theta$ and taking the real part:

$$\text{P.I.} = \text{R.P.} \frac{1}{4} (\cos 2x + i \sin 2x) \left(\frac{x^2}{4} + i \left(\frac{x}{8} - \frac{x^3}{3} \right) \right) = \frac{1}{4} \left(\frac{x^2}{4} \cos 2x - \left(\frac{x}{8} - \frac{x^3}{3} \right) \sin 2x \right).$$

General solution:

$$y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{4} \left(\frac{x^2}{4} \cos 2x - \left(\frac{x}{8} - \frac{x^3}{3} \right) \sin 2x \right).$$

Problem 7.15. Solve $(D^2 + 1)y = x \sin x$.

Solution. **C.F.** A.E.: $m^2 + 1 = 0 \Rightarrow m = \pm i$ ($\alpha = 0, \beta = 1$). **C.F.** $= c_1 \cos x + c_2 \sin x$.

P.I. ($\sin x \rightarrow$ I.P.)

$$\begin{aligned}
 \text{P.I.} &= \frac{1}{D^2 + 1} x \sin x \\
 &= \text{I.P. } e^{ix} \frac{1}{(D + i)^2 + 1} x \\
 &= \text{I.P. } e^{ix} \frac{1}{D^2 + i^2 + 2Di + 1} x \\
 &= \text{I.P. } e^{ix} \frac{1}{D^2 + 2Di} x \quad (\text{Type 3}). \\
 &= \text{I.P. } e^{ix} \frac{1}{2Di \left(\frac{D^2}{2Di} + 1 \right)} x = \text{I.P. } e^{ix} \frac{1}{2Di \left(1 + \frac{D}{2i} \right)} x = \text{I.P. } e^{ix} \frac{1}{2Di} \left(1 - \frac{Di}{2} \right)^{-1} x. \\
 &= \text{I.P. } e^{ix} \frac{1}{2Di} \left(1 + \frac{Di}{2} + \frac{D^2 i^2}{4} + \dots \right) x = \text{I.P. } e^{ix} \frac{1}{2Di} \left(1 + \frac{Di}{2} - \frac{D^2}{4} \right) x \\
 &= \text{I.P. } e^{ix} \frac{1}{2Di} \left(x + \frac{i}{2}(1) - 0 \right) = \text{I.P. } e^{ix} \left(-\frac{i}{2} \right) \frac{1}{D} \left(x + \frac{i}{2} \right) \\
 &= \text{I.P. } e^{ix} \left(-\frac{i}{2} \right) \left(\int x dx + \frac{i}{2} \int dx \right) = \text{I.P. } (\cos x + i \sin x) \left(-\frac{i}{2} \right) \left(\frac{x^2}{2} + \frac{ix}{2} \right) \\
 &= \text{I.P. } \frac{1}{2} (\cos x + i \sin x) \left(\frac{x}{2} - \frac{x^2}{2} i \right) \quad (\text{multiplying } -i \text{ inside}).
 \end{aligned}$$

Taking the imaginary part (do not include i in the answer):

$$\text{P.I.} = \frac{1}{2} \left(-\cos x \cdot \frac{x^2}{2} + \sin x \cdot \frac{x}{2} \right) = \frac{1}{2} \left(\frac{x \sin x}{2} - \frac{x^2 \cos x}{2} \right) = \frac{x \sin x}{4} - \frac{x^2 \cos x}{4}.$$

General solution:

$$y = c_1 \cos x + c_2 \sin x + \frac{1}{4} (x \sin x - x^2 \cos x).$$

Type 6: $f(x) = e^{ax}(\cos bx \text{ or } \sin bx) x^n$ (**product of all three**).

$$\text{P.I.} = \frac{1}{\phi(D)} e^{ax} (\cos bx \text{ or } \sin bx) x^n.$$

Replace D by $D + a$ (take the exponential term out):

$$\text{P.I.} = e^{ax} \frac{1}{\psi(D)} (\cos bx \text{ or } \sin bx) x^n,$$

and then proceed by Type 5.

Problem 7.16. Solve $(D^2 + 4D + 4)y = 8x^2e^{2x} \sin 2x$.

Solution. **C.F.** A.E.: $m^2 + 4m + 4 = 0 \Rightarrow (m + 2)^2 = 0 \Rightarrow m = -2, -2$. **C.F.** $= (c_1 + c_2x)e^{-2x}$.

P.I. Replace D by $D + 2$ and take the exponential out of the operator:

$$\text{P.I.} = \frac{1}{D^2 + 4D + 4} 8x^2e^{2x} \sin 2x = 8e^{2x} \frac{1}{(D + 2)^2 + 4(D + 2) + 4} x^2 \sin 2x.$$

The denominator is $D^2 + 4 + 4D + 4D + 8 + 4 = D^2 + 8D + 16 = (D + 4)^2$:

$$\text{P.I.} = 8e^{2x} \frac{1}{(D + 4)^2} x^2 \sin 2x \quad (\text{Type 5; } \sin \rightarrow \text{I.P.}) = \text{I.P. } 8e^{2x} \frac{1}{(D + 4)^2} e^{i2x} x^2.$$

Replace D by $D + 2i$:

$$= \text{I.P. } 8e^{2x} e^{i2x} \frac{1}{(D + 2i + 4)^2} x^2 \quad (\text{Type 3; take the least power of } D \text{ as common factor})$$

$$= \text{I.P. } 8e^{2x} e^{i2x} \frac{1}{(4 + 2i)^2 \left(1 + \frac{D}{4 + 2i}\right)^2} x^2 = \text{I.P. } 8e^{2x} e^{i2x} \frac{1}{(4 + 2i)^2} \left(1 + \frac{D}{4 + 2i}\right)^{-2} x^2.$$

Using $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$:

$$= \text{I.P. } 8e^{2x} e^{i2x} \frac{1}{(4 + 2i)^2} \left(1 - \frac{2D}{4 + 2i} + \frac{3D^2}{(4 + 2i)^2}\right) x^2.$$

Since $4 + 2i = 2(2 + i)$, $(4 + 2i)^2 = 2^2(2 + i)^2$:

$$= \text{I.P. } \frac{8e^{2x} e^{i2x}}{2^2(2 + i)^2} \left(1 - \frac{D}{2 + i} + \frac{3D^2}{4(2 + i)^2}\right) x^2 = \text{I.P. } \frac{2e^{2x} e^{i2x}}{(2 + i)^2} \left(x^2 - \frac{2x}{2 + i} + \frac{3(2)}{4(2 + i)^2}\right).$$

Now $(2 + i)^2 = 4 + i^2 + 4i = 3 + 4i$, so

$$= \text{I.P. } \frac{2e^{2x} e^{i2x}}{3 + 4i} \left(x^2 - \frac{2x}{2 + i} + \frac{3}{2(2 + i)^2}\right).$$

Rationalising: $\frac{1}{3 + 4i} = \frac{3 - 4i}{9 + 16} = \frac{3 - 4i}{25}$, $\frac{1}{2 + i} = \frac{2 - i}{4 + 1} = \frac{2 - i}{5}$, $\frac{1}{(2 + i)^2} = \frac{3 - 4i}{25}$:

$$\begin{aligned} &= \text{I.P. } \frac{2e^{2x}(3 - 4i)}{25} e^{i2x} \left(x^2 - \frac{2(2 - i)x}{5} + \frac{3}{2} \cdot \frac{3 - 4i}{25}\right) \\ &= \text{I.P. } \frac{2e^{2x}(3 - 4i)}{25} e^{i2x} \left(x^2 - \frac{4x}{5} + \frac{2ix}{5} + \frac{9}{50} - \frac{6i}{25}\right) \\ &= \text{I.P. } \frac{2e^{2x}}{25} (\cos 2x + i \sin 2x)(3 - 4i) \left(\left(x^2 - \frac{4x}{5} + \frac{9}{50}\right) + i \left(\frac{2x}{5} - \frac{6}{25}\right)\right). \end{aligned}$$

Multiplying $(3 - 4i)$ inside:

$$\begin{aligned} &(3 - 4i) \left(\left(x^2 - \frac{4x}{5} + \frac{9}{50}\right) + i \left(\frac{2x}{5} - \frac{6}{25}\right)\right) \\ &= \left[3 \left(x^2 - \frac{4x}{5} + \frac{9}{50}\right) + 4 \left(\frac{2x}{5} - \frac{6}{25}\right)\right] + i \left[3 \left(\frac{2x}{5} - \frac{6}{25}\right) - 4 \left(x^2 - \frac{4x}{5} + \frac{9}{50}\right)\right] \\ &= \left[3x^2 - \frac{12x}{5} + \frac{27}{50} + \frac{8x}{5} - \frac{24}{25}\right] + i \left[\frac{6x}{5} - \frac{18}{25} - 4x^2 + \frac{16x}{5} - \frac{36}{50}\right] \\ &= \left(3x^2 - \frac{4x}{5} - \frac{21}{50}\right) + i \left(-4x^2 + \frac{22x}{5} - \frac{72}{50}\right). \end{aligned}$$

Hence

$$\begin{aligned} \text{P.I.} &= \text{I.P.} \cdot \frac{2e^{2x}}{25} (\cos 2x + i \sin 2x) \left(\left(3x^2 - \frac{4x}{5} - \frac{21}{50} \right) + i \left(-4x^2 + \frac{22x}{5} - \frac{72}{50} \right) \right), \\ \text{P.I.} &= \frac{2e^{2x}}{25} \left[\cos 2x \left(-4x^2 + \frac{22x}{5} - \frac{72}{50} \right) + \sin 2x \left(3x^2 - \frac{4x}{5} - \frac{21}{50} \right) \right]. \end{aligned}$$

General solution:⁴

$$y = (c_1 + c_2x)e^{-2x} + \frac{2e^{2x}}{25} \left[\cos 2x \left(-4x^2 + \frac{22x}{5} - \frac{72}{50} \right) + \sin 2x \left(3x^2 - \frac{4x}{5} - \frac{21}{50} \right) \right].$$

Exercise problems

Solve the following differential equations:

- (i) $(D^4 - 2D^2 + 1)y = 40 \cosh x$
- (ii) $(D^2 + 9)y = \cos^3 x$
- (iii) $(D^2 - 5D + 6)y = x^2 + 3x - 1$
- (iv) $(D^2 - 3D + 2)y = 1 + xe^x$ (worked below as $(1+x)e^x$)
- (v) $(D^2 + 3D + 2)y = e^{2x} \sin x$
- (vi) $(D^2 + 1)y = x \cos x$
- (vii) $(D^2 - 2D + 1)y = xe^x \cos x$

Problem 7.17. Solve $(D^4 - 2D^2 + 1)y = 40 \cosh x$.

Solution. C.F. A.E.: $m^4 - 2m^2 + 1 = 0 \Rightarrow (m^2 - 1)^2 = 0 \Rightarrow m^2 = 1, 1 \Rightarrow m = \pm 1, \pm 1$.

$$\text{C.F.} = (c_1 + c_2x)e^x + (c_3 + c_4x)e^{-x}.$$

P.I. Using $\cosh x = \frac{e^x + e^{-x}}{2}$:

$$\text{P.I.} = \frac{1}{D^4 - 2D^2 + 1} 40 \frac{e^x + e^{-x}}{2} = 20 \left(\frac{1}{D^4 - 2D^2 + 1} e^x + \frac{1}{D^4 - 2D^2 + 1} e^{-x} \right) \quad (\text{Type 1}).$$

Replace D by 1 and -1:

$$= 20 \left(\frac{e^x}{1 - 2 + 1} + \frac{e^{-x}}{1 - 2 + 1} \right) = 20 \left(\frac{e^x}{0} + \frac{e^{-x}}{0} \right) \quad (\text{denominator} = 0).$$

Put x in the numerator and differentiate the denominator with respect to D:

$$= 20 \left(\frac{xe^x}{4D^3 - 4D} + \frac{xe^{-x}}{4D^3 - 4D} \right) \xrightarrow{D \rightarrow 1, -1} 20 \left(\frac{xe^x}{4 - 4} + \frac{xe^{-x}}{-4 + 4} \right) \quad (\text{denominator} = 0).$$

Again put x in the numerator and differentiate the denominator:

$$= 20 \left(\frac{x^2e^x}{12D^2 - 4} + \frac{x^2e^{-x}}{12D^2 - 4} \right) \xrightarrow{D \rightarrow 1, -1} 20 \left(\frac{x^2e^x}{8} + \frac{x^2e^{-x}}{8} \right).$$

$$\text{P.I.} = \frac{5}{2}x^2e^x + \frac{5}{2}x^2e^{-x} = 5x^2 \left(\frac{e^x + e^{-x}}{2} \right) = 5x^2 \cosh x.$$

General solution:

$$y = (c_1 + c_2x)e^x + (c_3 + c_4x)e^{-x} + 5x^2 \cosh x.$$

⁴In the notes, $\frac{6x}{5} + \frac{16x}{5}$ is written as $\frac{21x}{5}$, and the factor e^{2x} is dropped from the final line; both are corrected here (the result has been checked by substitution).

Problem 7.18. Solve $(D^2 + 9)y = \cos^3 x$.

Solution. **C.F.** A.E.: $m^2 + 9 = 0 \Rightarrow m = \pm 3i$ ($\alpha = 0, \beta = 3$). **C.F.** $= c_1 \cos 3x + c_2 \sin 3x$.

P.I. (Type 2) Using $\cos 3x = 4 \cos^3 x - 3 \cos x$, i.e. $\cos^3 x = \frac{\cos 3x + 3 \cos x}{4}$:

$$\text{P.I.} = \frac{1}{4} \cdot \frac{\cos 3x + 3 \cos x}{D^2 + 9} = \frac{1}{4} \left(\frac{\cos 3x}{D^2 + 9} + 3 \frac{\cos x}{D^2 + 9} \right).$$

Replace D^2 by $-a^2$:

$$= \frac{1}{4} \left(\frac{\cos 3x}{-9 + 9} + 3 \frac{\cos x}{-1 + 9} \right).$$

The first denominator is 0: put x in the numerator and differentiate the denominator with respect to D :

$$\begin{aligned} &= \frac{1}{4} \left(\frac{x \cos 3x}{2D} + \frac{3}{8} \cos x \right) = \frac{1}{4} \left(\frac{x D \cos 3x}{2D^2} + \frac{3}{8} \cos x \right) \xrightarrow{D^2 \rightarrow -9} \frac{1}{4} \left(\frac{x D \cos 3x}{-18} + \frac{3}{8} \cos x \right) \\ &= \frac{1}{4} \left(\frac{x}{-18} (-3 \sin 3x) + \frac{3}{8} \cos x \right) = \frac{1}{4} \left(\frac{x \sin 3x}{6} + \frac{3}{8} \cos x \right). \end{aligned}$$

$$\text{P.I.} = \frac{3}{32} \cos x + \frac{x \sin 3x}{24}.$$

General solution:

$$y = c_1 \cos 3x + c_2 \sin 3x + \frac{3}{32} \cos x + \frac{x \sin 3x}{24}.$$

Problem 7.19. Solve $(D^2 - 5D + 6)y = x^2 + 3x - 1$.

Solution. **C.F.** A.E.: $m^2 - 5m + 6 = 0 \Rightarrow m = 2, 3$. **C.F.** $= c_1 e^{2x} + c_2 e^{3x}$.

P.I. (Type 3; take 6 common)

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 - 5D + 6} (x^2 + 3x - 1) \\ &= \frac{1}{6 \left(\frac{D^2}{6} - \frac{5D}{6} + 1 \right)} (x^2 + 3x - 1) \\ &= \frac{1}{6} \left(1 + \frac{D^2 - 5D}{6} \right)^{-1} (x^2 + 3x - 1) \\ &= \frac{1}{6} \left(1 - \left(\frac{D^2 - 5D}{6} \right) + \left(\frac{D^2 - 5D}{6} \right)^2 \right) (x^2 + 3x - 1) \\ &= \frac{1}{6} \left(1 - \frac{D^2}{6} + \frac{5D}{6} + \frac{25D^2}{36} \right) (x^2 + 3x - 1) \\ &= \frac{1}{6} \left(1 + \frac{5}{6}D + \frac{19}{36}D^2 \right) (x^2 + 3x - 1) \\ &= \frac{1}{6} \left(x^2 + 3x - 1 + \frac{5}{6}(2x + 3) + \frac{19}{36}(2) \right) \\ &= \frac{1}{6} \left(x^2 + 3x - 1 + \frac{5}{3}x + \frac{5}{2} + \frac{19}{18} \right) \\ &= \frac{1}{6} \left(x^2 + \frac{14x}{3} + \frac{-18 + 45 + 19}{18} \right) \end{aligned}$$

$$= \frac{1}{6} \left(x^2 + \frac{14x}{3} + \frac{23}{9} \right).$$

General solution:

$$y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{6} \left(x^2 + \frac{14x}{3} + \frac{23}{9} \right).$$

Problem 7.20. Solve $(D^2 - 3D + 2)y = (1 + x)e^x$.

Solution. **C.F.** A.E.: $m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$. **C.F.** $= c_1 e^x + c_2 e^{2x}$.

P.I. (Type 4) Replace D by D + 1:

$$\begin{aligned} \text{P.I.} &= e^x \frac{1}{(D+1)^2 - 3(D+1) + 2} (1+x) \\ &= e^x \frac{1}{D^2 + 1 + 2D - 3D - 3 + 2} (1+x) \\ &= e^x \frac{1}{D^2 - D} (1+x) \quad (\text{Type 3}). \end{aligned}$$

Taking D out:

$$= e^x \frac{1}{D(D-1)} (1+x) = e^x \frac{1}{D} \cdot \frac{-1}{(1-D)} (1+x) = e^x \frac{1}{D} (-(1-D)^{-1}) (1+x).$$

Using $(1-x)^{-1} = 1 + x + x^2 + \dots$:

$$\begin{aligned} &= e^x \frac{1}{D} (-(1 + D + D^2)) (1+x) \\ &= e^x \frac{1}{D} (-D^2 - D - 1) (1+x) \\ &= e^x \frac{1}{D} ((0+0) - (0+1) - (1+x)) \\ &= e^x \frac{1}{D} (-x - 2) \\ &= e^x \left(-\int x \, dx - \int 2 \, dx \right) \\ &= e^x \left(-\frac{x^2}{2} - 2x \right). \end{aligned}$$

$$\text{P.I.} = -e^x \left(\frac{x^2}{2} + 2x \right).$$

General solution:

$$y = c_1 e^x + c_2 e^{2x} - e^x \left(\frac{x^2}{2} + 2x \right).$$

Problem 7.21. Solve $(D^2 + 3D + 2)y = e^{2x} \sin x$.

Solution. **C.F.** A.E.: $m^2 + 3m + 2 = 0 \Rightarrow (m + 1)(m + 2) = 0 \Rightarrow m = -1, -2$. **C.F.** = $c_1 e^{-x} + c_2 e^{-2x}$.

P.I. (Type 4) Replace D by D + 2:

$$\begin{aligned} \text{P.I.} &= e^{2x} \frac{1}{(D+2)^2 + 3(D+2) + 2} \sin x \\ &= e^{2x} \frac{1}{D^2 + 4 + 4D + 3D + 6 + 2} \sin x \\ &= e^{2x} \frac{1}{D^2 + 7D + 12} \sin x \quad (\text{Type 2}). \end{aligned}$$

Replace D^2 by -1^2 :

$$\begin{aligned} &= e^{2x} \frac{1}{-1 + 7D + 12} \sin x = e^{2x} \frac{1}{7D + 11} \sin x = e^{2x} \frac{7D - 11}{49D^2 - 121} \sin x \\ &\quad \xrightarrow{D^2 \rightarrow -1} e^{2x} \frac{7D - 11}{-49 - 121} \sin x \\ &= -\frac{e^{2x}}{170} (7D - 11) \sin x = -\frac{1}{170} e^{2x} (7 \cos x - 11 \sin x). \end{aligned}$$

General solution:

$$y = c_1 e^{-x} + c_2 e^{-2x} - \frac{1}{170} e^{2x} (7 \cos x - 11 \sin x).$$

Problem 7.22. Solve $(D^2 + 1)y = x \cos x$.

Solution. **C.F.** A.E.: $m^2 + 1 = 0 \Rightarrow m = \pm i$ ($\alpha = 0, \beta = 1$). **C.F.** = $c_1 \cos x + c_2 \sin x$.

P.I. (Type 5; $\cos x \rightarrow$ R.P.) Replace D by D + i:

$$\begin{aligned} \text{P.I.} &= \text{R.P.} \frac{1}{D^2 + 1} x e^{ix} \\ &= \text{R.P.} e^{ix} \frac{1}{(D+i)^2 + 1} x \\ &= \text{R.P.} e^{ix} \frac{1}{D^2 + i^2 + 2Di + 1} x \\ &= \text{R.P.} e^{ix} \frac{1}{D^2 + 2Di} x \quad (\text{Type 3}). \end{aligned}$$

Since $\frac{D^2}{2Di} + 1 = \frac{D}{2i} + 1 = 1 - \frac{Di}{2}$:

$$\begin{aligned} &= \text{R.P.} e^{ix} \frac{1}{2Di \left(1 - \frac{Di}{2}\right)} x = \text{R.P.} e^{ix} \frac{1}{2Di} \left(1 - \frac{Di}{2}\right)^{-1} x = \text{R.P.} e^{ix} \frac{1}{2Di} \left(1 + \frac{Di}{2} + \frac{D^2 i^2}{4}\right) x \\ &= \text{R.P.} e^{ix} \frac{1}{2Di} \left(x + \frac{i}{2} + 0\right) = \text{R.P.} e^{ix} \left(-\frac{i}{2}\right) \left(\int x dx + \frac{i}{2} \int dx\right) = \text{R.P.} e^{ix} \left(-\frac{i}{2}\right) \left(\frac{x^2}{2} + \frac{ix}{2}\right) \\ &= \text{R.P.} (\cos x + i \sin x) \left(\frac{x}{4} - \frac{ix^2}{4}\right). \end{aligned}$$

Taking the real part:

$$\text{P.I.} = \frac{x \cos x}{4} + \frac{x^2 \sin x}{4}.$$

General solution:⁵

$$y = c_1 \cos x + c_2 \sin x + \frac{x \cos x}{4} + \frac{x^2 \sin x}{4}.$$

Problem 7.23. Solve $(D^2 - 2D + 1)y = xe^x \cos x$.

Solution. **C.F.** A.E.: $m^2 - 2m + 1 = 0 \Rightarrow m = 1, 1$. **C.F.** $= (c_1 + c_2x)e^x$.

P.I. Replace D by D + 1:

$$\text{P.I.} = e^x \frac{1}{(D+1)^2 - 2(D+1) + 1} x \cos x = e^x \frac{1}{D^2} x \cos x \quad (\text{Type 5}) = e^x \text{R.P.} \frac{1}{D^2} x e^{ix}.$$

Replace D by D + i:

$$\begin{aligned} &= \text{R.P.} e^x e^{ix} \frac{1}{(D+i)^2} x = \text{R.P.} e^x e^{ix} \frac{1}{D^2 - 1 + 2Di} x = \text{R.P.} e^x e^{ix} \frac{1}{-1(1 - (D^2 + 2Di))} x \\ &= \text{R.P.} (-e^x e^{ix}) (1 - (D^2 + 2Di))^{-1} x = \text{R.P.} (-e^x e^{ix}) (1 + (D^2 + 2Di) + (D^2 + 2Di)^2 + \dots) x \\ &= \text{R.P.} (-e^x e^{ix}) (1 + D^2 + 2Di + 4D^2 i^2 + \dots) x = \text{R.P.} (-e^x e^{ix}) (1 + 2Di - 3D^2) x \\ &= \text{R.P.} (-e^x e^{ix}) (x + 2i) = \text{R.P.} (-e^x) (\cos x + i \sin x)(x + 2i). \end{aligned}$$

Taking the real part: $(\cos x + i \sin x)(x + 2i)$ has real part $x \cos x - 2 \sin x$, so

$$\text{P.I.} = -e^x (x \cos x - 2 \sin x) = -xe^x \cos x + 2e^x \sin x.$$

General solution:

$$y = (c_1 + c_2x)e^x + e^x(2 \sin x - x \cos x).$$

Type 7: general $f(x)$. The R.H.S. $f(x)$ is not in the form of the previous six types; it may be any combination of functions. To find the P.I.,

$$\begin{aligned} \text{P.I.} &= \frac{1}{\phi(D)} f(x) \\ &= \frac{1}{(D - \alpha_1)(D - \alpha_2) \dots (D - \alpha_n)} f(x) \\ &= \left(\frac{A}{D - \alpha_1} + \frac{B}{D - \alpha_2} + \dots + \frac{K}{D - \alpha_n} \right) f(x) \end{aligned}$$

(partial fractions). In the above expression consider one factor:

$$\begin{aligned} \frac{1}{D - \alpha} f(x) = y &\Rightarrow (D - \alpha)y = f(x) \Rightarrow \frac{dy}{dx} - \alpha y = f(x) \quad (\text{linear equation}), \\ y e^{\int -\alpha dx} = \int e^{\int -\alpha dx} f(x) dx &\Rightarrow y e^{-\alpha x} = \int e^{-\alpha x} f(x) dx \Rightarrow y = e^{\alpha x} \int e^{-\alpha x} f(x) dx. \end{aligned}$$

$$\frac{1}{D - \alpha} f(x) = e^{\alpha x} \int e^{-\alpha x} f(x) dx$$

With the help of the above result we can find the required P.I.

⁵The notes' final line reads $\frac{x \cos x}{2}$; the working (and substitution) gives $\frac{x \cos x}{4}$.

Problem 7.24. Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = e^{-x} \sec^3 x$.

Solution. $(D^2 + 2D + 2)y = e^{-x} \sec^3 x$.

C.F. A.E.: $m^2 + 2m + 2 = 0 \Rightarrow m = \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm 2i}{2} = -1 \pm i$ ($\alpha = -1, \beta = 1$).

$$\text{C.F.} = e^{-x}(c_1 \cos x + c_2 \sin x).$$

P.I. Replace D by $D - 1$ (exponential):

$$\begin{aligned} \text{P.I.} &= e^{-x} \frac{1}{(D-1)^2 + 2(D-1) + 2} \sec^3 x \\ &= e^{-x} \frac{1}{D^2 + 1 - 2D + 2D - 2 + 2} \sec^3 x \\ &= e^{-x} \frac{1}{D^2 + 1} \sec^3 x. \end{aligned}$$

Since $D^2 + 1 = 0 \Rightarrow D = \pm i$, $D^2 + 1 = (D + i)(D - i)$:

$$\text{P.I.} = e^{-x} \frac{1}{(D + i)(D - i)} \sec^3 x.$$

Partial fractions: $\frac{1}{(D + i)(D - i)} = \frac{A}{D + i} + \frac{B}{D - i}$, i.e. $1 = A(D - i) + B(D + i)$.

$$D = i : 1 = 2iB \Rightarrow B = \frac{1}{2i} = -\frac{i}{2}; \quad D = -i : 1 = -2iA \Rightarrow A = -\frac{1}{2i} = \frac{i}{2}.$$

$$\text{P.I.} = e^{-x} \left(\frac{i}{2} \cdot \frac{1}{D + i} - \frac{i}{2} \cdot \frac{1}{D - i} \right) \sec^3 x = e^{-x} \left(\underbrace{\frac{i}{2} \frac{1}{D + i} \sec^3 x}_{\text{(II)}} - \underbrace{\frac{i}{2} \frac{1}{D - i} \sec^3 x}_{\text{(I)}} \right). \quad (1)$$

(I) ($\alpha = i$), using $\frac{1}{D - \alpha} f(x) = e^{\alpha x} \int e^{-\alpha x} f(x) dx$ and $e^{i\theta} = \cos \theta + i \sin \theta$:

$$\begin{aligned} \frac{1}{D - i} \sec^3 x &= e^{ix} \int e^{-ix} \sec^3 x dx \\ &= e^{ix} \int (\cos x - i \sin x) \sec^3 x dx \\ &= e^{ix} \left[\int \cos x \sec^3 x dx - i \int \sin x \sec^3 x dx \right] \\ &= e^{ix} \left[\int \sec^2 x dx - i \int \tan x \sec^2 x dx \right]. \end{aligned}$$

With $t = \tan x$, $dt = \sec^2 x dx$: $\int \tan x \sec^2 x dx = \int t dt = \frac{t^2}{2} = \frac{\tan^2 x}{2}$. So

$$\frac{1}{D - i} \sec^3 x = e^{ix} \left[\tan x - i \frac{\tan^2 x}{2} \right]. \quad (2)$$

(II) ($\alpha = -i$): similarly,

$$\frac{1}{D + i} \sec^3 x = e^{-ix} \int e^{ix} \sec^3 x dx = e^{-ix} \int (\cos x + i \sin x) \sec^3 x dx = e^{-ix} \left(\tan x + i \frac{\tan^2 x}{2} \right). \quad (3)$$

Substituting (2) and (3) in (1):

$$\begin{aligned} \text{P.I.} &= e^{-x} \left(\frac{i}{2} e^{-ix} \left(\tan x + \frac{i \tan^2 x}{2} \right) - \frac{i}{2} e^{ix} \left(\tan x - \frac{i \tan^2 x}{2} \right) \right) \\ &= e^{-x} \left(\tan x \left(\frac{i}{2} e^{-ix} - \frac{i}{2} e^{ix} \right) + \frac{i \tan^2 x}{2} \left(\frac{i}{2} e^{-ix} + \frac{i}{2} e^{ix} \right) \right) \\ &= e^{-x} \left(\tan x \cdot (-i) \left(\frac{e^{ix} - e^{-ix}}{2} \right) + \frac{i \tan^2 x}{2} \cdot i \left(\frac{e^{ix} + e^{-ix}}{2} \right) \right). \end{aligned}$$

Using $\frac{e^{ix} - e^{-ix}}{2} = i \sin x$ and $\frac{e^{ix} + e^{-ix}}{2} = \cos x$:

$$\text{P.I.} = e^{-x} \left(-i \tan x \cdot i \sin x - \frac{\tan^2 x}{2} \cos x \right) = e^{-x} \left(\sin x \tan x - \frac{1}{2} \tan^2 x \cos x \right).$$

General solution:

$$y = e^{-x}(c_1 \cos x + c_2 \sin x) + e^{-x} \left(\sin x \tan x - \frac{1}{2} \tan^2 x \cos x \right).$$

Problem 7.25. Solve $(D^2 - 2D + 2)y = e^x \tan x$.

Solution. **C.F.** A.E.: $m^2 - 2m + 2 = 0 \Rightarrow m = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$ ($\alpha = 1, \beta = 1$).

C.F. $= e^x(c_1 \cos x + c_2 \sin x)$.

P.I. Replace D by $D + 1$:

$$\begin{aligned} \text{P.I.} &= e^x \frac{1}{(D+1)^2 - 2(D+1) + 2} \tan x \\ &= e^x \frac{1}{D^2 + 1 + 2D - 2D - 2 + 2} \tan x \\ &= e^x \frac{1}{D^2 + 1} \tan x \\ &= e^x \frac{1}{(D+i)(D-i)} \tan x. \end{aligned}$$

As before, $A = \frac{i}{2}, B = -\frac{i}{2}$:

$$\text{P.I.} = e^x \left(\underbrace{\frac{i}{2} \frac{1}{D+i} \tan x}_{\text{(I)}} - \underbrace{\frac{i}{2} \frac{1}{D-i} \tan x}_{\text{(II)}} \right). \quad (1)$$

(I) ($\alpha = -i$):

$$\begin{aligned} \frac{1}{D+i} \tan x &= e^{-ix} \int e^{ix} \tan x \, dx \\ &= e^{-ix} \int (\cos x + i \sin x) \tan x \, dx \\ &= e^{-ix} \left[\int \sin x \, dx + i \int \frac{\sin^2 x}{\cos x} \, dx \right] \end{aligned}$$

$$\begin{aligned}
&= e^{-ix} \left[-\cos x + i \int \frac{1 - \cos^2 x}{\cos x} dx \right] \\
&= e^{-ix} \left[-\cos x + i \left(\int \sec x dx - \int \cos x dx \right) \right] \\
&= e^{-ix} [-\cos x + i \log |\sec x + \tan x| - i \sin x].
\end{aligned}$$

(II) ($\alpha = i$): similarly,

$$\frac{1}{D-i} \tan x = e^{ix} [-\cos x - i \log |\sec x + \tan x| + i \sin x].$$

Writing $\ell = \log |\sec x + \tan x|$ and substituting in (1):

$$\begin{aligned}
\text{P.I.} &= e^x \left[\frac{i}{2} e^{-ix} (-\cos x + i\ell - i \sin x) - \frac{i}{2} e^{ix} (-\cos x - i\ell + i \sin x) \right] \\
&= e^x \left[-\cos x \left(\frac{i}{2} e^{-ix} - \frac{i}{2} e^{ix} \right) + i(\ell - \sin x) \left(\frac{i}{2} e^{-ix} + \frac{i}{2} e^{ix} \right) \right] \\
&= e^x \left[-\cos x (-i) \left(\frac{e^{ix} - e^{-ix}}{2} \right) + i(\ell - \sin x) (i) \left(\frac{e^{ix} + e^{-ix}}{2} \right) \right] \\
&= e^x [i \cos x \cdot i \sin x - (\ell - \sin x) \cos x] = e^x [-\sin x \cos x - \ell \cos x + \sin x \cos x].
\end{aligned}$$

$$\text{P.I.} = -e^x \cos x \log |\sec x + \tan x|.$$

General solution:⁶

$$y = e^x (c_1 \cos x + c_2 \sin x) - e^x \cos x \log |\sec x + \tan x|.$$

Problems for practice

Solve (answers are given):

- $(D^2 + 4)y = 3x \sin x.$ $y = c_1 \cos 2x + c_2 \sin 2x + x \sin x - \frac{2}{3} \cos x$
- $\frac{d^2 y}{dx^2} - y = x \sin 3x + \cos x.$ $y = c_1 e^x + c_2 e^{-x} - \frac{1}{10} \left(\frac{3}{5} \cos 3x + x \sin 3x + 5 \cos x \right)$
- $\frac{d^2 y}{dx^2} - y = x \sin x + e^x + x^2 e^x.$ $y = c_1 e^x + c_2 e^{-x} - \frac{1}{2} (x \sin x + \cos x) + \frac{x e^x}{12} (2x^2 - 3x + 9)$
- $(D^4 + 2D^2 + 1)y = x^2 \cos x.$ $y = (c_1 + c_3 x) \cos x + (c_2 + c_4 x) \sin x + \frac{1}{12} (x^3 \sin x - \frac{1}{4} (x^4 - 9x^2) \cos x)$
- $\frac{d^2 y}{dx^2} + a^2 y = \sec ax.$ $y = c_1 \cos ax + c_2 \sin ax + \frac{x}{a} \sin ax + \frac{1}{a^2} \cos ax \log \cos ax$
- $\frac{d^2 y}{dx^2} + y = \operatorname{cosec} x.$ $y = c_1 \cos x + c_2 \sin x - x \cos x + \sin x \log \sin x$
- $(D^2 + 4)y = \tan 2x.$ $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4} \cos 2x \log (\sec 2x + \tan 2x)$
- $\frac{d^2 y}{dx^2} + y = x - \cot x.$ $y = c_1 \cos x + c_2 \sin x + x - \sin x \log (\operatorname{cosec} x - \cot x)$

⁶In the notes, $\frac{1}{D+i} \tan x$ is computed with e^{-ix} inside the integral (it should be e^{ix}), which flips the sign of the final P.I.; the corrected sign has been checked by substitution.

Solutions

Problem 7.26. Solve $(D^2 + 4)y = 3x \sin x$.

Solution. **C.F.** A.E.: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$ ($\alpha = 0, \beta = 2$). C.F. = $c_1 \cos 2x + c_2 \sin 2x$.

P.I. (Type 5; $\sin x \rightarrow$ I.P.) Replace D by $D + i$:

$$\begin{aligned} \text{P.I.} &= \text{I.P.} \frac{1}{D^2 + 4} 3x e^{ix} \\ &= \text{I.P.} e^{ix} \frac{1}{(D + i)^2 + 4} 3x \\ &= \text{I.P.} e^{ix} \frac{1}{D^2 + i^2 + 2Di + 4} 3x \\ &= \text{I.P.} e^{ix} \frac{1}{D^2 + 2Di + 3} 3x. \end{aligned}$$

Taking 3 common (Type 3):

$$\begin{aligned} &= \text{I.P.} e^{ix} \frac{1}{3 \left(\frac{D^2 + 2Di}{3} + 1 \right)} 3x = \text{I.P.} e^{ix} \left(1 + \frac{D^2 + 2Di}{3} \right)^{-1} x \\ &= \text{I.P.} e^{ix} \left(1 - \left(\frac{D^2 + 2Di}{3} \right) + \frac{(D^2 + 2Di)^2}{9} \right) x = \text{I.P.} e^{ix} \left(1 - \frac{2Di}{3} - \frac{D^2}{3} + \frac{4D^2 i^2}{9} \right) x \\ &= \text{I.P.} e^{ix} \left(1 - \frac{2Di}{3} - \frac{7D^2}{9} \right) x = \text{I.P.} e^{ix} \left(x - \frac{2i}{3} - \frac{7}{9}(0) \right) = \text{I.P.} (\cos x + i \sin x) \left(x - \frac{2i}{3} \right). \end{aligned}$$

$$\text{P.I.} = x \sin x - \frac{2}{3} \cos x.$$

General solution: $y = c_1 \cos 2x + c_2 \sin 2x + x \sin x - \frac{2}{3} \cos x.$

Problem 7.27. Solve $\frac{d^2 y}{dx^2} - y = x \sin 3x + \cos x$.

Solution. $(D^2 - 1)y = x \sin 3x + \cos x$.

C.F. A.E.: $m^2 - 1 = 0 \Rightarrow m = \pm 1$. C.F. = $c_1 e^x + c_2 e^{-x}$.

P.I.

$$\text{P.I.} = \underbrace{\frac{1}{D^2 - 1} x \sin 3x}_{(1) \text{ Type 5}} + \underbrace{\frac{1}{D^2 - 1} \cos x}_{(2) \text{ Type 2}}.$$

(1) Replace D by $D + 3i$:

$$\begin{aligned} \frac{1}{D^2 - 1} x \sin 3x &= \text{I.P.} e^{i3x} \frac{1}{(D + 3i)^2 - 1} x \\ &= \text{I.P.} e^{i3x} \frac{1}{D^2 + 9i^2 + 6Di - 1} x \\ &= \text{I.P.} e^{i3x} \frac{1}{D^2 + 6Di - 10} x \\ &= \text{I.P.} \frac{e^{i3x}}{-10} \left(1 + \frac{D^2 + 6Di}{-10} \right)^{-1} x \end{aligned}$$

$$\begin{aligned}
&= \text{I.P.} \frac{e^{i3x}}{-10} \left(1 + \frac{D^2 + 6Di}{10} + \dots \right) x \\
&= \text{I.P.} \frac{e^{i3x}}{-10} \left(x + \frac{6i}{10} \right) \\
&= -\frac{1}{10} \text{I.P.} (\cos 3x + i \sin 3x) \left(x + \frac{3i}{5} \right) \\
&= -\frac{1}{10} \left(x \sin 3x + \frac{3}{5} \cos 3x \right).
\end{aligned}$$

(2) Replace D^2 by -1 : $\frac{\cos x}{-1-1} = -\frac{\cos x}{2}$.

$$\text{P.I.} = -\frac{1}{10} \left(\frac{3}{5} \cos 3x + x \sin 3x \right) - \frac{\cos x}{2} = -\frac{1}{10} \left(\frac{3}{5} \cos 3x + x \sin 3x + 5 \cos x \right).$$

General solution:⁷

$$y = c_1 e^x + c_2 e^{-x} - \frac{1}{10} \left(\frac{3}{5} \cos 3x + x \sin 3x + 5 \cos x \right).$$

Problem 7.28. Solve $\frac{d^2 y}{dx^2} - y = x \sin x + e^x + x^2 e^x$.

Solution. $(D^2 - 1)y = x \sin x + e^x(x^2 + 1)$. **C.F.** A.E.: $m^2 - 1 = 0 \Rightarrow m = \pm 1$. **C.F.** = $c_1 e^x + c_2 e^{-x}$.

P.I.

$$\text{P.I.} = \underbrace{\frac{1}{D^2 - 1} x \sin x}_{(1)} + \underbrace{\frac{1}{D^2 - 1} e^x (x^2 + 1)}_{(2)}.$$

(1) Type 5; replace D by $D + i$:

$$\begin{aligned}
\frac{1}{D^2 - 1} x \sin x &= \text{I.P.} e^{ix} \frac{1}{(D + i)^2 - 1} x \\
&= \text{I.P.} e^{ix} \frac{1}{D^2 + 2Di - 2} x \\
&= \text{I.P.} \frac{e^{ix}}{-2} \left(1 - \frac{D^2 + 2Di}{2} \right)^{-1} x \\
&= \text{I.P.} e^{ix} \left(-\frac{1}{2} \right) \left(1 + \frac{D^2 + 2Di}{2} + \frac{(D^2 + 2Di)^2}{4} \right) x \\
&= \text{I.P.} e^{ix} \left(-\frac{1}{2} \right) \left(1 + \frac{2Di}{2} + \frac{D^2}{2} + \frac{4D^2 i^2}{4} \right) x \\
&= \text{I.P.} \left(-\frac{1}{2} e^{ix} \right) (x + i) \\
&= \text{I.P.} \left(-\frac{1}{2} \right) (\cos x + i \sin x) (x + i) \\
&= -\frac{1}{2} (\cos x + x \sin x).
\end{aligned}$$

⁷The notes set up part (1) and leave it incomplete; it is completed here, and agrees with the answer given in the notes.

(2) Type 4; replace D by D + 1:

$$\begin{aligned}
 \frac{1}{D^2 - 1} e^x (x^2 + 1) &= e^x \frac{1}{(D + 1)^2 - 1} (x^2 + 1) \\
 &= e^x \frac{1}{D^2 + 2D} (x^2 + 1) \\
 &= e^x \frac{1}{2D \left(\frac{D}{2} + 1 \right)} (x^2 + 1) \\
 &= e^x \frac{1}{2D} \left(1 + \frac{D}{2} \right)^{-1} (x^2 + 1) \\
 &= e^x \frac{1}{2D} \left(1 - \frac{D}{2} + \frac{D^2}{4} \right) (x^2 + 1) \\
 &= e^x \frac{1}{2D} \left((x^2 + 1) - \frac{1}{2}(2x) + \frac{1}{4}(2) \right) \\
 &= e^x \frac{1}{2D} \left(x^2 - x + \frac{3}{2} \right) \\
 &= \frac{e^x}{2} \int \left(x^2 - x + \frac{3}{2} \right) dx \\
 &= \frac{e^x}{2} \left(\frac{x^3}{3} - \frac{x^2}{2} + \frac{3x}{2} \right) \\
 &= \frac{e^x}{12} (2x^3 - 3x^2 + 9x) \\
 &= \frac{e^x x}{12} (2x^2 - 3x + 9).
 \end{aligned}$$

General solution:

$$y = c_1 e^x + c_2 e^{-x} + \frac{e^x x}{12} (2x^2 - 3x + 9) - \frac{1}{2} (\cos x + x \sin x).$$

Problem 7.29. Solve $(D^4 + 2D^2 + 1)y = x^2 \cos x$.

Solution. **C.F.** A.E.: $m^4 + 2m^2 + 1 = 0 \Rightarrow (m^2 + 1)^2 = 0 \Rightarrow m^2 = -1, -1 \Rightarrow m = \pm i, \pm i$ ($\alpha = 0, \beta = 1$, repeated).

$$\text{C.F.} = c_1 \cos x + c_2 \sin x + x(c_3 \cos x + c_4 \sin x) = (c_1 + c_3 x) \cos x + (c_2 + c_4 x) \sin x.$$

P.I. $D^4 + 2D^2 + 1 = (D^2 + 1)^2$. Replace D by D + i:

$$\begin{aligned}
 \text{P.I.} &= \text{R.P.} \frac{1}{(D^2 + 1)^2} x^2 e^{ix} \\
 &= \text{R.P.} e^{ix} \frac{1}{((D + i)^2 + 1)^2} x^2 \\
 &= \text{R.P.} e^{ix} \frac{1}{(D^2 + 2Di)^2} x^2 \\
 &= \text{R.P.} e^{ix} \frac{1}{(2Di)^2 \left(\frac{D^2}{2Di} + 1 \right)^2} x^2 \\
 &= \text{R.P.} e^{ix} \frac{1}{4i^2 D^2} \cdot \frac{1}{\left(1 + \frac{D}{2i} \right)^2} x^2
 \end{aligned}$$

$$= \text{R.P. } e^{ix} \frac{1}{-4D^2} \left(1 - \frac{Di}{2}\right)^{-2} x^2.$$

Using $(1 - x)^{-2} = 1 + 2x + 3x^2 + \dots$ (neglecting higher powers):

$$\begin{aligned} &= \text{R.P. } e^{ix} \left(-\frac{1}{4D^2}\right) \left(1 + \frac{2Di}{2} + 3\left(\frac{Di}{2}\right)^2\right) x^2 = \text{R.P. } \left(-\frac{e^{ix}}{4}\right) \frac{1}{D^2} \left(1 + Di - \frac{3}{4}D^2\right) x^2 \\ &= \text{R.P. } \left(-\frac{e^{ix}}{4}\right) \frac{1}{D^2} \left(x^2 + 2xi - \frac{3}{4}(2)\right) = \text{R.P. } \left(-\frac{e^{ix}}{4}\right) \frac{1}{D} \left(\frac{x^3}{3} + \frac{2ix^2}{2} - \frac{3}{2}x\right) \\ &= \text{R.P. } \left(-\frac{e^{ix}}{4}\right) \left(\frac{x^4}{3 \cdot 4} + \frac{ix^3}{3} - \frac{3}{2} \cdot \frac{x^2}{2}\right) = \text{R.P. } \left(-\frac{1}{4}\right) (\cos x + i \sin x) \left(\left(\frac{x^4}{12} - \frac{3x^2}{4}\right) + i\frac{x^3}{3}\right) \\ &= -\frac{1}{4} \left(\cos x \left(\frac{x^4}{12} - \frac{3x^2}{4}\right) + i^2 \sin x \frac{x^3}{3}\right) = -\frac{1}{4} \left(\cos x \frac{x^4 - 9x^2}{12} - \frac{x^3 \sin x}{3}\right) \\ &= -\frac{1}{4} \cdot \frac{\cos x (x^4 - 9x^2)}{12} + \frac{1}{12} x^3 \sin x = \frac{1}{12} \left(x^3 \sin x - \frac{1}{4} \cos x (x^4 - 9x^2)\right). \end{aligned}$$

General solution:

$$y = (c_1 + c_3x) \cos x + (c_2 + c_4x) \sin x + \frac{1}{12} \left(x^3 \sin x - \frac{1}{4} \cos x (x^4 - 9x^2)\right).$$

Problem 7.30. Solve $\frac{d^2y}{dx^2} + a^2y = \sec ax$.

Solution. $(D^2 + a^2)y = \sec ax$. **C.F.** A.E.: $m^2 + a^2 = 0 \Rightarrow m = \pm ai$ ($\alpha = 0, \beta = a$).
C.F. = $c_1 \cos ax + c_2 \sin ax$.

P.I. (Type 7)

$$\text{P.I.} = \frac{1}{D^2 + a^2} \sec ax = \frac{1}{(D + ai)(D - ai)} \sec ax.$$

$$\frac{1}{(D + ai)(D - ai)} = \frac{A}{D + ai} + \frac{B}{D - ai}; \quad 1 = A(D - ai) + B(D + ai). \quad D = ai: 1 = 2aiB \Rightarrow B = \frac{1}{2ai} = -\frac{i}{2a}; \quad D = -ai: 1 = -2aiA \Rightarrow A = \frac{i}{2a}.$$

$$\text{P.I.} = \frac{i}{2a} \underbrace{\frac{1}{D + ai} \sec ax}_{(1)} - \frac{i}{2a} \underbrace{\frac{1}{D - ai} \sec ax}_{(2)}.$$

(1)

$$\begin{aligned} \frac{1}{D + ai} \sec ax &= e^{-aix} \int e^{aix} \sec ax \, dx \\ &= e^{-aix} \int (\cos ax + i \sin ax) \frac{1}{\cos ax} \, dx \\ &= e^{-aix} \int (1 + i \tan ax) \, dx \\ &= e^{-aix} \left(\int dx + i \int \tan ax \, dx \right) \\ &= e^{-aix} \left(x + \frac{i}{a} \log |\sec ax| \right) \end{aligned}$$

$$\begin{aligned}
&= (\cos ax - i \sin ax) \left(x + \frac{i}{a} \log(\sec ax) \right) \\
&= x \cos ax - ix \sin ax + \frac{i}{a} \log(\sec ax) \cos ax - \frac{i^2}{a} \log(\sec ax) \sin ax \\
&= x \cos ax - ix \sin ax - \frac{i}{a} \log(\cos ax) \cos ax - \frac{1}{a} \log(\cos ax) \sin ax.
\end{aligned}$$

(Here we used $\log \sec ax = -\log \cos ax$.) (2) Similarly,

$$\frac{1}{D - ai} \sec ax = x \cos ax + ix \sin ax + \frac{i}{a} \log(\cos ax) \cos ax - \frac{1}{a} \log(\cos ax) \sin ax.$$

Hence

$$\begin{aligned}
\text{P.I.} &= \frac{i}{2a} \left(x \cos ax - ix \sin ax - \frac{i}{a} \log(\cos ax) \cos ax - \frac{1}{a} \log(\cos ax) \sin ax \right) \\
&\quad - \frac{i}{2a} \left(x \cos ax + ix \sin ax + \frac{i}{a} \log(\cos ax) \cos ax - \frac{1}{a} \log(\cos ax) \sin ax \right) \\
&= -\frac{i^2 x \sin ax}{2a} - \frac{i^2}{2a^2} \log(\cos ax) \cos ax - \frac{i^2 x \sin ax}{2a} - \frac{i^2}{2a^2} \log(\cos ax) \cos ax \\
&= \frac{x \sin ax}{2a} + \frac{x \sin ax}{2a} + \frac{1}{2a^2} \log(\cos ax) \cos ax + \frac{1}{2a^2} \log(\cos ax) \cos ax \\
&= \frac{x \sin ax}{a} + \frac{1}{a^2} \log(\cos ax) \cos ax.
\end{aligned}$$

General solution:

$$y = c_1 \cos ax + c_2 \sin ax + \frac{x}{a} \sin ax + \frac{1}{a^2} \log(\cos ax) \cos ax.$$

Problem 7.31. Solve $(D^2 + 1)y = \operatorname{cosec} x$.

Solution. **C.F.** A.E.: $m^2 + 1 = 0 \Rightarrow m = \pm i$ ($\alpha = 0, \beta = 1$). **C.F.** $= c_1 \cos x + c_2 \sin x$.

P.I. (Type 7) With $A = \frac{i}{2}$, $B = -\frac{i}{2}$ as before,

$$\text{P.I.} = \frac{1}{(D + i)(D - i)} \operatorname{cosec} x = \frac{i}{2} \underbrace{\frac{1}{D + i} \operatorname{cosec} x}_{\text{(II)}} - \frac{i}{2} \underbrace{\frac{1}{D - i} \operatorname{cosec} x}_{\text{(I)}}.$$

(I)

$$\begin{aligned}
\frac{1}{D - i} \operatorname{cosec} x &= e^{ix} \int e^{-ix} \operatorname{cosec} x \, dx \\
&= e^{ix} \int (\cos x - i \sin x) \operatorname{cosec} x \, dx \\
&= e^{ix} \left[\int \frac{\cos x}{\sin x} \, dx - i \int \frac{\sin x}{\sin x} \, dx \right] \\
&= e^{ix} \left(\int \cot x \, dx - i \int dx \right) \\
&= e^{ix} (\log(\sin x) - ix) \\
&= (\cos x + i \sin x) (\log(\sin x) - ix)
\end{aligned}$$

$$\begin{aligned}
&= \cos x \log(\sin x) + i \sin x \log(\sin x) - ix \cos x - i^2 x \sin x \\
&= \log(\sin x) \cos x + i \log(\sin x) \sin x - ix \cos x + x \sin x.
\end{aligned}$$

(II) Similarly,

$$\frac{1}{D+i} \operatorname{cosec} x = \log(\sin x) \cos x - i \log(\sin x) \sin x + ix \cos x + x \sin x.$$

$$\begin{aligned}
\text{P.I.} &= \frac{i}{2} (\log(\sin x) \cos x - i \log(\sin x) \sin x + ix \cos x + x \sin x) \\
&\quad - \frac{i}{2} (\log(\sin x) \cos x + i \log(\sin x) \sin x - ix \cos x + x \sin x) \\
&= -\frac{i^2}{2} \log(\sin x) \sin x + \frac{i^2}{2} x \cos x - \frac{i^2}{2} \log(\sin x) \sin x + \frac{i^2}{2} x \cos x \\
&= \log(\sin x) \sin x - x \cos x.
\end{aligned}$$

General solution:

$$y = c_1 \cos x + c_2 \sin x + \log(\sin x) \sin x - x \cos x.$$

Problem 7.32. Solve $(D^2 + 4)y = \tan 2x$.

Solution. C.F. A.E.: $m^2 + 4 = 0 \Rightarrow m = \pm 2i$ ($\alpha = 0, \beta = 2$). C.F. = $c_1 \cos 2x + c_2 \sin 2x$.

P.I. (Type 7)

$$\text{P.I.} = \frac{1}{(D+2i)(D-2i)} \tan 2x; \quad 1 = A(D-2i) + B(D+2i).$$

$$D = 2i: 1 = 4iB \Rightarrow B = \frac{1}{4i} = -\frac{i}{4}; \quad D = -2i: 1 = -4iA \Rightarrow A = \frac{i}{4}.$$

$$\text{P.I.} = \frac{i}{4} \underbrace{\frac{1}{D+2i} \tan 2x}_{\text{(II)}} - \frac{i}{4} \underbrace{\frac{1}{D-2i} \tan 2x}_{\text{(I)}}.$$

(I) Using $\sin^2 2x = 1 - \cos^2 2x$:

$$\begin{aligned}
\frac{1}{D-2i} \tan 2x &= e^{2ix} \int e^{-2ix} \tan 2x \, dx \\
&= e^{2ix} \int (\cos 2x - i \sin 2x) \tan 2x \, dx \\
&= e^{2ix} \left[\int \cos 2x \frac{\sin 2x}{\cos 2x} \, dx - i \int \sin 2x \frac{\sin 2x}{\cos 2x} \, dx \right] \\
&= e^{2ix} \left[\int \sin 2x \, dx - i \int \frac{1}{\cos 2x} \, dx + i \int \frac{\cos^2 2x}{\cos 2x} \, dx \right] \\
&= e^{2ix} \left[\int \sin 2x \, dx - i \int \sec 2x \, dx + i \int \cos 2x \, dx \right] \\
&= e^{2ix} \left(-\frac{\cos 2x}{2} - \frac{i}{2} \log |\sec 2x + \tan 2x| + \frac{i}{2} \sin 2x \right).
\end{aligned}$$

Writing $L = \log(\sec 2x + \tan 2x)$ and $e^{2ix} = \cos 2x + i \sin 2x$:

$$\frac{1}{D-2i} \tan 2x = (\cos 2x + i \sin 2x) \left(-\frac{\cos 2x}{2} - \frac{i}{2} L + \frac{i}{2} \sin 2x \right)$$

$$\begin{aligned}
&= -\frac{\cos^2 2x}{2} - \frac{i}{2}L \cos 2x + \frac{i}{2} \sin 2x \cos 2x - \frac{i}{2} \sin 2x \cos 2x - \frac{i^2}{2}L \sin 2x + \frac{i^2}{2} \sin^2 2x \\
&= -\frac{1}{2}(\cos^2 2x + \sin^2 2x) - \frac{i}{2}L \cos 2x + \frac{1}{2}L \sin 2x = -\frac{1}{2} - \frac{i}{2}L \cos 2x + \frac{1}{2}L \sin 2x.
\end{aligned}$$

(II) Similarly,

$$\frac{1}{D+2i} \tan 2x = -\frac{1}{2} + \frac{i}{2}L \cos 2x + \frac{1}{2}L \sin 2x.$$

$$\begin{aligned}
\text{P.I.} &= \frac{i}{4} \left(-\frac{1}{2} + \frac{i}{2}L \cos 2x + \frac{1}{2}L \sin 2x \right) - \frac{i}{4} \left(-\frac{1}{2} - \frac{i}{2}L \cos 2x + \frac{1}{2}L \sin 2x \right) \\
&= \frac{i}{4} \left(\frac{i}{2}L \cos 2x \right) - \frac{i}{4} \left(-\frac{i}{2}L \cos 2x \right) \\
&= -\frac{1}{8}L \cos 2x - \frac{1}{8}L \cos 2x \\
&= -\frac{1}{4} \log(\sec 2x + \tan 2x) \cos 2x.
\end{aligned}$$

General solution:

$$y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4} \log(\sec 2x + \tan 2x) \cos 2x.$$

Problem 7.33. Solve $(D^2 + 1)y = x - \cot x$.

Solution. **C.F.** A.E.: $m^2 + 1 = 0 \Rightarrow m = \pm i$. **C.F.** $= c_1 \cos x + c_2 \sin x$.

P.I.

$$\text{P.I.} = \underbrace{\frac{1}{D^2 + 1}x}_{\text{(II)}} - \underbrace{\frac{1}{D^2 + 1}\cot x}_{\text{(I)}}.$$

(I) Type 7, with $A = \frac{i}{2}$, $B = -\frac{i}{2}$:

$$\frac{1}{D^2 + 1} \cot x = \frac{i}{2} \frac{1}{D + i} \cot x - \frac{i}{2} \frac{1}{D - i} \cot x.$$

Using $\cos^2 x = 1 - \sin^2 x$:

$$\begin{aligned}
\frac{1}{D - i} \cot x &= e^{ix} \int e^{-ix} \cot x \, dx \\
&= e^{ix} \int (\cos x - i \sin x) \cot x \, dx \\
&= e^{ix} \left[\int \cos x \frac{\cos x}{\sin x} \, dx - i \int \sin x \frac{\cos x}{\sin x} \, dx \right] \\
&= e^{ix} \left(\int \frac{1}{\sin x} \, dx - \int \frac{\sin^2 x}{\sin x} \, dx - i \int \cos x \, dx \right) \\
&= e^{ix} \left(\int \operatorname{cosec} x \, dx - \int \sin x \, dx - i \int \cos x \, dx \right) \\
&= e^{ix} (\log(\operatorname{cosec} x - \cot x) + \cos x - i \sin x) \\
&= (\cos x + i \sin x) (\log(\operatorname{cosec} x - \cot x) + \cos x - i \sin x) \\
&= \cos x \log(\operatorname{cosec} x - \cot x) + \cos^2 x - i \sin x \cos x
\end{aligned}$$

$$\begin{aligned}
& + i \sin x \log(\operatorname{cosec} x - \cot x) + i \sin x \cos x + \sin^2 x \\
& = \cos x \log(\operatorname{cosec} x - \cot x) + i \sin x \log(\operatorname{cosec} x - \cot x) + 1.
\end{aligned}$$

Similarly, $\frac{1}{D+i} \cot x = \cos x \log(\operatorname{cosec} x - \cot x) - i \sin x \log(\operatorname{cosec} x - \cot x) + 1$. Hence

$$\begin{aligned}
\frac{1}{D^2+1} \cot x &= \frac{i}{2} (\cos x \log(\operatorname{cosec} x - \cot x) - i \sin x \log(\operatorname{cosec} x - \cot x) + 1) \\
&\quad - \frac{i}{2} (\cos x \log(\operatorname{cosec} x - \cot x) + i \sin x \log(\operatorname{cosec} x - \cot x) + 1) \\
&= \sin x \log(\operatorname{cosec} x - \cot x).
\end{aligned}$$

(II) Type 3:

$$\frac{1}{D^2+1} x = (1+D^2)^{-1} x = (1-D^2+\dots)x = x-0=x.$$

$$\text{P.I.} = x - \sin x \log(\operatorname{cosec} x - \cot x).$$

General solution:

$$y = c_1 \cos x + c_2 \sin x + x - \sin x \log(\operatorname{cosec} x - \cot x).$$

8 Homogeneous Linear Equations with Variable Coefficients

8.1 Euler's (Cauchy's) homogeneous linear differential equation

A differential equation of the form

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = f(x),$$

where $a_0, a_1, \dots, a_n$ are constants, is known as *Euler's homogeneous differential equation*. It can also be written as

$$(a_0 x^n D^n + a_1 x^{n-1} D^{n-1} + \dots + a_{n-1} x D + a_n) y = f(x). \quad (1)$$

To convert the equation into one with constant coefficients, put $x = e^z$, i.e. $z = \log_e x$, so that $\frac{dz}{dx} = \frac{1}{x}$. Then

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{dy}{dz} \cdot \frac{1}{x} \Rightarrow x \frac{dy}{dx} = \frac{dy}{dz} \Rightarrow x D y = \theta y, \quad D = \frac{d}{dx}, \quad \theta = \frac{d}{dz},$$

$$\boxed{x D = \theta}.$$

Now

$$\begin{aligned}
\frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dz} \right) = -\frac{1}{x^2} \frac{dy}{dz} + \frac{1}{x} \frac{d}{dz} \left(\frac{dy}{dz} \right) \frac{dz}{dx} \quad (\text{product rule}) \\
&= -\frac{1}{x^2} \frac{dy}{dz} + \frac{1}{x^2} \frac{d^2 y}{dz^2}, \\
x^2 \frac{d^2 y}{dx^2} &= -\frac{dy}{dz} + \frac{d^2 y}{dz^2} \Rightarrow x^2 D^2 y = -\theta y + \theta^2 y = \theta(\theta-1)y,
\end{aligned}$$

$$x D = \theta, \quad x^2 D^2 = \theta(\theta-1), \quad x^3 D^3 = \theta(\theta-1)(\theta-2), \dots$$

By the above transformation, the given differential equation (1) is converted into a differential equation with constant coefficients, and hence we can solve it with the help of the previous types.

Problems

Problem 8.1. Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \log x \sin(\log x)$.

Solution. $(x^2 D^2 + xD + 1)y = \log x \sin(\log x)$. (1)

Put $x = e^z$ ($z = \log x$); then $xD = \theta$, $x^2 D^2 = \theta(\theta - 1)$, where $\theta = \frac{d}{dz}$, $D = \frac{d}{dx}$. Equation (1) becomes

$$(\theta(\theta - 1) + \theta + 1)y = z \sin z \Rightarrow (\theta^2 - \theta + \theta + 1)y = z \sin z \Rightarrow (\theta^2 + 1)y = z \sin z.$$

C.F. A.E.: $m^2 + 1 = 0 \Rightarrow m = \pm i$ ($\alpha = 0$, $\beta = 1$). C.F. = $c_1 \cos z + c_2 \sin z$.

P.I. (Type 5) Replace θ by $\theta + i$:

$$\begin{aligned} \text{P.I.} &= \frac{1}{\theta^2 + 1} z \sin z \\ &= \text{I.P.} \frac{1}{\theta^2 + 1} z e^{iz} \\ &= \text{I.P.} e^{iz} \frac{1}{(\theta + i)^2 + 1} z \\ &= \text{I.P.} e^{iz} \frac{1}{\theta^2 + i^2 + 2\theta i + 1} z \\ &= \text{I.P.} e^{iz} \frac{1}{\theta^2 + 2\theta i} z \\ &= \text{I.P.} e^{iz} \frac{1}{2\theta i \left(\frac{\theta^2}{2\theta i} + 1 \right)} z \\ &= \text{I.P.} e^{iz} \frac{1}{2\theta i} \left(1 - \frac{\theta i}{2} \right)^{-1} z \quad \left(\frac{\theta}{2i} = -\frac{\theta i}{2} \right) \\ &= \text{I.P.} e^{iz} \frac{1}{2\theta i} \left(1 + \frac{\theta i}{2} + \frac{\theta^2 i^2}{4} \right) z \\ &= \text{I.P.} e^{iz} \frac{1}{2\theta i} \left(z + \frac{i}{2} + 0 \right) \\ &= \text{I.P.} e^{iz} \frac{1}{\theta} \left(\frac{z}{2i} + \frac{1}{4} \right) \\ &= \text{I.P.} e^{iz} \left(\frac{1}{2i} \int z dz + \frac{1}{4} \int dz \right) \\ &= \text{I.P.} e^{iz} \left(\frac{1}{2i} \cdot \frac{z^2}{2} + \frac{1}{4} z \right) \\ &= \text{I.P.} e^{iz} \left(\frac{z^2}{4i} + \frac{z}{4} \right) \\ &= \text{I.P.} (\cos z + i \sin z) \left(-\frac{iz^2}{4} + \frac{z}{4} \right) \\ &= -\frac{z^2 \cos z}{4} + \frac{z \sin z}{4}. \end{aligned}$$

$$y = \text{C.F.} + \text{P.I.} = c_1 \cos z + c_2 \sin z + \frac{1}{4} (z \sin z - z^2 \cos z).$$

Replacing z by $\log x$:

$$y = c_1 \cos(\log x) + c_2 \sin(\log x) + \frac{1}{4} (\log x \sin(\log x) - (\log x)^2 \cos(\log x)).$$

Problem 8.2. Solve $(x^2D^2 + 4xD + 2)y = e^x$.

Solution. Put $x = e^z$: $x^2D^2 = \theta(\theta - 1)$, $xD = \theta$ ($D = \frac{d}{dx}$, $\theta = \frac{d}{dz}$).

$$(\theta(\theta - 1) + 4\theta + 2)y = e^{e^z} \Rightarrow (\theta^2 - \theta + 4\theta + 2)y = e^{e^z} \Rightarrow (\theta^2 + 3\theta + 2)y = e^{e^z}.$$

C.F. A.E.: $m^2 + 3m + 2 = 0 \Rightarrow m = -1, -2$. C.F. = $c_1e^{-z} + c_2e^{-2z}$.

P.I. (Type 7)

$$\text{P.I.} = \frac{1}{\theta^2 + 3\theta + 2}e^{e^z} = \frac{1}{(\theta + 1)(\theta + 2)}e^{e^z}.$$

$$\frac{1}{(\theta + 1)(\theta + 2)} = \frac{A}{\theta + 1} + \frac{B}{\theta + 2}; 1 = A(\theta + 2) + B(\theta + 1); \theta = -1 \Rightarrow A = 1; \theta = -2 \Rightarrow 1 = B(-1) \Rightarrow B = -1.$$

$$\text{P.I.} = \underbrace{\frac{1}{\theta + 1}e^{e^z}}_{(a)} - \underbrace{\frac{1}{\theta + 2}e^{e^z}}_{(b)}.$$

(a) With $e^z = t$, $e^z dz = dt$:

$$\frac{1}{\theta + 1}e^{e^z} = e^{-z} \int e^ze^{e^z} dz = e^{-z} \int e^t dt = e^{-z}e^t = e^{-z}e^{e^z}.$$

(b) With the same substitution, and integration by parts ($\int uv = u \int v - \int(u' \int v)$):

$$\begin{aligned} \frac{1}{\theta + 2}e^{e^z} &= e^{-2z} \int e^{2z}e^{e^z} dz = e^{-2z} \int e^z \cdot e^ze^{e^z} dz = e^{-2z} \int te^t dt \\ &= e^{-2z} \left(t \int e^t dt - \int \left(\frac{dt}{dt} \int e^t dt \right) dt \right) = e^{-2z} \left(te^t - \int e^t dt \right) = e^{-2z}(te^t - e^t) \\ &= e^{-2z}(e^ze^{e^z} - e^{e^z}) = e^{-z}e^{e^z} - e^{-2z}e^{e^z}. \end{aligned}$$

$$\text{P.I.} = e^{-z}e^{e^z} - e^{-z}e^{e^z} + e^{-2z}e^{e^z} = e^{-2z}e^{e^z}.$$

$$y = \text{C.F.} + \text{P.I.} = c_1e^{-z} + c_2e^{-2z} + e^{-2z}e^{e^z}.$$

With $z = \log x$:

$$y = c_1e^{-\log x} + c_2e^{-2\log x} + e^{-2\log x}e^{e^{\log x}} = c_1e^{\log x^{-1}} + c_2e^{\log x^{-2}} + e^{\log x^{-2}}e^x = c_1x^{-1} + c_2x^{-2} + x^{-2}e^x,$$

$$y = \frac{c_1}{x} + \frac{c_2}{x^2} + \frac{e^x}{x^2}.$$

Problem 8.3. The radial displacement u in a rotating disc at a distance r from the axis is given by

$$r^2 \frac{d^2u}{dr^2} + r \frac{du}{dr} - u + kr^3 = 0,$$

where k is a constant. Solve the above equation under the conditions $u = 0$ when $r = 0$ and $r = a$.

Solution.

$$r^2 \frac{d^2u}{dr^2} + r \frac{du}{dr} - u = -kr^3 \Rightarrow (r^2D^2 + rD - 1)u = -kr^3 \quad \left(D = \frac{d}{dr} \right).$$

Put $r = e^z$: $r^2 D^2 = \theta(\theta - 1)$, $rD = \theta$.

$$(\theta(\theta - 1) + \theta - 1)u = -ke^{3z} \Rightarrow (\theta^2 - \theta + \theta - 1)u = -ke^{3z} \Rightarrow (\theta^2 - 1)u = -ke^{3z}.$$

C.F. A.E.: $m^2 - 1 = 0 \Rightarrow m = \pm 1$. C.F. = $c_1 e^z + c_2 e^{-z}$.

P.I. (Type 1; replace θ by 3)

$$\text{P.I.} = \frac{1}{\theta^2 - 1} (-ke^{3z}) = \frac{-ke^{3z}}{9 - 1} = -\frac{ke^{3z}}{8}.$$

$$u = c_1 e^z + c_2 e^{-z} - \frac{k}{8} e^{3z}.$$

With $z = \log r$ ($e^z = r$):

$$u = c_1 r + \frac{c_2}{r} - \frac{kr^3}{8}.$$

*Applying the conditions.*⁸ For u to be finite ($u = 0$) at $r = 0$ we need $c_2 = 0$. At $r = a$, $u = 0$:

$$c_1 a - \frac{ka^3}{8} = 0 \Rightarrow c_1 = \frac{ka^2}{8}.$$

$$\boxed{u = \frac{kr}{8} (a^2 - r^2)}.$$

Problem 8.4. Solve $(x^2 D^2 + xD + 1)y = \sin(\log x^2)$.

Solution. Put $x = e^z$: $x^2 D^2 = \theta(\theta - 1)$, $xD = \theta$; also $\log x^2 = 2 \log x = 2z$.

$$(\theta(\theta - 1) + \theta + 1)y = \sin(2z) \Rightarrow (\theta^2 - \theta + \theta + 1)y = \sin 2z \Rightarrow (\theta^2 + 1)y = \sin 2z.$$

C.F. A.E.: $m^2 + 1 = 0 \Rightarrow m = \pm i$. C.F. = $c_1 \cos z + c_2 \sin z$.

P.I. (Type 2; replace θ^2 by -2^2)

$$\text{P.I.} = \frac{1}{\theta^2 + 1} \sin 2z = \frac{1}{-4 + 1} \sin 2z = \frac{\sin 2z}{-3}.$$

$$y = c_1 \cos z + c_2 \sin z - \frac{1}{3} \sin 2z.$$

With $z = \log x$:

$$y = c_1 \cos(\log x) + c_2 \sin(\log x) - \frac{1}{3} \sin(2 \log x), \quad \text{i.e.}$$

$$\boxed{y = c_1 \cos(\log x) + c_2 \sin(\log x) - \frac{1}{3} \sin(\log x^2)}.$$

⁸The notes stop at the general solution; the conditions stated in the problem are applied here.

8.2 Legendre's linear differential equation

An equation of the form

$$a_0(ax+b)^n \frac{d^n y}{dx^n} + a_1(ax+b)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_n y = f(x)$$

is reduced to one with constant coefficients by putting $ax+b = e^z$, i.e. $z = \log(ax+b)$. Then

$$(ax+b)D = a\theta, \quad (ax+b)^2 D^2 = a^2 \theta(\theta-1), \quad \dots \quad \left(\theta = \frac{d}{dz} \right).$$

Problem 8.5. Solve $(3x+2)^2 \frac{d^2 y}{dx^2} + (3x+2) \frac{dy}{dx} + 7y = 3x^2 + 2x + 1$.

Solution. Put $3x+2 = e^z$, so that $x = \frac{e^z - 2}{3}$; then $(3x+2)D = 3\theta$ and $(3x+2)^2 D^2 = 3^2 \theta(\theta-1)$. the left-hand side becomes

$$(9\theta(\theta-1) + 3\theta + 7)y.$$

The right-hand side:

$$\begin{aligned} 3x^2 + 2x + 1 &= 3 \left[\frac{e^z - 2}{3} \right]^2 + 2 \left[\frac{e^z - 2}{3} \right] + 1 = 3 \left(\frac{e^{2z} + 4 - 4e^z}{9} \right) + \frac{2}{3}e^z - \frac{4}{3} + 1 \\ &= \frac{e^{2z}}{3} + \frac{4}{3} - \frac{4}{3}e^z + \frac{2}{3}e^z - \frac{4}{3} + 1 = \frac{e^{2z}}{3} - \frac{2}{3}e^z + 1 = \frac{1}{3}(e^{2z} - 2e^z + 3). \end{aligned}$$

So

$$(9\theta^2 - 9\theta + 3\theta + 7)y = \frac{1}{3}(e^{2z} - 2e^z + 3) \Rightarrow (9\theta^2 - 6\theta + 7)y = \frac{1}{3}(e^{2z} - 2e^z + 3).$$

C.F. A.E.: $9m^2 - 6m + 7 = 0$,

$$m = \frac{6 \pm \sqrt{36 - 252}}{18} = \frac{6 \pm \sqrt{-216}}{18} = \frac{6 \pm 6\sqrt{6}i}{18} = \frac{1}{3} \pm i \frac{\sqrt{6}}{3}.$$

$$\text{C.F.} = e^{z/3} \left(c_1 \cos \frac{\sqrt{6}}{3} z + c_2 \sin \frac{\sqrt{6}}{3} z \right).$$

P.I. (Type 1)

$$\text{P.I.} = \frac{1}{9\theta^2 - 6\theta + 7} \cdot \frac{1}{3}(e^{2z} - 2e^z + 3) = \frac{1}{3} \frac{e^{2z}}{9\theta^2 - 6\theta + 7} - \frac{2}{3} \frac{e^z}{9\theta^2 - 6\theta + 7} + \frac{e^{0z}}{9\theta^2 - 6\theta + 7}.$$

Replace θ by 2, 1 and 0 respectively:

$$= \frac{1}{3} \frac{e^{2z}}{36 - 12 + 7} - \frac{2}{3} \frac{e^z}{9 - 6 + 7} + \frac{1}{0 - 0 + 7} = \frac{1}{3} \cdot \frac{e^{2z}}{31} - \frac{2}{3} \cdot \frac{e^z}{10} + \frac{1}{7}.$$

Since $3x+2 = e^z$, $\log(3x+2) = z$:

$$\text{P.I.} = \frac{1}{3} \cdot \frac{(3x+2)^2}{31} - \frac{2}{3} \cdot \frac{(3x+2)}{10} + \frac{1}{7}.$$

General solution:

$$y = (3x+2)^{1/3} \left[c_1 \cos \left(\frac{\sqrt{6}}{3} \log(3x+2) \right) + c_2 \sin \left(\frac{\sqrt{6}}{3} \log(3x+2) \right) \right] + \frac{(3x+2)^2}{93} - \frac{3x+2}{15} + \frac{1}{7}.$$

9 Simultaneous Linear Differential Equations

Here x and y are functions of t and $D = \frac{d}{dt}$. We eliminate one of the dependent variables by operating suitably on the equations, solve the resulting equation (C.F. + P.I.), and then find the other variable from one of the given equations.

Problem 9.1. Solve the simultaneous equations (i) $\frac{dx}{dt} + \frac{dy}{dt} + 3x = \sin t$; (ii) $\frac{dx}{dt} + y - x = \cos t$.

Solution. With $D = \frac{d}{dt}$:

$$\begin{aligned} \text{(i)} \quad Dx + Dy + 3x &= \sin t \\ \Rightarrow (D + 3)x + Dy &= \sin t; \\ \text{(ii)} \quad Dx + y - x &= \cos t \\ \Rightarrow (D - 1)x + y &= \cos t. \end{aligned}$$

Operate on (ii) by D (using $D \cos t = \frac{d}{dt} \cos t = -\sin t$) and subtract from (i):

$$\begin{array}{rcl} (D + 3)x + Dy & = & \sin t \\ D(D - 1)x + Dy & = & D \cos t = -\sin t \\ \hline (D + 3)x - D(D - 1)x & = & \sin t + \sin t \end{array}$$

$$(D + 3)x - (D^2 - D)x = 2 \sin t \Rightarrow Dx + 3x - D^2x + Dx = 2 \sin t \Rightarrow (-D^2 + 2D + 3)x = 2 \sin t. \quad (1)$$

C.F. A.E.: $-m^2 + 2m + 3 = 0 \Rightarrow m^2 - 2m - 3 = 0 \Rightarrow (m - 3)(m + 1) = 0 \Rightarrow m = 3, -1$.
C.F. $= c_1 e^{3t} + c_2 e^{-t}$.

P.I. (Type 2; replace D^2 by $-a^2 = -1$)

$$\begin{aligned} \text{P.I.} &= \frac{1}{-D^2 + 2D + 3} 2 \sin t \\ &= \frac{2 \sin t}{-(-1) + 2D + 3} \\ &= \frac{2 \sin t}{2D + 4} \\ &= \frac{\sin t}{D + 2} \\ &= \frac{(D - 2) \sin t}{(D + 2)(D - 2)} \\ &= \frac{(D - 2) \sin t}{D^2 - 4} \\ &\xrightarrow{D^2 \rightarrow -1} \frac{(D - 2) \sin t}{-5} \\ &= -\frac{1}{5} (D \sin t - 2 \sin t) \\ &= -\frac{1}{5} (\cos t - 2 \sin t) \\ &= \frac{1}{5} (2 \sin t - \cos t). \end{aligned}$$

Solution of (1):

$$x = c_1 e^{3t} + c_2 e^{-t} + \frac{1}{5} (2 \sin t - \cos t).$$

To get y , from (ii): $y = \cos t - (D - 1)x$.

$$\begin{aligned}
 y &= \cos t - (D - 1) \left(c_1 e^{3t} + c_2 e^{-t} + \frac{1}{5}(2 \sin t - \cos t) \right) \\
 &= \cos t - \left[3c_1 e^{3t} - c_2 e^{-t} + \frac{1}{5}(2 \cos t + \sin t) - \left(c_1 e^{3t} + c_2 e^{-t} + \frac{1}{5}(2 \sin t - \cos t) \right) \right] \\
 &= \cos t - \left[2c_1 e^{3t} - 2c_2 e^{-t} + \frac{1}{5}(2 \cos t + \sin t - 2 \sin t + \cos t) \right] \\
 &= \cos t - \left[2c_1 e^{3t} - 2c_2 e^{-t} + \frac{1}{5}(3 \cos t - \sin t) \right] \\
 &= \cos t - \frac{3}{5} \cos t + \frac{1}{5} \sin t - 2c_1 e^{3t} + 2c_2 e^{-t} \\
 &= \frac{2}{5} \cos t + \frac{1}{5} \sin t - 2c_1 e^{3t} + 2c_2 e^{-t}.
 \end{aligned}$$

$$y = -2c_1 e^{3t} + 2c_2 e^{-t} + \frac{1}{5}(2 \cos t + \sin t).^9$$

Problem 9.2. Solve (i) $\frac{dx}{dt} + 4x + 3y = t$; (ii) $\frac{dy}{dt} + 2x + 5y = e^t$.

Solution. $(D + 4)x + 3y = t$ (i); $2x + (D + 5)y = e^t$ (ii).

Multiply (i) by 2 and operate on (ii) by $(D + 4)$, then subtract:

$$\begin{array}{rcl}
 2(D + 4)x + 6y & = & 2t \\
 2(D + 4)x + (D + 4)(D + 5)y & = & (D + 4)e^t \\
 \hline
 6y - (D^2 + 9D + 20)y & = & 2t - (e^t + 4e^t)
 \end{array}$$

$$[6 - (D^2 + 9D + 20)]y = 2t - 5e^t \Rightarrow (-D^2 - 9D - 14)y = 2t - 5e^t \Rightarrow (D^2 + 9D + 14)y = 5e^t - 2t. \quad (3)$$

$$\text{C.F. A.E.: } m^2 + 9m + 14 = 0 \Rightarrow (m + 7)(m + 2) = 0 \Rightarrow m = -7, -2. \quad \text{C.F.} = c_1 e^{-2t} + c_2 e^{-7t}.$$

P.I.

$$\text{P.I.} = 5 \underbrace{\frac{1}{D^2 + 9D + 14}}_{\text{Type 1: (I)}} e^t - 2 \underbrace{\frac{1}{D^2 + 9D + 14}}_{\text{Type 3: (II)}} t.$$

$$\text{(I) Replace } D \text{ by } 1: \text{ P.I.}_1 = \frac{5e^t}{1 + 9 + 14} = \frac{5e^t}{24}.$$

(II)

$$\begin{aligned}
 \text{P.I.}_2 &= 2 \frac{1}{14 \left(\frac{D^2 + 9D}{14} + 1 \right)} t \\
 &= \frac{2}{14} \left(1 + \frac{D^2 + 9D}{14} \right)^{-1} t \\
 &= \frac{2}{14} \left(1 - \left(\frac{D^2 + 9D}{14} \right) + \left(\frac{D^2 + 9D}{14} \right)^2 \right) t \\
 &= \frac{2}{14} \left(1 - \frac{D^2}{14} + \frac{81D^2}{196} - \frac{9D}{14} \right) t
 \end{aligned}$$

⁹The notes end with $\frac{1}{5}(2 \cos t - \sin t)$; since $\cos t - \frac{3}{5} \cos t + \frac{1}{5} \sin t = \frac{2}{5} \cos t + \frac{1}{5} \sin t$, the sign of $\sin t$ is corrected here (checked by substitution in both equations).

$$\begin{aligned}
&= \frac{2}{14} \left(1 + \frac{67D^2}{196} - \frac{9D}{14} \right) t \\
&= \frac{2}{14} \left(t + 0 - \frac{9}{14} \right) \\
&= \frac{1}{7} \left(t - \frac{9}{14} \right) \\
&= \frac{14t - 9}{98}.
\end{aligned}$$

$$\text{P.I.} = \frac{5e^t}{24} - \frac{14t - 9}{98} = \frac{5e^t}{24} - \left(\frac{t}{7} - \frac{9}{98} \right).$$

$$y = c_1 e^{-2t} + c_2 e^{-7t} + \frac{5e^t}{24} - \frac{t}{7} + \frac{9}{98}.$$

Solving for x . Use y in (ii): $2x + (D + 5)y = e^t$.

$$2x + \left(-2c_1 e^{-2t} - 7c_2 e^{-7t} + \frac{5}{24} e^t - \frac{1}{7} \right) + \left(5c_1 e^{-2t} + 5c_2 e^{-7t} + \frac{25}{24} e^t - \frac{5t}{7} + \frac{45}{98} \right) = e^t,$$

$$2x + 3c_1 e^{-2t} - 2c_2 e^{-7t} + \frac{30}{24} e^t - \frac{5t}{7} + \frac{31}{98} = e^t \quad \left(-\frac{1}{7} + \frac{45}{98} = \frac{31}{98} \right),$$

$$2x + 3c_1 e^{-2t} - 2c_2 e^{-7t} + \frac{e^t}{4} - \frac{5t}{7} + \frac{31}{98} = 0 \quad \left(\frac{30}{24} e^t - e^t = -\frac{e^t}{4} \right).$$

$$x = -\frac{1}{2} \left(3c_1 e^{-2t} - 2c_2 e^{-7t} + \frac{e^t}{4} - \frac{5t}{7} + \frac{31}{98} \right) = -\frac{3}{2} c_1 e^{-2t} + c_2 e^{-7t} - \frac{e^t}{8} + \frac{5t}{14} - \frac{31}{196},$$

$$x = -\frac{3}{2} c_1 e^{-2t} + c_2 e^{-7t} - \frac{e^t}{8} + \frac{1}{196} (70t - 31).$$

Problem 9.3. The equations of motion of a particle are given by $\frac{dx}{dt} + \omega y = 0$, $\frac{dy}{dt} - \omega x = 0$. Find the path of the particle and show that it is a circle.

Solution. $Dx + \omega y = 0$ (i); $Dy - \omega x = 0$ (ii). Multiply (i) by ω and operate on (ii) by D , then add:

$$\begin{array}{rcl}
D\omega x + \omega^2 y & = & 0 \\
-D\omega x + D^2 y & = & 0 \\
\hline
(\omega^2 + D^2)y & = & 0
\end{array}$$

C.F. A.E.: $m^2 + \omega^2 = 0 \Rightarrow m = \pm i\omega$. R.H.S. = 0 $\Rightarrow$ P.I. = 0, so

$$y = c_1 \cos \omega t + c_2 \sin \omega t.$$

From (ii), $Dy = \omega x$:

$$D(c_1 \cos \omega t + c_2 \sin \omega t) = \omega x \Rightarrow -c_1 \omega \sin \omega t + c_2 \omega \cos \omega t = \omega x \Rightarrow x = c_2 \cos \omega t - c_1 \sin \omega t.$$

Squaring and adding:¹⁰

$$x^2 + y^2 = (c_2 \cos \omega t - c_1 \sin \omega t)^2 + (c_1 \cos \omega t + c_2 \sin \omega t)^2 = c_1^2 + c_2^2,$$

which is a circle with centre at the origin and radius $\sqrt{c_1^2 + c_2^2}$.

¹⁰The notes stop after finding x and y ; the elimination of t is added to show the path is a circle.

10 Method of Variation of Parameters

Under this method we can find the solution of the second-order ODE

$$y'' + Py' + Qy = f(x).$$

The complementary function of the above equation will be of the form $au + bv$, where a, b are arbitrary constants and u, v are functions of x . By the procedure of the method of variation of parameters, we can write the general solution of the above equation as

$$y = \phi u + \psi v, \quad \phi = \int \frac{-v f(x)}{uv' - u'v} dx, \quad \psi = \int \frac{u f(x)}{uv' - u'v} dx.$$

Problem 10.1. Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \sin(e^x)$ by the method of variation of parameters.

Solution. $(D^2 + 3D + 2)y = \sin(e^x)$. **C.F.** A.E.: $m^2 + 3m + 2 = 0 \Rightarrow m = -1, -2$. C.F. = $ae^{-x} + be^{-2x} = au + bv$.

$$\begin{aligned} u &= e^{-x}, \quad v = e^{-2x}; \\ u' &= -e^{-x}, \quad v' = -2e^{-2x}; \\ uv' - u'v &= e^{-x}(-2e^{-2x}) - (-e^{-x})e^{-2x} = -2e^{-3x} + e^{-3x} = -e^{-3x}. \end{aligned}$$

$$\phi = \int \frac{-v f(x)}{uv' - u'v} dx = \int \frac{-e^{-2x} \sin(e^x)}{-e^{-3x}} dx = \int e^x \sin(e^x) dx.$$

With $e^x = t$, $e^x dx = dt$: $\phi = \int \sin t dt = -\cos t + c_1 = -\cos(e^x) + c_1$.

$$\begin{aligned} \psi &= \int \frac{u f(x)}{uv' - u'v} dx \\ &= \int \frac{e^{-x} \sin(e^x)}{-e^{-3x}} dx \\ &= - \int e^{2x} \sin(e^x) dx = - \int t \sin t dt \quad (e^x = t, e^x dx = dt). \end{aligned}$$

By parts:

$$\begin{aligned} \psi &= - \left[t \int \sin t dt - \int \left(\frac{dt}{dt} \int \sin t dt \right) dt \right] \\ &= - \left[t(-\cos t) - \int (-\cos t) dt \right] \\ &= - [-t \cos t + \sin t] + c_2, \end{aligned}$$

$$\psi = t \cos t - \sin t + c_2 = e^x \cos(e^x) - \sin(e^x) + c_2.$$

General solution:

$$\begin{aligned} y &= \phi u + \psi v = (-\cos(e^x) + c_1) e^{-x} + (e^x \cos(e^x) - \sin(e^x) + c_2) e^{-2x} \\ &= -\cos(e^x) e^{-x} + c_1 e^{-x} + e^{-x} \cos(e^x) - e^{-2x} \sin(e^x) + c_2 e^{-2x}, \end{aligned}$$

$$\boxed{y = c_1 e^{-x} + c_2 e^{-2x} - e^{-2x} \sin(e^x)}.$$

Problem 10.2. Solve $(D^2 + 2D + 1)y = 4e^{-x} \log x$.

Solution. **C.F.** A.E.: $m^2 + 2m + 1 = 0 \Rightarrow m = -1, -1$. **C.F.** $= ae^{-x} + be^{-x}$.

$$u = e^{-x}, v = xe^{-x}; \quad u' = -e^{-x}, v' = e^{-x} - xe^{-x};$$

$$uv' - u'v = e^{-x}(e^{-x} - xe^{-x}) - (-e^{-x})(xe^{-x}) = e^{-2x} - xe^{-2x} + xe^{-2x} = e^{-2x}.$$

$$\begin{aligned} \phi &= \int \frac{-v f(x)}{uv' - u'v} dx = \int \frac{-xe^{-x} \cdot 4e^{-x} \log x}{e^{-2x}} dx = -4 \int x \log x dx \\ &= -4 \left[\log x \int x dx - \int \left(\frac{d}{dx}(\log x) \int x dx \right) dx \right] = -4 \left[\log x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx \right] \\ &= -4 \left[\frac{x^2 \log x}{2} - \frac{1}{2} \cdot \frac{x^2}{2} \right] \Rightarrow \phi = -2x^2 \log x + x^2 + c_1. \end{aligned}$$

$$\begin{aligned} \psi &= \int \frac{u f(x)}{uv' - u'v} dx = \int \frac{e^{-x} \cdot 4e^{-x} \log x}{e^{-2x}} dx = 4 \int \log x dx = 4 \int \log x \cdot 1 dx \\ &= 4 \left[\log x \int dx - \int \left(\frac{d}{dx}(\log x) \int dx \right) dx \right] = 4 \left[x \log x - \int \frac{1}{x} \cdot x dx \right] = 4[x \log x - x] + c_2 \\ &\Rightarrow \psi = 4x(\log x - 1) + c_2. \end{aligned}$$

General solution:

$$\begin{aligned} y &= \phi u + \psi v = (-2x^2 \log x + x^2 + c_1) e^{-x} + (4x \log x - 4x + c_2) x e^{-x} \\ &= -2x^2 e^{-x} \log x + x^2 e^{-x} + c_1 e^{-x} + 4x^2 \log x e^{-x} - 4x^2 e^{-x} + c_2 x e^{-x}, \end{aligned}$$

$$\boxed{y = c_1 e^{-x} + c_2 x e^{-x} + 2x^2 e^{-x} \log x - 3x^2 e^{-x}}.^{11}$$

Problem 10.3. Solve $(D^2 + 1)y = \sec x$.

Solution. **C.F.** A.E.: $m^2 + 1 = 0 \Rightarrow m = \pm i$. **C.F.** $= a \cos x + b \sin x$.

$$u = \cos x, v = \sin x; \quad u' = -\sin x, v' = \cos x;$$

$$uv' - u'v = \cos x \cos x - (-\sin x \sin x) = \cos^2 x + \sin^2 x = 1.$$

$$\begin{aligned} \phi &= \int \frac{-v f(x)}{uv' - u'v} dx = - \int \frac{\sin x \sec x}{1} dx = - \int \sin x \cdot \frac{1}{\cos x} dx = - \int \tan x dx = -\log(\sec x) + c_1, \\ \psi &= \int \frac{u f(x)}{uv' - u'v} dx = \int \frac{\cos x \sec x}{1} dx = \int \cos x \cdot \frac{1}{\cos x} dx = \int dx = x + c_2. \end{aligned}$$

General solution:

$$\begin{aligned} y &= \phi u + \psi v \\ &= (-\log(\sec x) + c_1) \cos x + (x + c_2) \sin x \\ &= -\log(\sec x) \cos x + c_1 \cos x + x \sin x + c_2 \sin x, \end{aligned}$$

$$\boxed{y = c_1 \cos x + c_2 \sin x + x \sin x + \cos x \log(\cos x)}.$$

¹¹The notes write $+3x^2 e^{-x}$; since $x^2 e^{-x} - 4x^2 e^{-x} = -3x^2 e^{-x}$, the sign is corrected here (checked by substitution).

Problem 10.4. Solve $(D^2 - 1)y = (1 + e^{-x})^{-2}$.

Solution. **C.F.** A.E.: $m^2 - 1 = 0 \Rightarrow m = \pm 1$. **C.F.** $= c_1 e^x + c_2 e^{-x}$.

$$u = e^x, \quad v = e^{-x}; \quad u' = e^x, \quad v' = -e^{-x}; \quad uv' - u'v = e^x(-e^{-x}) - (e^x)(e^{-x}) = -1 - 1 = -2.$$

$$f(x) = \left(1 + \frac{1}{e^x}\right)^{-2} = \left(\frac{e^x + 1}{e^x}\right)^{-2} = \left(\frac{e^x}{e^x + 1}\right)^2 = \frac{e^{2x}}{(e^x + 1)^2}.$$

With $e^x + 1 = t$, $e^x dx = dt$:

$$\phi = \int \frac{-v f(x)}{uv' - u'v} dx = \int \frac{-e^{-x} \frac{e^{2x}}{(e^x + 1)^2}}{-2} dx = \frac{1}{2} \int \frac{e^x}{(e^x + 1)^2} dx = \frac{1}{2} \int \frac{dt}{t^2} = \frac{1}{2} \cdot \frac{t^{-1}}{-1},$$

$$\phi = -\frac{1}{2(e^x + 1)} + c_1.$$

With $e^x + 1 = t$, $e^x dx = dt$, $e^{2x} = (t - 1)^2$:

$$\begin{aligned} \psi &= \int \frac{u f(x)}{uv' - u'v} dx = \int \frac{e^x \frac{e^{2x}}{(e^x + 1)^2}}{-2} dx = -\frac{1}{2} \int \frac{e^{2x} \cdot e^x}{(e^x + 1)^2} dx = -\frac{1}{2} \int \frac{(t - 1)^2}{t^2} dt \\ &= -\frac{1}{2} \int \frac{t^2 + 1 - 2t}{t^2} dt = -\frac{1}{2} \left[\int 1 dt + \int \frac{1}{t^2} dt - \int \frac{2}{t} dt \right] = -\frac{1}{2} \left[t - \frac{1}{t} - 2 \log t \right], \end{aligned}$$

$$\psi = -\frac{1}{2}(e^x + 1) + \frac{1}{2(e^x + 1)} + \log(e^x + 1) + c_2.$$

General solution:

$$\begin{aligned} y &= \phi u + \psi v = \left(-\frac{1}{2(e^x + 1)} + c_1\right) e^x + \left(-\frac{1}{2}(e^x + 1) + \frac{1}{2(e^x + 1)} + \log(e^x + 1) + c_2\right) e^{-x} \\ &= -\frac{e^x}{2(e^x + 1)} + c_1 e^x - \frac{1}{2}(e^x + 1)e^{-x} + \frac{e^{-x}}{2(e^x + 1)} + e^{-x} \log(e^x + 1) + c_2 e^{-x}, \end{aligned}$$

$$y = c_1 e^x + c_2 e^{-x} + \frac{e^{-x} - e^x}{2(e^x + 1)} + e^{-x} \left(\log(e^x + 1) - \frac{e^x + 1}{2} \right).^{12}$$

Problem 10.5. Solve $\frac{d^2 y}{dx^2} - 3\frac{dy}{dx} + 2y = \frac{e^x}{1 + e^x}$.

Solution. $(D^2 - 3D + 2)y = \frac{e^x}{1 + e^x}$. **C.F.** A.E.: $m^2 - 3m + 2 = 0 \Rightarrow m = 2, 1$. **C.F.** $= c_1 e^x + c_2 e^{2x}$.

$$u = e^x, \quad v = e^{2x}; \quad u' = e^x, \quad v' = 2e^{2x}; \quad uv' - u'v = e^x \cdot 2e^{2x} - e^x \cdot e^{2x} = 2e^{3x} - e^{3x} = e^{3x}.$$

$$\phi = \int \frac{-v f(x)}{uv' - u'v} dx = - \int \frac{e^{2x} \cdot \frac{e^x}{1 + e^x}}{e^{3x}} dx = - \int \frac{1}{1 + e^x} dx = - \int \frac{e^x}{1 + e^x} \cdot \frac{1}{e^x} dx.$$

With $1 + e^x = t$ ($e^x = t - 1$, $e^x dx = dt$):

$$\phi = - \int \frac{dt}{t(t - 1)}.$$

¹²In the notes the sign of ϕ is lost ($-vf/(uv' - u'v) = +\frac{1}{2}e^{-x}f$), giving $\frac{e^x + e^{-x}}{2(e^x + 1)}$; the corrected form above has been checked by substitution.

Partial fractions: $\frac{1}{t(t-1)} = \frac{A}{t-1} + \frac{B}{t}$, $At + B(t-1) = 1$; $t = 0 \Rightarrow B = -1$; $t = 1 \Rightarrow A = 1$.

$$\phi = - \int \left(\frac{1}{t-1} - \frac{1}{t} \right) dt = -(\log(t-1) - \log t) = \log t - \log(t-1) = \log(1+e^x) - \log(e^x),$$

$$\phi = \log \left(\frac{1+e^x}{e^x} \right) + c_1.$$

$$\psi = \int \frac{u f(x)}{uv' - u'v} dx = \int \frac{e^x \cdot \frac{e^x}{1+e^x}}{e^{3x}} dx = \int \frac{e^{-x}}{1+e^x} dx = \int \frac{dt}{(t-1)^2 t} \quad (1+e^x = t, e^x dx = dt).$$

Partial fractions: $\frac{1}{t(t-1)^2} = \frac{A}{t} + \frac{B}{t-1} + \frac{C}{(t-1)^2}$, i.e. $1 = A(t-1)^2 + Bt(t-1) + Ct$.

$$t = 1 : 1 = C; \quad t = 0 : 1 = A; \quad t = 2 : 1 = A(1) + 2B + 2C = 1 + 2B + 2 \Rightarrow B = -1.$$

$$\begin{aligned} \psi &= \int \left(\frac{1}{t} - \frac{1}{t-1} + \frac{1}{(t-1)^2} \right) dt = \log t - \log(t-1) + \frac{(t-1)^{-1}}{-1} + c_2 \\ &= \log(1+e^x) - \log(e^x) - \frac{1}{e^x} + c_2 = \log \left(\frac{1+e^x}{e^x} \right) - \frac{1}{e^x} + c_2. \end{aligned}$$

General solution:

$$\begin{aligned} y &= \phi u + \psi v = \left(\log \left(\frac{1+e^x}{e^x} \right) + c_1 \right) e^x + \left(\log \left(\frac{1+e^x}{e^x} \right) - \frac{1}{e^x} + c_2 \right) e^{2x} \\ &= \log \left(\frac{1+e^x}{e^x} \right) e^x + c_1 e^x + \log \left(\frac{1+e^x}{e^x} \right) e^{2x} - \frac{e^{2x}}{e^x} + c_2 e^{2x}, \end{aligned}$$

$$\boxed{y = c_1 e^x + c_2 e^{2x} - e^x + \log \left(\frac{1+e^x}{e^x} \right) (e^x + e^{2x})}.$$

11 Orthogonal Trajectories

- For a family of curves $f(x, y, \alpha) = 0$, where α is a parameter, we can find another family of curves $g(x, y, \beta) = 0$ (β a parameter) such that each and every member of these two families intersect orthogonally. They are called *orthogonal trajectories* of each other.
- If $m_1 = \frac{dy}{dx}$ is the slope of the family f and $m_2 = \frac{dy}{dx}$ is the slope of the family g , then $m_1 m_2 = -1$.
- Therefore, to identify the orthogonal trajectories g of the given family f , we use the relation

$$m_2 = -\frac{1}{m_1}, \quad \text{i.e.} \quad \left(\frac{dy}{dx} \right)_{(g)} = -\frac{1}{\left(\frac{dy}{dx} \right)_{(f)}}.$$

- Solving the above differential equation, we get the family of curves g .

Note. In the case of polar form $r = f(\theta)$, the slope is taken as $m_1 = r \frac{d\theta}{dr}$.

11.1 Problems

Problem 11.1 (Cartesian form). Find the orthogonal trajectories of the family of concentric circles $x^2 + y^2 = a^2$.

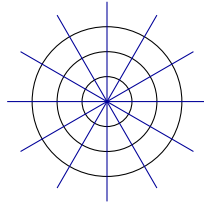
Solution. Differentiating $x^2 + y^2 = a^2$ (1) with respect to x :

$$2x + 2y \frac{dy}{dx} = 0 \Rightarrow 2y \frac{dy}{dx} = -2x \Rightarrow \frac{dy}{dx} = -\frac{x}{y}, \quad m_1 = -\frac{x}{y}.$$

Let m_2 be the slope of the orthogonal trajectories:

$$m_2 = -\frac{1}{m_1} = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y}{x} \Rightarrow \int \frac{dy}{y} = \int \frac{dx}{x} \Rightarrow \log y = \log x + \log m \Rightarrow \boxed{y = mx},$$

m being a parameter. The orthogonal trajectories form a family of straight lines passing through the origin (radial lines).



Problem 11.2. Find the orthogonal trajectories of the family of curves $xy = k$.

Solution. Differentiating with respect to x : $x \frac{dy}{dx} + y = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x} = m_1$.

$$m_2 = -\frac{1}{m_1} = \frac{x}{y} \Rightarrow \frac{dy}{dx} = \frac{x}{y} \Rightarrow \int y \, dy = \int x \, dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + c \Rightarrow y^2 = x^2 + 2c,$$

$$\boxed{x^2 - y^2 = -2c} \quad (\text{or } x^2 - y^2 = a^2, \quad a^2 = -2c),$$

a family of rectangular hyperbolas.

Problem 11.3 (Polar form). Find the orthogonal trajectories of the family of curves $r = a \sin \theta$.

Solution. Differentiating with respect to θ : $\frac{dr}{d\theta} = a \cos \theta \Rightarrow \frac{d\theta}{dr} = \frac{1}{a \cos \theta}$.

$$m_1 = r \frac{d\theta}{dr} = a \sin \theta \cdot \frac{1}{a \cos \theta} = \tan \theta; \quad m_2 = -\frac{1}{m_1} = -\frac{1}{\tan \theta} = -\cot \theta.$$

$$r \frac{d\theta}{dr} = -\cot \theta \Rightarrow -\frac{d\theta}{\cot \theta} = \frac{dr}{r} \Rightarrow -\int \tan \theta \, d\theta = \int \frac{dr}{r} \Rightarrow -\log |\sec \theta| = \log r + \log c^{-1}.$$

$$\log c + \log(\cos \theta) = \log r \Rightarrow \boxed{r = c \cos \theta}.$$

Problem 11.4. Find the orthogonal trajectories of the family of curves $r = a \cos \theta$.

Solution. $\frac{dr}{d\theta} = -a \sin \theta \Rightarrow \frac{d\theta}{dr} = -\frac{1}{a \sin \theta}$.

$$m_1 = r \frac{d\theta}{dr} = a \cos \theta \cdot \left(-\frac{1}{a \sin \theta} \right) = -\cot \theta; \quad m_2 = -\frac{1}{m_1} = \frac{1}{\cot \theta} = \tan \theta.$$

$$r \frac{d\theta}{dr} = \tan \theta \Rightarrow \int \cot \theta d\theta = \int \frac{dr}{r} \Rightarrow \log(\sin \theta) + \log c = \log r \Rightarrow \boxed{r = c \sin \theta}.$$

11.2 Self-orthogonal families

Note. • A family of curves is said to be *self-orthogonal* if the family of orthogonal trajectories is the family itself.

- To identify such a family, construct the differential equation of the given family by eliminating the parameter. If, on replacing $\frac{dy}{dx}$ by $-\frac{1}{dy/dx}$ (i.e. p by $-\frac{1}{p}$), we get the same differential equation, then both the given family and its orthogonal trajectories have the same differential equation, and hence the given family is self-orthogonal.

Problem 11.5. Prove that the family of curves $\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$ is self-orthogonal, λ being the parameter.

Solution.

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1. \quad (1)$$

Differentiating with respect to x :

$$\frac{2x}{a^2 + \lambda} + \frac{2y}{b^2 + \lambda} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{2x}{a^2 + \lambda} \cdot \frac{b^2 + \lambda}{2y} \Rightarrow p = -\frac{b^2 + \lambda}{a^2 + \lambda} \cdot \frac{x}{y}.$$

To eliminate λ :

$$\frac{yp}{b^2 + \lambda} = -\frac{x}{a^2 + \lambda} \Rightarrow yp(a^2 + \lambda) = -x(b^2 + \lambda) \Rightarrow ypa^2 + yp\lambda + xb^2 + x\lambda = 0$$

$$\Rightarrow \lambda(yp + x) + xb^2 + ypa^2 = 0 \Rightarrow \lambda = \frac{-xb^2 - ypa^2}{yp + x}.$$

Now

$$a^2 + \lambda = a^2 - \frac{xb^2 + ypa^2}{yp + x} = \frac{ypa^2 + xa^2 - xb^2 - ypa^2}{yp + x} = \frac{x(a^2 - b^2)}{yp + x}, \quad (2)$$

$$b^2 + \lambda = b^2 - \frac{xb^2 + ypa^2}{yp + x} = \frac{ypb^2 + xb^2 - xb^2 - ypa^2}{yp + x} = \frac{yp(b^2 - a^2)}{yp + x}. \quad (3)$$

Substituting (2) and (3) in (1):

$$\begin{aligned} x^2 \left(\frac{yp + x}{x(a^2 - b^2)} \right) + y^2 \left(\frac{yp + x}{yp(b^2 - a^2)} \right) &= 1 \Rightarrow x \left(\frac{yp + x}{a^2 - b^2} \right) + y \left(\frac{-(x + yp)}{p(a^2 - b^2)} \right) = 1 \\ \Rightarrow x(x + yp) - \frac{y}{p}(x + yp) &= a^2 - b^2 \Rightarrow (x + yp) \left(x - \frac{y}{p} \right) = a^2 - b^2, \end{aligned} \quad (4)$$

which is the differential equation of the given family (1).

To get the differential equation of the orthogonal trajectories of (1), replace p by $-\frac{1}{p}$ in (4):

$$\left(x - \frac{y}{p}\right)(x - y(-p)) = a^2 - b^2 \Rightarrow \left(x - \frac{y}{p}\right)(x + yp) = a^2 - b^2. \quad (5)$$

Since (4) = (5), the family is self-orthogonal.

Problem 11.6. Prove that the system of parabolas $y^2 = 4a(x + a)$ is self-orthogonal, a being the parameter.

Solution. Differentiating $y^2 = 4a(x + a)$ with respect to x :

$$2y \frac{dy}{dx} = 4a(1 + 0) \Rightarrow 2yp = 4a \Rightarrow a = \frac{yp}{2}.$$

Substituting $a = \frac{yp}{2}$:

$$y^2 = 4 \cdot \frac{yp}{2} \left(x + \frac{yp}{2}\right) = 2yp \left(\frac{2x + yp}{2}\right) = yp(2x + yp) \Rightarrow y^2 = y^2 p^2 + 2xyp \Rightarrow y = yp^2 + 2xp. \quad (1)$$

Replace p by $-\frac{1}{p}$ in $y^2 = yp(2x + yp)$:

$$\begin{aligned} y^2 &= -\frac{y}{p} \left(2x - \frac{y}{p}\right) = -\frac{y}{p} \left(\frac{2xp - y}{p}\right) = -\frac{1}{p^2} (2xpy - y^2) \\ &\Rightarrow -y^2 p^2 = 2xpy - y^2 \\ &\Rightarrow y^2 = 2xyp + y^2 p^2, \\ &\Rightarrow y = 2xp + yp^2. \end{aligned} \quad (2)$$

Since (1) = (2), the family is self-orthogonal.

12 Applications of Differential Equations

12.1 Newton's law of cooling

- The rate of change of temperature of a body is directly proportional to the difference between the temperature of the body and the surrounding temperature.
- That is, if T is the temperature of a body at any time t and T_s is the surrounding temperature, then

$$\frac{dT}{dt} = k(T_s - T) \quad \text{or} \quad \frac{dT}{dt} = -k(T - T_s),$$

where k is the proportionality constant.

Solving the above differential equation:

$$\begin{aligned} \int \frac{dT}{T - T_s} &= - \int k dt \Rightarrow \log |T - T_s| = -kt + c \Rightarrow T - T_s = e^{-kt} e^c \Rightarrow T - T_s = Ae^{-kt}, \\ T &= T_s + Ae^{-kt}. \end{aligned} \quad (1)$$

Suppose T_0 is the initial temperature of the body at $t = 0$: $T_0 = T_s + A \Rightarrow A = T_0 - T_s$. Then (1) becomes

$$T = T_s + (T_0 - T_s)e^{-kt}.$$

Problem 12.1. A body at temperature 40°C is kept in a surrounding of constant temperature 20°C . It is observed that the body's temperature falls to 35°C in 10 minutes. Find how much more time it will take for the body to attain a temperature of 30°C .

Solution. By Newton's law of cooling, $T = T_s + (T_0 - T_s)e^{-kt}$, with $T_0 = 40^\circ\text{C}$, $T_s = 20^\circ\text{C}$; $T = 35^\circ\text{C}$ at $t = 10$ min:

$$35 = 20 + (40 - 20)e^{-k \cdot 10} \Rightarrow 15 = 20e^{-10k} \Rightarrow 0.75 = e^{-10k}.$$

Taking natural logarithms: $\ln 0.75 = -10k \Rightarrow -0.28768 = -10k \Rightarrow k = 0.028768$.

Now for $T = 30^\circ\text{C}$:

$$30 = 20 + (40 - 20)e^{-0.028768t} \Rightarrow 10 = 20e^{-0.028768t} \Rightarrow 0.5 = e^{-0.028768t}.$$

$$\ln 0.5 = -0.028768t \Rightarrow -0.693147 = -0.028768t \Rightarrow t = 24.1 \text{ min.}$$

So the body attains 30°C at $t \approx 24.1$ min, i.e. about $24.1 - 10 = 14.1$ minutes more after it reaches 35°C .

12.2 Law of natural growth and decay

The rate of growth of a substance x at time t is directly proportional to the quantity present at that time:

$$\frac{dx}{dt} = kx \quad (k : \text{proportionality constant}).$$

Solving:

$$\int \frac{dx}{x} = k \int dt \Rightarrow \log x = kt + c \Rightarrow x = e^{kt}e^c \Rightarrow x(t) = Ae^{kt}. \quad (1)$$

If x_0 is the amount of substance at $t = 0$, (1) gives $x_0 = A$, so

$$x(t) = x_0e^{kt} \quad (\text{growth}), \quad x(t) = x_0e^{-kt} \quad (\text{decay}),$$

since, similarly, the rate of decay of a substance at any time t is directly proportional to the negative of its quantity: $\frac{dx}{dt} = -kx$. Here x_0 is the initial amount of substance.

Problem 12.2. A bacterial population B is known to have a rate of growth proportional to B itself. If between noon and 2 p.m. the population triples, at what time, no controls being exerted, should B become 100 times what it was at noon?

Solution. Law of natural growth: $B = B_0e^{kt}$, where B_0 is the population at $t = 0$ (noon). At $t = 2$ (2 p.m.), $B = 3B_0$:

$$3B_0 = B_0e^{2k} \Rightarrow 3 = e^{2k} \Rightarrow \ln 3 = 2k \Rightarrow k = 0.5493.$$

For $B = 100B_0$:

$$100B_0 = B_0e^{0.5493t} \Rightarrow 100 = e^{0.5493t} \Rightarrow \ln 100 = 0.5493t \Rightarrow t = \frac{4.6052}{0.5493} = 8.38 \text{ hours.}$$

So B becomes 100 times its noon value about 8.38 hours after noon, i.e. at about 8:23 p.m.¹³

¹³The notes write the answer as 8:38 p.m.; 8.38 hours is 8 hours 23 minutes.

Problem 12.3. Radium decomposes at a rate proportional to the quantity of radium present. Suppose that it is found that in 25 years approximately 1.1% of a certain quantity of radium has decomposed. Determine approximately how long it will take for one-half of the original amount of radium to decompose.

Solution. Law of natural decay: $x = x_0 e^{-kt}$. At $t = 25$ years, $x = x_0 - 1.1\%$ of $x_0 = x_0 - \frac{1.1}{100}x_0 = x_0 - 0.011x_0 = 0.989x_0$:

$$0.989x_0 = x_0 e^{-25k} \Rightarrow 0.989 = e^{-25k} \Rightarrow \ln 0.989 = -25k \Rightarrow k = 4.424 \times 10^{-4}.$$

For half the original amount to decompose, $x = x_0 - 50\%$ of $x_0 = \frac{x_0}{2}$:

$$\frac{x_0}{2} = x_0 e^{-4.424 \times 10^{-4}t} \Rightarrow 0.5 = e^{-4.424 \times 10^{-4}t} \Rightarrow \log_e 0.5 = -4.424 \times 10^{-4}t \Rightarrow t \approx 1566.7 \text{ years.}$$

Homework

1. Oil is heated to 70°C . It cools to 50°C after 6 min. Calculate the time taken to cool from 50°C to 40°C , given that the surrounding temperature is 25°C . (Ans: about 5 min)
2. If the half-life of uranium is 1500 years, (i) find the percentage of the original amount that will remain after 4500 years; (ii) find in how many years only $\frac{1}{10}$ of the original amount will remain. (Ans: 12.5%, i.e. $\frac{1}{8}$; about 4985 years)
3. In a certain bacteria culture the rate of increase in the number of bacteria is proportional to the number present. (i) If the number triples in 5 hours, how many will be present in 10 hours? (ii) When will the number present be 10 times the number initially present? (Ans: 9 times; 10.48 hours)

Problem 12.4. Oil is heated to 70°C and cools to 50°C after 6 min. Calculate the time taken to cool from 50°C to 40°C , the surrounding temperature being 25°C .

Solution. (i) $T_0 = 70^\circ\text{C}$, $T = 50^\circ\text{C}$ at $t = 6$ min, $T_s = 25^\circ\text{C}$. From $T = T_s + (T_0 - T_s)e^{-kt}$:

$$50 = 25 + (70 - 25)e^{-6k} \Rightarrow 25 = 45e^{-6k} \Rightarrow 0.5556 = e^{-6k} \Rightarrow -0.5878 = -6k \Rightarrow k = 0.09796.$$

(ii) Now take $T_0 = 50^\circ\text{C}$, $T = 40^\circ\text{C}$, $t = ?$:

$$40 = 25 + (50 - 25)e^{-0.09796t} \Rightarrow 15 = 25e^{-0.09796t} \Rightarrow 0.6 = e^{-0.09796t}.$$

Taking $\ln$: $-0.5108 = -0.09796t \Rightarrow t = 5.2$ min.

Problem 12.5. The half-life of uranium is 1500 years. (i) Find the percentage of the original amount that remains after 4500 years. (ii) In how many years will only $\frac{1}{10}$ of the original amount remain?

Solution. At $t = 1500$ years, $x = \frac{x_0}{2}$. From $x = x_0 e^{-kt}$:

$$\frac{x_0}{2} = x_0 e^{-1500k} \Rightarrow 0.5 = e^{-1500k} \Rightarrow -0.693 = -1500k \Rightarrow k = 4.62 \times 10^{-4}.$$

(i) $t = 4500$ years:

$$x = x_0 e^{-4500 \times 4.62 \times 10^{-4}} = x_0 e^{-2.079} = \frac{1}{7.99} x_0 \approx \frac{1}{8} x_0,$$

i.e. 12.5% of the original amount remains.

(ii) $x = \frac{x_0}{10}$:

$$\frac{x_0}{10} = x_0 e^{-4.62 \times 10^{-4} t} \Rightarrow 0.1 = e^{-4.62 \times 10^{-4} t} \Rightarrow -2.3025 = -4.62 \times 10^{-4} t \Rightarrow t \approx 4984 \text{ years.}$$

Problem 12.6. In a bacteria culture the number triples in 5 hours. (i) How many will be present in 10 hours? (ii) When will the number be 10 times the initial number?

Solution. (i) $x = 3x_0$ at $t = 5$ h. From $x = x_0 e^{kt}$:

$$3x_0 = x_0 e^{5k} \Rightarrow 3 = e^{5k} \Rightarrow 1.0986 = 5k \Rightarrow k = 0.2197.$$

At $t = 10$ h: $x = x_0 e^{0.2197 \times 10} = x_0(8.995) \approx 9x_0$.

(ii) $x = 10x_0$:

$$10x_0 = x_0 e^{0.2197t} \Rightarrow \ln 10 = 0.2197t \Rightarrow 2.3025 = 0.2197t \Rightarrow t = 10.48 \text{ hours.}$$

12.3 Electric circuits

In this section we identify the current $I(t)$ in a circuit at any time t by solving the respective differential equation:

(i) **R - L circuit:** $\frac{dI}{dt} + \frac{R}{L} I = \frac{E(t)}{L}$, where R is the resistance, L the inductance and $E(t)$ the voltage in the circuit.

(ii) **R - C circuit:** the charge Q on the capacitor is given by $\frac{dQ}{dt} + \frac{1}{RC} Q = \frac{E(t)}{R}$, and $I = \frac{dQ}{dt}$ (C : capacitance).

(iii) **R - L - C circuit:** $\frac{dI}{dt} + \frac{R}{L} I + \frac{1}{LC} Q = \frac{E(t)}{L}$, or, with $I = \frac{dQ}{dt}$,

$$\frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} Q = \frac{E(t)}{L}.$$

(iv) **L - C circuit** (remove the R term): $\frac{d^2Q}{dt^2} + \frac{1}{LC} Q = \frac{E(t)}{L}$.

Problem 12.7. A voltage Ee^{-at} is applied at time $t = 0$ to a circuit containing inductance L and resistance R . Show that the current at any time t is $\frac{E}{R - aL} (e^{-at} - e^{-Rt/L})$.

Solution. R - L circuit:

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E(t)}{L} = \frac{Ee^{-at}}{L},$$

which is linear ($\frac{dy}{dx} + Py = Q$ with solution $ye^{\int P dx} = \int e^{\int P dx} Q dx + c$):

$$Ie^{\int \frac{R}{L} dt} = \int e^{\int \frac{R}{L} dt} \frac{E}{L} e^{-at} dt + c \Rightarrow Ie^{\frac{R}{L}t} = \frac{E}{L} \int e^{\frac{R}{L}t} e^{-at} dt + c = \frac{E}{L} \int e^{(\frac{R}{L}-a)t} dt + c,$$

$$Ie^{\frac{R}{L}t} = \frac{E}{L} \cdot \frac{e^{(\frac{R}{L}-a)t}}{\frac{R}{L} - a} + c.$$

At $t = 0$, $I = 0$:

$$0 = \frac{E}{L} \cdot \frac{1}{\frac{R}{L} - a} + c \Rightarrow c = -\frac{E}{L} \cdot \frac{1}{\frac{R}{L} - a}.$$

$$Ie^{\frac{R}{L}t} = \frac{E}{L} \cdot \frac{e^{(\frac{R}{L}-a)t}}{\frac{R}{L} - a} - \frac{E}{L} \cdot \frac{1}{\frac{R}{L} - a} = \frac{E}{L} \cdot \frac{1}{\frac{R}{L} - a} [e^{(\frac{R}{L}-a)t} - 1] = \frac{E}{L} \cdot \frac{L}{R - aL} [e^{(\frac{R}{L}-a)t} - 1],$$

$$I = \frac{E}{R - aL} [e^{(\frac{R}{L}-a)t} - 1] \Rightarrow \boxed{I = \frac{E}{R - aL} (e^{-at} - e^{-\frac{R}{L}t})}.$$

Problem 12.8. Find the current at any time $t > 0$ in a circuit having in series a constant electromotive force 40 V, a resistor 10 Ω and an inductor 0.2 H, given that the initial current is 0.

Solution. R - L circuit:

$$\frac{dI}{dt} + \frac{R}{L} I = \frac{E(t)}{L} \Rightarrow \frac{dI}{dt} + \frac{10}{0.2} I = \frac{40}{0.2} \Rightarrow \frac{dI}{dt} + 50I = 200,$$

linear with $P = 50$, $Q = 200$:

$$Ie^{\int 50 dt} = \int e^{\int 50 dt} 200 dt + c \Rightarrow Ie^{50t} = \int e^{50t} 200 dt + c = 200 \cdot \frac{e^{50t}}{50} + c \Rightarrow Ie^{50t} = 4e^{50t} + c.$$

At $t = 0$, $I = 0$: $0 = 4 + c \Rightarrow c = -4$.

$$Ie^{50t} = 4e^{50t} - 4 \Rightarrow I = 4 - 4e^{-50t} \Rightarrow \boxed{I = 4(1 - e^{-50t})}.$$

Problem 12.9. A capacitor $C = 0.01$ F in series with a resistor $R = 20 \Omega$ is charged from a battery $E = 10$ V. Assuming that the capacitor is initially completely uncharged, determine the charge $Q(t)$ and the current $I(t)$.

Solution. R - C circuit:

$$\frac{dQ}{dt} + \frac{1}{RC} Q = \frac{E(t)}{R} \Rightarrow \frac{dQ}{dt} + \frac{1}{20 \times 0.01} Q = \frac{10}{20} \Rightarrow \frac{dQ}{dt} + 5Q = 0.5,$$

with $P = 5$, $Q = 0.5$:

$$Qe^{\int 5 dt} = \int e^{\int 5 dt} (0.5) dt + c \Rightarrow Qe^{5t} = 0.5 \int e^{5t} dt + c = 0.5 \frac{e^{5t}}{5} + c \Rightarrow Qe^{5t} = 0.1e^{5t} + c.$$

At $t = 0$, $Q = 0$: $0 = 0.1 + c \Rightarrow c = -0.1$.

$$Qe^{5t} = 0.1e^{5t} - 0.1 \Rightarrow Q = 0.1 - 0.1e^{-5t} \Rightarrow \boxed{Q = 0.1(1 - e^{-5t})}.$$

$$I = \frac{dQ}{dt} = 0.1(0 + 5e^{-5t}) \Rightarrow \boxed{I = 0.5e^{-5t}}.$$

(Also, the voltage across the capacitor is $V = \frac{Q}{C}$.)

Problem 12.10. A circuit consists of an inductance of 0.05 H, a resistance of 5Ω and a condenser of capacitance 0.0004 F. If $Q = I = 0$ at time $t = 0$, find $Q(t)$ and $I(t)$ when $E = 110$ V.

Solution. R - L - C circuit:

$$\frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} Q = \frac{E}{L} \Rightarrow \frac{d^2Q}{dt^2} + \frac{5}{0.05} \frac{dQ}{dt} + \frac{1}{0.05 \times 0.0004} Q = \frac{110}{0.05},$$

$$\frac{d^2Q}{dt^2} + 100 \frac{dQ}{dt} + 50000 Q = 2200 \Rightarrow (D^2 + 100D + 50000)Q = 2200.$$

C.F. A.E.: $m^2 + 100m + 50000 = 0$,

$$m = \frac{-100 \pm \sqrt{100^2 - 4(1)(50000)}}{2} = \frac{-100 \pm \sqrt{-190000}}{2} = \frac{-100 \pm 100\sqrt{19}i}{2} = -50 \pm i 50\sqrt{19}.$$

$$\text{C.F.} = e^{-50t} (c_1 \cos 50\sqrt{19}t + c_2 \sin 50\sqrt{19}t).$$

P.I. Replace D by 0 ($2200 = 2200e^{0t}$):

$$\text{P.I.} = \frac{1}{D^2 + 100D + 50000} 2200e^{0t} = \frac{2200}{50000} = 0.044.$$

$$Q = e^{-50t} (c_1 \cos 50\sqrt{19}t + c_2 \sin 50\sqrt{19}t) + 0.044.$$

$$I = \frac{dQ}{dt} = e^{-50t} (-c_1 50\sqrt{19} \sin 50\sqrt{19}t + c_2 50\sqrt{19} \cos 50\sqrt{19}t) - 50e^{-50t} (c_1 \cos 50\sqrt{19}t + c_2 \sin 50\sqrt{19}t).$$

Applying the initial conditions.¹⁴ $Q(0) = 0$: $c_1 + 0.044 = 0 \Rightarrow c_1 = -0.044$. $I(0) = 0$: $50\sqrt{19}c_2 - 50c_1 = 0 \Rightarrow c_2 = \frac{c_1}{\sqrt{19}} = -\frac{0.044}{\sqrt{19}}$. Hence

$$Q = 0.044 - 0.044 e^{-50t} \left(\cos 50\sqrt{19}t + \frac{1}{\sqrt{19}} \sin 50\sqrt{19}t \right),$$

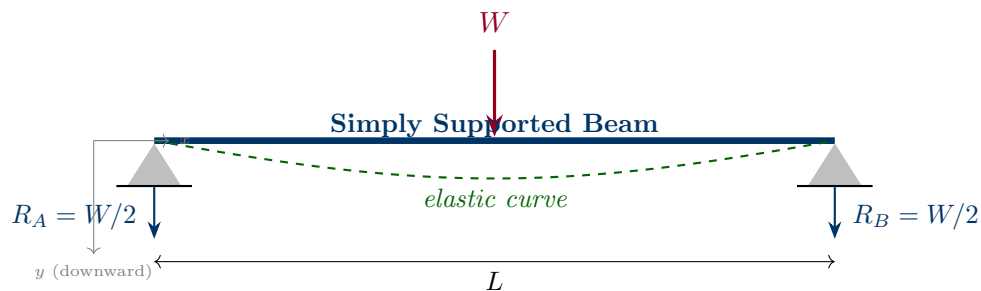
$$I = \frac{dQ}{dt} = \frac{0.044 \times 50 \times 20}{\sqrt{19}} e^{-50t} \sin 50\sqrt{19}t = \frac{44}{\sqrt{19}} e^{-50t} \sin 50\sqrt{19}t.$$

12.4 Bending of Beams

The deflection of loaded beams is an important engineering application of higher-order linear ODEs with constant coefficients. The elastic curve $y(x)$ of a beam satisfies a linear ODE of order 2 (in terms of the bending moment) or order 4 (in terms of the load), which is solved by successive integration together with the boundary conditions at the supports.

12.4.1 Introduction to Beam Deflection

A **beam** is a structural member that resists forces and bending moments. When a load is applied, the beam bends and its axis deforms into a curve called the **elastic curve** or **deflection curve**.



Key Physical Quantities

Symbol	Quantity	Unit
E	Young's modulus (material stiffness)	N/m ² (Pa)
I	Second moment of area (cross-section shape)	m ⁴
EI	Flexural rigidity	N·m ²
$y(x)$	Deflection at position x (downward +ve)	m
$M(x)$	Bending moment at x	N·m
$w(x)$	Distributed load intensity	N/m
W	Point load	N

12.4.2 The Governing Differential Equation

Euler–Bernoulli Beam Equation The fundamental relationship between curvature and bending moment, under the *small-deflection* assumption $|y'| \ll 1$, gives:

¹⁴The notes stop after writing Q and I in terms of c_1, c_2 ; the conditions $Q = I = 0$ at $t = 0$ are applied here.

Moment–Curvature Relation (2nd-order ODE):

$$EI \frac{d^2 y}{dx^2} = M(x)$$

Load–Deflection Relation (4th-order ODE):

$$EI \frac{d^4 y}{dx^4} = w(x)$$

The complete chain of derivatives is:

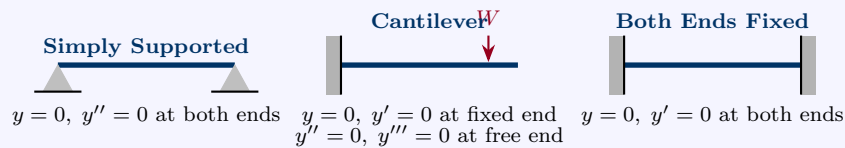
$$EI y'''' = w(x) \xrightarrow{\int} EI y''' = S(x) \xrightarrow{\int} EI y'' = M(x) \xrightarrow{\int} EI y' = \theta(x) \xrightarrow{\int} EI y = \text{deflection}$$

where $S(x)$ is the shear force and $\theta(x) = dy/dx$ is the slope.

Sign Convention: y is measured *downward* (positive in the direction of load). Bending moment $M > 0$ causes *sagging* (concave up).

12.4.3 Boundary Conditions

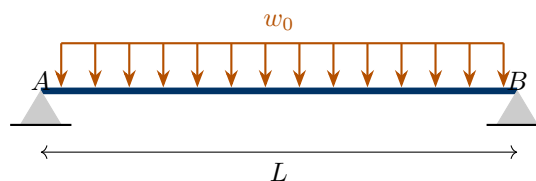
The number of boundary conditions (BCs) required equals the order of the ODE used.

Standard Boundary Conditions

Support type	y	y'	y''	y'''
Pinned / Roller (simply supported)	0	free	0	free
Fixed (clamped) end	0	0	free	free
Free end	free	free	0	0

12.4.4 Standard Solved Examples (Concepts Illustrated)

Example 1 — Simply Supported Beam with UDL **Setup:** Beam of length L , simply supported at both ends, uniform load w_0 per unit length (downward).



Governing ODE (4th order): $EI y'''' = w_0$

Boundary conditions: $y(0) = 0$, $y''(0) = 0$, $y(L) = 0$, $y''(L) = 0$.

Solution: Integrate four times:

$$\begin{aligned} EI y''' &= w_0 x + C_1 \\ EI y'' &= \frac{w_0}{2} x^2 + C_1 x + C_2 \\ EI y' &= \frac{w_0}{6} x^3 + \frac{C_1}{2} x^2 + C_2 x + C_3 \\ EI y &= \frac{w_0}{24} x^4 + \frac{C_1}{6} x^3 + \frac{C_2}{2} x^2 + C_3 x + C_4 \end{aligned}$$

Apply BCs: $y(0) = 0 \Rightarrow C_4 = 0$; $y''(0) = 0 \Rightarrow C_2 = 0$.

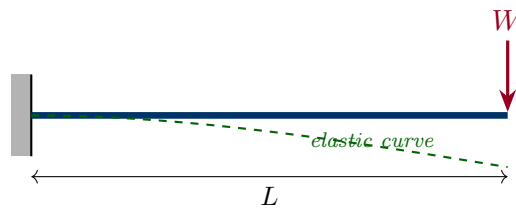
$y''(L) = 0$: $\frac{w_0}{2} L^2 + C_1 L = 0 \Rightarrow C_1 = -\frac{w_0 L}{2}$.

$y(L) = 0$: $\frac{w_0 L^4}{24} - \frac{w_0 L^4}{12} + C_3 L = 0 \Rightarrow C_3 = \frac{w_0 L^3}{24}$.

$$y(x) = \frac{w_0}{24EI} (x^4 - 2Lx^3 + L^3x)$$

Maximum deflection at $x = L/2$: $\delta_{\max} = \frac{5w_0 L^4}{384 EI}$

Example 2 — Cantilever with Point Load at Free End **Setup:** Clamped at $x = 0$, free at $x = L$, point load W at $x = L$.



Bending moment: $M(x) = -W(L - x)$ (hogging, sign adjusted for downward y). Using $EI y'' = W(L - x)$:

Integrate: $EI y' = W\left(Lx - \frac{x^2}{2}\right) + C_1$. BC $y'(0) = 0 \Rightarrow C_1 = 0$.

Integrate: $EI y = W\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right) + C_2$. BC $y(0) = 0 \Rightarrow C_2 = 0$.

$$y(x) = \frac{W}{6EI} (3Lx^2 - x^3) \quad \delta_{\max} = \frac{WL^3}{3EI} \text{ at } x = L$$

12.4.5 Summary of Important Results

Maximum Deflection Formulae		
Beam & Load	Location	$\delta_{\max}$
Simply supported, UDL w_0	Centre ($x = L/2$)	$\frac{5w_0L^4}{384EI}$
Simply supported, central W	Centre	$\frac{WL^3}{48EI}$
Cantilever, point W at tip	Free end	$\frac{WL^3}{3EI}$
Cantilever, UDL w_0	Free end	$\frac{w_0L^4}{8EI}$
Both ends fixed, UDL w_0	Centre	$\frac{w_0L^4}{384EI}$

12.4.6 Short Questions (2-Mark Type)

Short Question 1. Write the Euler–Bernoulli beam equation relating deflection $y(x)$ to bending moment $M(x)$ and to distributed load $w(x)$.

Solution. The two standard forms are:

$$EI \frac{d^2y}{dx^2} = M(x) \quad (\text{moment form, 2nd-order ODE})$$

$$EI \frac{d^4y}{dx^4} = w(x) \quad (\text{load form, 4th-order ODE})$$

where E = Young's modulus, I = second moment of area, and $y(x)$ = downward deflection. The chain of derivatives connects load $\rightarrow$ shear $\rightarrow$ moment $\rightarrow$ slope $\rightarrow$ deflection.

Short Question 2. State the boundary conditions for (a) a simply supported beam and (b) a cantilever (clamped–free) beam, each of length L .

Solution. (a) **Simply Supported** (supported at $x = 0$ and $x = L$, using the 4th-order ODE):

$$y(0) = 0, \quad y''(0) = 0, \quad y(L) = 0, \quad y''(L) = 0.$$

Reason: Zero deflection and zero bending moment at each pin/roller support.

(b) **Cantilever** (clamped at $x = 0$, free at $x = L$):

$$y(0) = 0, \quad y'(0) = 0, \quad y''(L) = 0, \quad y'''(L) = 0.$$

Reason: Fixed end has zero deflection and zero slope; free end has zero bending moment and zero shear force.

Short Question 3. For a cantilever beam of length L with a point load W at the free end, the deflection is $y(x) = \frac{W}{6EI}(3Lx^2 - x^3)$. Find the maximum deflection.

Solution. The deflection is maximum at the free end, i.e. at $x = L$:

$$y(L) = \frac{W}{6EI}(3L \cdot L^2 - L^3) = \frac{W}{6EI}(3L^3 - L^3) = \frac{W}{6EI} \cdot 2L^3.$$

$$\delta_{\max} = \frac{WL^3}{3EI}.$$

Short Question 4. The deflection of a simply supported beam under UDL w_0 is $y(x) = \frac{w_0}{24EI}(x^4 - 2Lx^3 + L^3x)$. Find the slope $\theta(x) = dy/dx$ and evaluate it at $x = 0$.

Solution. Differentiate:

$$EI\theta = \frac{w_0}{24}(4x^3 - 6Lx^2 + L^3),$$

$$\theta(x) = \frac{w_0}{24EI}(4x^3 - 6Lx^2 + L^3).$$

At $x = 0$:

$$\theta(0) = \frac{w_0L^3}{24EI}.$$

This is the maximum slope (at the supports) for this symmetric problem.

Short Question 5. Identify the type of ODE used to model a beam and explain why the constants of integration equal the number of boundary conditions needed.

Solution. **Type:** The beam equation $EI y'' = M(x)$ is a **2nd-order linear ODE** with constant coefficients ($EI = \text{const}$); the load form $EI y^{(4)} = w(x)$ is a **4th-order** one.

Constants of integration:

- Each integration introduces one arbitrary constant.
- Integrating a 2nd-order ODE twice gives 2 constants $\Rightarrow$ 2 BCs needed.
- Integrating a 4th-order ODE four times gives 4 constants $\Rightarrow$ 4 BCs needed.

The BCs (deflection, slope, moment, shear at the supports) uniquely determine all constants, giving the unique solution for the elastic curve.

12.4.7 Long Problems (8-Mark Type)

Problem 1 — Simply Supported Beam, Uniformly Distributed Load. A simply supported beam of length L carries a uniformly distributed load w_0 per unit length over its entire span. Using the 4th-order ODE $EI \frac{d^4y}{dx^4} = w_0$, find the deflection curve $y(x)$ and the maximum deflection. (*Assume constant EI .*)

Solution. **Step 1: Set up.** Boundary conditions (simply supported at $x = 0$ and $x = L$):

$$y(0) = 0, \quad y''(0) = 0, \quad y(L) = 0, \quad y''(L) = 0.$$

Step 2: Integrate four times.

$$EI y''' = w_0 x + C_1 \quad (1)$$

$$EI y'' = \frac{w_0}{2} x^2 + C_1 x + C_2 \quad (2)$$

$$EI y' = \frac{w_0}{6} x^3 + \frac{C_1}{2} x^2 + C_2 x + C_3 \quad (3)$$

$$EI y = \frac{w_0}{24} x^4 + \frac{C_1}{6} x^3 + \frac{C_2}{2} x^2 + C_3 x + C_4 \quad (4)$$

Step 3: Apply boundary conditions.

$$BC1: y(0) = 0 \text{ in (4)} \Rightarrow C_4 = 0.$$

$$BC2: y''(0) = 0 \text{ in (2)} \Rightarrow C_2 = 0.$$

$$BC3: y''(L) = 0 \text{ in (2):}$$

$$\frac{w_0 L^2}{2} + C_1 L = 0 \Rightarrow C_1 = -\frac{w_0 L}{2}.$$

$$BC4: y(L) = 0 \text{ in (4) (with } C_2 = C_4 = 0, C_1 = -w_0 L/2):$$

$$\frac{w_0 L^4}{24} - \frac{w_0 L}{2} \cdot \frac{L^3}{6} + C_3 L = 0 \Rightarrow C_3 = \frac{w_0 L^3}{24}.$$

Step 4: Deflection curve.

$$y(x) = \frac{w_0}{24EI} (x^4 - 2Lx^3 + L^3x)$$

Step 5: Maximum deflection.

By symmetry, maximum occurs at $x = L/2$:

$$y\left(\frac{L}{2}\right) = \frac{w_0}{24EI} \left(\frac{L^4}{16} - 2L \cdot \frac{L^3}{8} + L^3 \cdot \frac{L}{2} \right) = \frac{w_0}{24EI} \left(\frac{L^4}{16} - \frac{L^4}{4} + \frac{L^4}{2} \right) = \frac{w_0}{24EI} \cdot \frac{5L^4}{16}.$$

$$\delta_{\max} = \frac{5w_0 L^4}{384 EI}$$

Problem 2 — Cantilever Beam with UDL. A cantilever beam of length L is clamped at $x = 0$ and free at $x = L$. It carries a uniformly distributed load w_0 per unit length. Using $EI y^{(4)} = w_0$, find the deflection $y(x)$ and the maximum deflection (at the free end).

Solution. **Boundary conditions:** $y(0) = 0, y'(0) = 0$ (clamped end); $y''(L) = 0, y'''(L) = 0$ (free end, no moment/shear).

Integrate four times:

$$EI y''' = w_0 x + C_1$$

$$EI y'' = \frac{w_0}{2} x^2 + C_1 x + C_2$$

$$EI y' = \frac{w_0}{6} x^3 + \frac{C_1}{2} x^2 + C_2 x + C_3$$

$$EI y = \frac{w_0}{24} x^4 + \frac{C_1}{6} x^3 + \frac{C_2}{2} x^2 + C_3 x + C_4$$

Apply BCs:

$$y'''(L) = 0: w_0 L + C_1 = 0 \Rightarrow C_1 = -w_0 L.$$

$$y''(L) = 0: \frac{w_0 L^2}{2} - w_0 L^2 + C_2 = 0 \Rightarrow C_2 = \frac{w_0 L^2}{2}.$$

$$y'(0) = 0: C_3 = 0.$$

$$y(0) = 0: C_4 = 0.$$

Deflection curve:

$$EI y = \frac{w_0}{24} x^4 - \frac{w_0 L}{6} x^3 + \frac{w_0 L^2}{4} x^2.$$

$$y(x) = \frac{w_0}{24EI} (x^4 - 4Lx^3 + 6L^2x^2)$$

Maximum deflection at $x = L$:

$$y(L) = \frac{w_0}{24EI} (L^4 - 4L^4 + 6L^4) = \frac{w_0}{24EI} \cdot 3L^4.$$

$$\delta_{\max} = \frac{w_0 L^4}{8EI}$$

Problem 3 — Simply Supported Beam with Central Point Load. A beam of length L is simply supported at $x = 0$ and $x = L$. A point load W acts at the centre $x = L/2$. For $0 \leq x \leq L/2$, the bending moment is $M(x) = \frac{W}{2}x$. Using $EI y'' = M(x)$, find the deflection for $0 \leq x \leq L/2$ and hence the maximum deflection.

Solution. **Governing ODE** for $0 \leq x \leq L/2$:

$$EI y'' = \frac{W}{2}x.$$

Integrate:

$$EI y' = \frac{W}{4}x^2 + A.$$

$$EI y = \frac{W}{12}x^3 + Ax + B.$$

Boundary conditions:

$$y(0) = 0 \Rightarrow B = 0.$$

By symmetry, the slope at mid-span is zero: $y'(L/2) = 0$:

$$\frac{W}{4} \cdot \frac{L^2}{4} + A = 0 \Rightarrow A = -\frac{WL^2}{16}.$$

Deflection for $0 \leq x \leq L/2$:

$$y(x) = \frac{W}{48EI} (4x^3 - 3L^2x)$$

Note: the standard form is written as $y(x) = \frac{W}{48EI}(4x^3 - 3L^2x)$; the negative sign means deflection is downward.

Maximum deflection at $x = L/2$:

$$y\left(\frac{L}{2}\right) = \frac{W}{48EI} \left(4 \cdot \frac{L^3}{8} - 3L^2 \cdot \frac{L}{2}\right) = \frac{W}{48EI} \left(\frac{L^3}{2} - \frac{3L^3}{2}\right) = \frac{W}{48EI} \cdot (-L^3).$$

$$\delta_{\max} = \frac{WL^3}{48EI} \quad (\text{magnitude})$$

Problem 4 — Cantilever with Linearly Varying (Triangular) Load. A cantilever beam of length L is clamped at $x = 0$ and free at $x = L$. The load intensity varies linearly from w_0 at the fixed end to 0 at the free end: $w(x) = w_0\left(1 - \frac{x}{L}\right)$. Using $EI y^{(4)} = w(x)$, find the deflection curve $y(x)$ and the maximum deflection at the free end.

Solution. **ODE:**

$$EI y^{(4)} = w_0 \left(1 - \frac{x}{L}\right).$$

Boundary conditions: $y(0) = 0$, $y'(0) = 0$ (clamped); $y''(L) = 0$, $y'''(L) = 0$ (free).

Integrate successively:

$$\begin{aligned} EI y''' &= w_0 \left(x - \frac{x^2}{2L}\right) + C_1 \\ EI y'' &= w_0 \left(\frac{x^2}{2} - \frac{x^3}{6L}\right) + C_1 x + C_2 \\ EI y' &= w_0 \left(\frac{x^3}{6} - \frac{x^4}{24L}\right) + \frac{C_1}{2} x^2 + C_2 x + C_3 \\ EI y &= w_0 \left(\frac{x^4}{24} - \frac{x^5}{120L}\right) + \frac{C_1}{6} x^3 + \frac{C_2}{2} x^2 + C_3 x + C_4 \end{aligned}$$

Apply free-end BCs:

$$y'''(L) = 0: w_0 \left(L - \frac{L}{2}\right) + C_1 = 0 \Rightarrow C_1 = -\frac{w_0 L}{2}.$$

$$y''(L) = 0: w_0 \left(\frac{L^2}{2} - \frac{L^2}{6}\right) - \frac{w_0 L}{2} \cdot L + C_2 = 0$$

$$\Rightarrow w_0 \frac{L^2}{3} - \frac{w_0 L^2}{2} + C_2 = 0 \Rightarrow C_2 = \frac{w_0 L^2}{6}.$$

Apply clamped-end BCs:

$$y'(0) = 0 \Rightarrow C_3 = 0; \quad y(0) = 0 \Rightarrow C_4 = 0.$$

Deflection curve:

$$y(x) = \frac{w_0}{EI} \left(\frac{x^4}{24} - \frac{x^5}{120L} - \frac{Lx^3}{12} + \frac{L^2x^2}{12}\right)$$

Maximum deflection at $x = L$:

$$y(L) = \frac{w_0}{EI} \left(\frac{L^4}{24} - \frac{L^4}{120} - \frac{L^4}{12} + \frac{L^4}{12}\right) = \frac{w_0 L^4}{EI} \left(\frac{1}{24} - \frac{1}{120}\right) = \frac{w_0 L^4}{EI} \cdot \frac{5-1}{120}.$$

$$\delta_{\max} = y(L) = \frac{w_0 L^4}{30EI}$$

Problem 5 — Fixed-Fixed Beam with UDL (Both Ends Clamped). A beam of length L is clamped at both ends. A uniform load w_0 acts throughout. Use $EI y^{(4)} = w_0$ and boundary conditions $y(0) = y'(0) = y(L) = y'(L) = 0$ to find the deflection curve and the maximum deflection.

Solution. **Boundary conditions:** $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y'(L) = 0$.

Integrate four times:

$$EI y''' = w_0 x + C_1 \quad (i)$$

$$EI y'' = \frac{w_0}{2} x^2 + C_1 x + C_2 \quad (ii)$$

$$EI y' = \frac{w_0}{6} x^3 + \frac{C_1}{2} x^2 + C_2 x + C_3 \quad (iii)$$

$$EI y = \frac{w_0}{24} x^4 + \frac{C_1}{6} x^3 + \frac{C_2}{2} x^2 + C_3 x + C_4 \quad (iv)$$

Apply $y(0) = 0$ in (iv): $C_4 = 0$.

Apply $y'(0) = 0$ in (iii): $C_3 = 0$.

Apply $y'(L) = 0$ in (iii):

$$\frac{w_0 L^3}{6} + \frac{C_1}{2} L^2 + C_2 L = 0. \quad (A)$$

Apply $y(L) = 0$ in (iv):

$$\frac{w_0 L^4}{24} + \frac{C_1 L^3}{6} + \frac{C_2 L^2}{2} = 0. \quad (B)$$

Solve (A) and (B):

$$\text{From (B): } C_1 L + 4C_2 = -\frac{w_0 L^2}{4}. \quad \text{From (A): } C_1 L + 2C_2 = -\frac{w_0 L^2}{3}.$$

$$\text{Subtract: } 2C_2 = \frac{w_0 L^2}{12} \Rightarrow C_2 = \frac{w_0 L^2}{24}.$$

$$\text{Then: } C_1 = \frac{-w_0 L^2/3 - 2C_2}{L} = \frac{-w_0 L^2/3 - w_0 L^2/12}{L} = -\frac{w_0 L}{4}.$$

Deflection curve:

$$EI y = \frac{w_0}{24} x^4 - \frac{w_0 L}{12} x^3 + \frac{w_0 L^2}{24} x^2 = \frac{w_0}{24} x^2 (x^2 - 2Lx + L^2) = \frac{w_0}{24} x^2 (x - L)^2.$$

$$y(x) = \frac{w_0}{24EI} x^2 (x - L)^2$$

Maximum deflection at $x = L/2$:

$$y\left(\frac{L}{2}\right) = \frac{w_0}{24EI} \cdot \frac{L^2}{4} \cdot \frac{L^2}{4} = \frac{w_0 L^4}{384EI}.$$

$$\delta_{\max} = \frac{w_0 L^4}{384EI}$$

Observation: The fixed-fixed beam deflects *eight times less* than a simply supported beam under the same load ($\delta_{\max}^{\text{SS}} = \frac{5w_0L^4}{384EI}$ vs. $\delta_{\max}^{\text{FF}} = \frac{w_0L^4}{384EI}$), demonstrating the stiffening effect of clamped supports.

12.4.8 Quick-Reference Summary

